OBSTETRICS GYNAECOLOGY

WORKBOOK

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
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DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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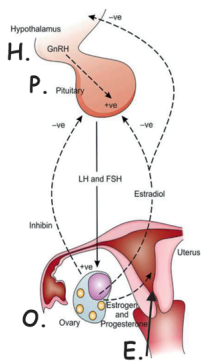
I

GYNAECOLOGY

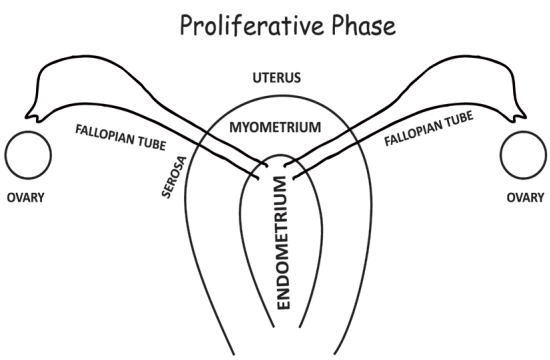
GYNAECOLOGY

Menstrual Cycle



DR. AMIT GUPTA AT 9 PM

H. P. O. E. Axis



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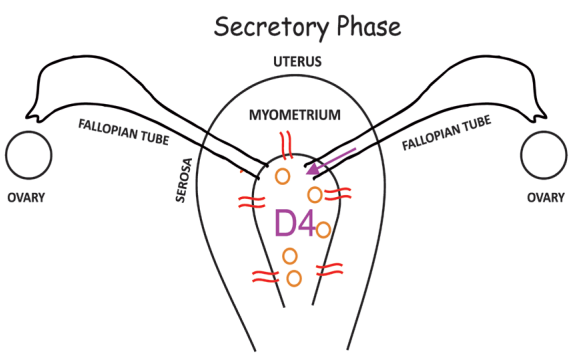

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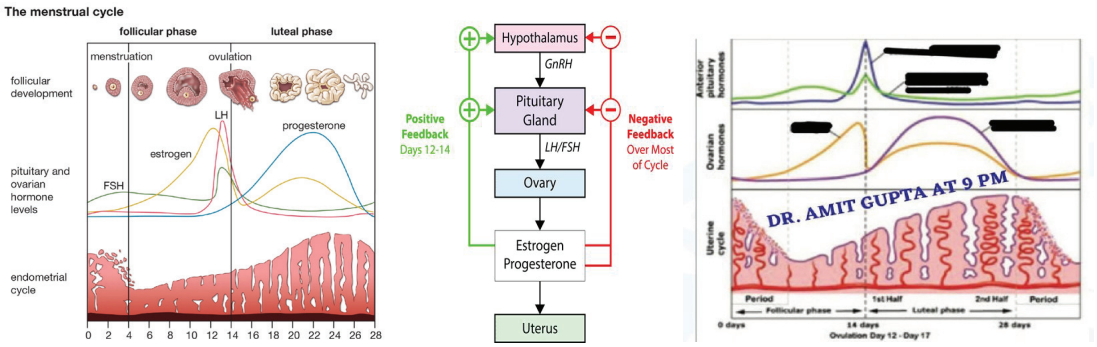


(H.P.O.E. Axis) Hypothalamo - Pituitary - Ovarian - Endometrial Axis

Meaning of Menstruation - A lady gets regular (cyclical or monthly) vaginal bleeding every month in reproductive age.

Purpose - Nature gives a chance for ..... every month. So endometrium (bed of Fetus) is prepared (proliferated) every month. If fertilization occur in this month , this endometrium should stay inside. If fertilization ..... occur in this month, this endometrium is expelled out in the form of vaginal bleeding (.....). This is why first symptom of pregnancy is .....

Active space

How does menstruation occur - Normal Hormonal functioning and coordination among hypothalamus-Pituitary-Ovary-Endometrium is responsible for normal menstruation. This is why normal functioning of hypothalamus-pituitary-ovary and endometrium is required for normal menstruation. This is why this hormonal axis is called Hypothalamo-

Pituitary-Ovarian -Endometrial axis (H.P.O.E. Axis)

Events in the functioning of H.P.O.E. Axis obviously start from .....
.....

Events -

Hypothalamus secretes GnRH - It acts on anterior pituitary

Anterior pituitary secretes FSH and LH

FSH is mainly responsible for follicular growth.

LH is mainly responsible for ovulation.

FSH acts on ovarian follicle and causes follicular growth (stimulation). FSH acts on granulosa cells of ovarian follicle which secrete estradiol and inhibin B. Ovarian follicle grows at the rate of 2 mm/day.

Follicular growth pattern

Estradiol has three actions- actions of estradiol

LH is increasing. LH surge - peak - ovulation.

Actions of LH

Ovulation

Corpus luteum

Estrogen proliferates endometrium for implantation. Progesterone hold it , increases blood supply to endometrium and prepares first nutrition for fertilized ovum so as the name is.

Active space

Maximum amount of serum progesterone is on day ..... postovulation because activity of corpus luteum is maximum on day ..... postovulation. If fertilization did not occur in this month (cycle), nature want to .... this endometrium out and ..... endometrium will be

proliferated (prepared) next month for another chance of fertilization and interstitial implantation. So this ..... is expelled out in the form of vaginal bleeding .....

Functions of FSH:-

1. To select a cohort (Group) of follicles every month for Growth.
   (Pre antral stages are not dependent on FSH)

2. Selection & Growth of dominant follicle = on day ..... of cycle

3. Final phase of ovum release (Collagen breakdown)

Functons of LH :-

1. Ovulation 2. Formation & maintenance of .....

3) Final growth of follicle

Estrogen in normal amounts ..... feedback to LH

Estrogen in higher amounts ..... feedback to LH (neuroendocrine phenomenon)

Progesterone in small amount ..... feedback to LH/FSH

Progesterone in high amount ..... feedback to LH/FSH

Summarisation of events :-

Hypothalamus Secretes
GnRH in pulsatile fashion
which acts on anterior pituitary

↓

Anterior Pituitary
Secretes FSH and LH

1 in ..... Frequency pulse → FSH

1 in ..... Frequency pulse → LH

Active space

Follicular growth pattern

initially many follicles start growing. as soon as one follicle attains .....size, other follicles start regressing

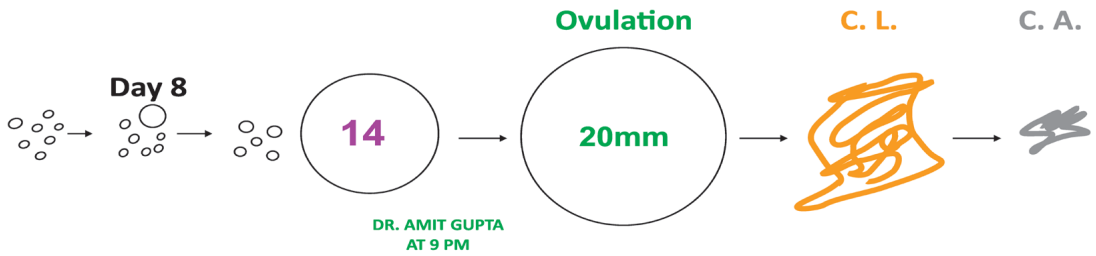
Selection & growth of dominant follicle ..... of cycle due to .....

Follicular growth / day .....

Size of dominant follicle .....

Size of mature follicle .....

Follicle ruptures at .....



FSH → Ovarian granulosa cells

Estradiol Inhibin-B - -Ve feedback to .....

Actions of estradiol

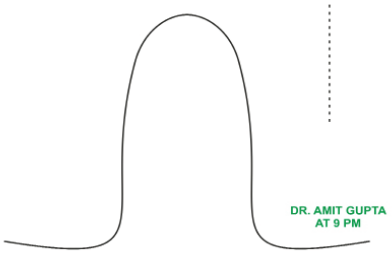
1. ↑ Luteinising hormone (positive feedback only before ovulation, otherwise negative feedback)
2. Proliferation of endometrial glands → Thickening
3. Negative feedback to FSH

Actions of LH

1. Ovulation via LH surge & LH peak

A. LH surge - due to .....

..... hours before ovulation



B. LH Peak - Attained ..... before ovulation

Active space

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When minimum amount of LH required for ovulation is attained

Minimum amount of LH required for ovulation .....

$1\ pg = 10^{-12} gm \quad 1\ ng = 10^{-9} gm$

Minimum Estradiol required for LH surge .....which should sustain for .....

2. LH acts on theca cells - Secrete Androgens

↓ Aromatisation (maximum in.....)

Esterone (E1)

3. Acts on Granulosa cells - Small Amount of Progesterone

↑ FSH & ↑ LH (Mid cycle .....) help in .....

Ovulation occur due to .....

Synthesis of Progesterone begins ..... before ovulation
LH surge - leutinization of Granulosa cells
Start producing Progesterone in small amounts

After ovulation, Ruptured follicle .....

at ovulation ..... released from ovary

Corpus Luteum is Supported by .....

Maximum size & activity of corpus luteum is on day .....

hCG is structurally & functionally almost similar to .....

Minimum levels of FSH / LH are on day .....

Pregnant Corpus Luteum is Supported by .....

Active space

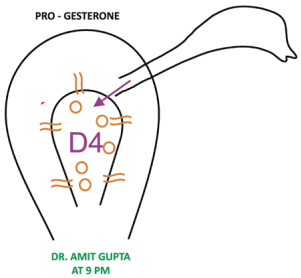
Corpus Luteum Secretes

3 Hormones in non pregnant female

1. Estradiol

2. Progesterone - .....
3. Inhibin-A
4. Hormones in pregnant female

.....



Actions of progesterone

1. Support / hold/ stabilise / maintain endometrium

2. Secretion of glycogen/ lipids which are stored in tiny- numerous vacuoles in liquid form.

   Earliest histological feature of secretory endometrium

   .....

   Late histological feature of secretory endometrium

   after 2-3 days vacuoles become heavy & tortuous and are called

   .....

3. Vasodilatation

   Because progesterone is smooth .....

4. Antiproliferative effect

- Activity of corpus luteum is maximum on day .... post ovulation.
- Levels of progesterone Maximum on Day ..... post ovulation ≥ .....

Negative feedback to LH

↓

Corpus luteum degenerates (Degenerated corpus luteum → Corpus Albicans)

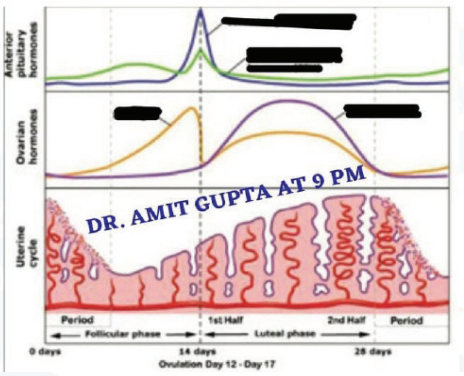
↓

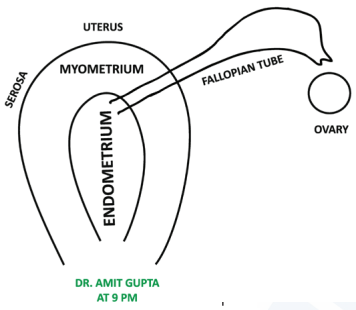
↓ Progesterone, ↓ Estrogen, ↓ Inhibin-A

Active space

Events due to ↓ progesterone

- Not able to hold endometrium
- Vasoconstriction
- Metalloproteinase





Menstruation

Endometrial shedding - Expelled out by .....

↓

Blood (Mainly .....)

Menstruation is - .....

A lady gets regular monthly vaginal bleeding due to .....

Hormone Acting on Postovulatory Endometrium is .....

Events due to fall of estrogen & Inhibin - A .....

Natural Estrogens - 4

- E1 → Esterone (from aromatisation)
- E2 → Estradiol - Main in reproductive age (pregnant or nonpregnant)
- E3 → Estriol-pregnancy (Synthesized by ..... utilising .....
- E4 → Esteterol - Pregnancy
- ..... may be taken as marker of fetal well being

↓

Maximum amount of estrogen in reproductive age .....

Maximum amount of estrogen in menopausal lady .....

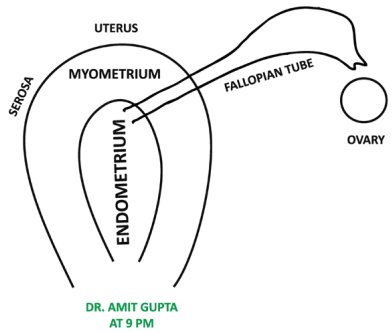
Most potent estrogen → Ethinyl Estradiol > Conjugated equine estrogen > E2 > E1 > E3 > E4

Active space

Most important hormone responsible for menstruation.....

1st half/ Estrogenic phase/ Proliferative phase/ Follicular phase

2nd half/ Progesteronic/Secretory / Luteal phase



Mittelshmerz (Mid cycle .....)

_____

Pre menstrual tension (syndrome)

- Approximately ..... days before menses

MC causes - .....

Manegement .....

Best time to do Breast Self-Exam

..... Menstruation

MC Cancer in females →

Day of ovulation → Cycle length - ..... because ..... half of cycle is .....

Eg. in ..... day cycle

Day of ovulation .....

Active space

Levels of progesterone are maximum on ..... day in ..... day cycle

Normal parameters - according to FIGO (International federation of Gynaecologists & obstetrician)

1. Cycle Length / Frequency .....
2. Amount of blood loss in single episode of menstruation ....
3. Duration of bleeding .....
4. Mean cycle length .....
5. Cycle to cycle variation .....

Menarche- ..... of menses

Menopause- ..... of menses

Reproductive period- period between menarche and menopause

Age of Menarche ..... (mean ..... years)

Active space

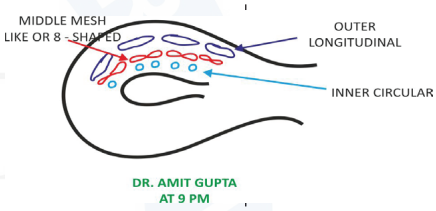
ACTIONS OF ESTROGEN AND PROGESTERONE

Actions of Estrogen and progesterone

Estrogen causes .....

Progesterone causes .....

Progesterone acts only on .....

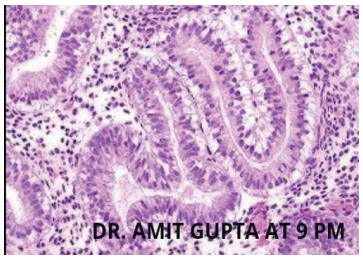


1. On myometrium
   - Estrogen ..... myometrial contraction,
     ..... no. of cells in myometrium
   - Progesterone ..... myometrium contraction
     ..... Estrogen in increase in cells
   - Uterine size ↑ Due to .....

2. On endometrium
   Estrogen ..... of endometrial glands
   Progesterone .....

Histology →

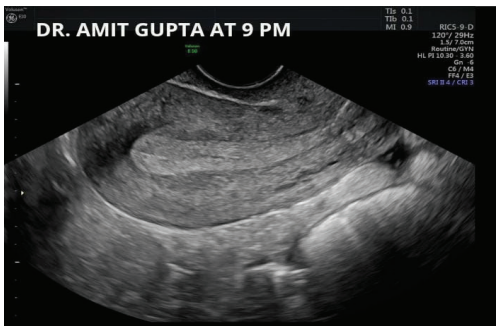
Proliferative



Early Secretory

Late Secretory

Active space



Proliferative endometrium .....

a. Stratum / zona .....

b. Stratum / zona .....

c. Stratum / zona .....

..... Not shed - off during menses because it is ... .

Secretory endometrium .....

3. On cervix

Estrogen→cervical mucus →copious ....., watery, profuse, alkaline,

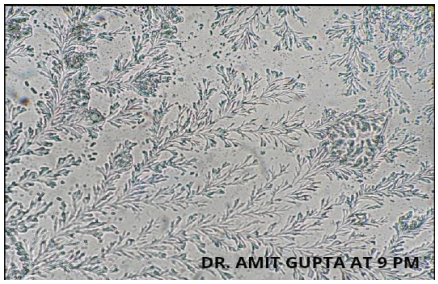
elastic-can be stretched upto.....and ..... break .....

Progesterone →....., scanty, non elastic, ...

Microscopy

Estrogen- .....

progestrone .....



Cervical mucus good for sperm penetration is .....

Cervical mucus impermeable for Microorganisms is .....

Active space

4. On vagina:

Epithelium

- Stratified squamous non keratinized epithelium

↓

3 types of cells

↓

Basal Parabasal	Intermediate cells	Superficial or cornified cells or Eosinophilic
Abundant when ..... hormonal predominance dominance - before puberty - after menopause	Abundant when progesterone - secretory phase - Pregnancy	Abundant when Estrogen - proliferative phase

Maturation index Ratio/100 cells

In I half → _____

In II Half → _____

In Pregnancy → _____

Before Puberty → _____

After ovulation-maturation index shifts from .....

Cornification index or eosinophilic index .....

Maximum in .....

Active space

5. On Breast

Estrogen .....

Progesterone .....

Breast size ↑ due to .....

Eg. in Precocious puberty

6. On Bone

Estrogen .....

Progesterone .....

7. On Fallopian tube

Estrogen .....

Progesterone .....

8. Coagulation

Estrogen .....

Progesterone .....

Active space

DEVELOPMENT OF MALE AND FEMALE

- Depends on presence/Absence of SRY complex or sex determining Region
- SRY complex is situated on Short arm of.....
- If SRY complex is present →.....
- Testis is made by .....
- Ovary is made by .....
- Testis appear at ..... weeks of Intrauterine life
- Ovary appear at ..... weeks of Intrauterine life

Male & female internal genitalia are differentiated by ..... of IUL

Male & female external genitalia can be differentiated by ..... of IUL

Testis -

1. Sertoli cells: secrete mullerian inhibiting factor or Anti - Mullerian hormone & Inhibin

↓

Suppression of mullerian ducts

↓

Remanant A. Appendix of testis or hydatid of morgagni

B. Prostatic utricle

Active space

2. Leydig cells :----------

hCG From placenta is first stimulus for
Secretion of testosterone in fetus

Testosterone in Females
- 50% from peripheral Conversion
- 25% from Adrenal gland
- 25% from ovary

Testosterone

↓

5 a reductase

↓

Dihydrotestosterone

↓

Male external genital organs

Wolffian duct

↓

Male internal genital organs
- Epididymis
- Seminal vesicle
- Vas deferens
- Ejaculatory duct

AMH : Testosterone are secreted by .....

If SRY complex is absent→ .....

1. No Sertoli cells: no .....

Mullerian duct .....

↓

Female internal genital organs .....

i. uterus

ii. cervix

iii. fallopian tube

iv. upper part of vagina

Active space

Lower part of vagina is formed by sinovaginal bulb of urogenital sinus

2. No Leydig cells.....No.....testosterone

Suppression of .....

Remanants .....

A. From upper part .....

B. From middle part .....

C. From lower part .....

No

.....

Active space

Puberty

(In hypothalamus) > (In adipose tissues) Stimulate release of .....
..... Hypothalamo - Pitutary - Ovarian axis
(H.P.O. Axis)

Puberty- stage of development of secondary sexual characters.

secondary sexual characters are governed by -
in males .....
in females.....

Age in female→ ---------- years (mean- --------- years)

Normal time interval between onset of puberty & menarche .....

Functions of Androgens in female :-

1. Pubic & axillary hair growth = Pubarche
2. Libido (Sexual drive)
3. Right amount of androgens required - to from Estrogen by granulosa cells

Sequence of changes -

1. Earliest sign ------ Growth Spurt
2. Earliest visible sign ------Thelarche (Breast Budding)
3. Pubarche ------ Pubic Hairs
4. Peak Height Velocity
5. Menarche

Tanner Staging → to Explain secondary sexual characters

Tanner Staging 1, 2 → Under Developed secondary sexual characters

Active space

Tanner Staging 4,5→Properly Developed Secondary sexual characters

Hair growth is not estrogen dependent pubertal change in females

Precocious puberty - If secondary sexual characters develop before ... . years

Types -

1. Central precocious puberty - premature activation of ..... LH/FSH .....

   - MC cause .....

   - Mx - GnRH Agonist in .....

Principle of management -

To increase activity of Hypothalamo - pituitary - ovarian axis - GnRH agonist in .....

To decrease activity of Hypothalamo - pituitary - ovarian axis - GnRH agonist in .....

2. Peripheral precocious puberty - ..... activation of ..... but sexhormones d/t Peripheral cause Eg. estrogen secreting tumor LH/FSH .....

Precocious menarche - If menses begin before .....years

- MC cause - .....
- Mx - GnRH Agonist in .....

Delayed puberty - If secondary sexual characters don't develop even ..... years of age

- MC cause -idiopathic constitutional delay
- Mx - GnRH Agonist in .....

Active space

ABNORMALITIES OF MENSTRUATION

Abnormalities of Menstruation

Atypical Uterine Bleeding - Any deviation from .....

FIGO - International federation of Gynaecologist & obstetrician

ACOG - American college of gynaecology

Types of AUB - According to FIGO

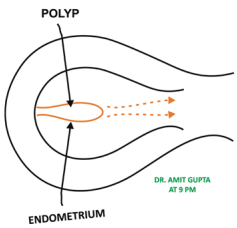
1. Oligomenorrhea → delayed or infrequent cycle (New Name)
   (Old Name)
   If menses after .....
   MC cause - polycystic ovarian syndrome

2. Polymenorrhea → Early/Frequent menses (New Name)
   (Old Name)
   If menses before .....
   MC cause - Pelvic inflammatory disease (PID)

3. Menorrhagia → (Old Name)
   New Categories
   A. Heavy Menstrual bleeding
      If amont of blood loss - MC Cause - Uterine Fibroid
   B. Prolonged bleeding
      If duration .....

Active space

4. Metrorrhagia → (Old Name)



New Categories

A. Intermenstrual Bleeding → Spotting P/V Between.....

MC cause

B. Irregular Cycle

If cycle to cycle variation .....

Menometrorrhagia - Excessive bleeding with .....

Mnemonic → Palm - coein classification for causes of AUB

		New Classification
P-	Polyp	AUB - .....
A-	Adenomyosis	AUB - .....
L-	Leiomyoma (Fibroid)	AUB - .....
M-	Malignancy	AUB - .....
C-	Coagulation disorder	AUB - .....
O-	Ovulatory disorder	AUB - .....
E-	Endometrial disorder	AUB - .....
I-	Iatrogenic	AUB - .....
N-	Not otherwise Classified	AUB - .....

DUB → Irregular Bleeding P/V ..... Organic pelvic pathology.

MC cause of AUB in pubertal girls is AUB-O due to immaturity of H.P.O. Axis

Mc cause of pubertal menorragia is anovulation

5. Cryptomenorrhea

Cyclical pain ..... bleeding P/V or

Active space

..... amenorrhoea or

..... bulging hymen on P/S exam or

..... bulge on P/A exam or Midline .....

Fullness in .....

Retention of .....

Secondary sexual characters.....

MC cause - ..... Hymen
..... Vaginal Septum
..... Atresia

Per - vaginal examination is contraindicated in .....

so per - speculam examination is done.

DR. AMIT GUPTA
AT 9:00 PM

Examination Protocal

First Step - Per- speculum(P/S) exam

↓

Tense bluish bulging .....

↓

Per-rectal examination(P/R) Exam

↓

Fullness in anterior compartment

Active space

↓

Ultrasound → Collected blood

USG

↓

1. Haematocolpos(Blood in .....)

↓

2. Haematometra (Blood in .....)

MC Cause of Haematometra .....

↓

3. Haematosalpinx (blood in .....)

↓

4. Haemoperitoneum(Blood in .....)

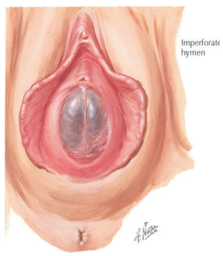
Complication of cryptomenorrhoea - .....

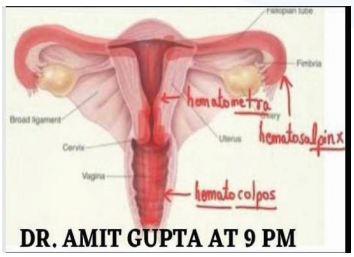
Mx 1. If due to Imperforate hymen

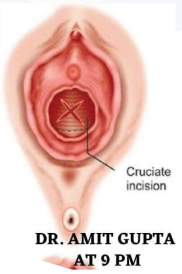
- X-Shaped or cruciate incision or ..... on hymen

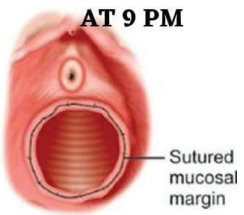
2. If due to transverse vaginal septum→MC→.....Part of vagina

.....









Active space

Hypomenorrhoea → (Old name)

New categories

A Light menstrual bleeding

If amount of blood loss .....

B Shortened bleeding

If duration of bleeding .....

MC Cause - Asherman Syndrome

Formation of synechiae or adhesions or..... in endometrium

Causes of asherman syndrome

Vigorous .... surface .....

..... basal endometrium is .....

MC Cause .....curettage

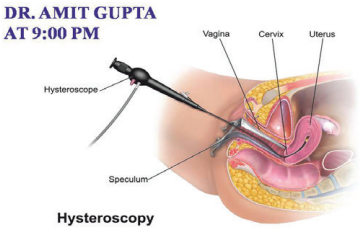
Post - Abortal curettage

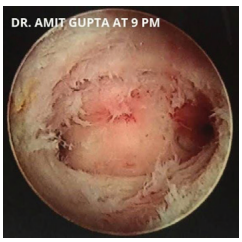
Endometrial TB

V. Rare → Schishtosomiasis

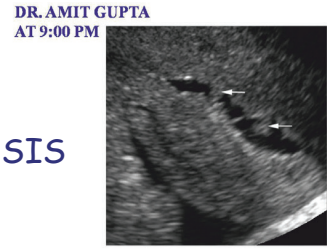
Gold standard to diagnose endometrial disorder

Hysteroscopy





SIS



HSG

Active space

MIST

Symptoms - MC seen .....
MC complication .....

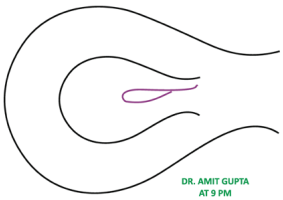
TAS → Trans - Abdominal Sonography
TVS → Trans - Vaginal SonographY
SIS → Saline Infusion Sonography
HSG → Hysterosalpingography

1st investigation .....
Better investigation .....
Screening test .....

Diagnosis – Definitive → Hysteroscopy >. Saline infusion sonography or sono-hysterography > Hysterosalpingography -Multiple..... filling defects (.....appearance)

Mx

- Hysteroscopic Adhesiolysis followed by Insertion of balloon catheter followed by Estrogen + Progesterone



Dysmenorrhea- pain at Menstruation

A. Primary dysmenorrhoea→ Pain at menses..... organic pelvic pathology
B. Secondary dysmenorrhoea→ pain at menses.....organic pelvic pathology
   - MCC – .....

If a female presents with dysmenorrhoea, what is the next step ....

Primary dysmenorrhoea or ..... Dysmenorrhoea

- Cyclical pain with

Active space

MC Cause - Due to excessive PGF2a → Excessive myometrial .....

.....

Primary dysmenorrhea depends on psychological factors.

Pain is maximum on .....

Pain starts with menses or just before menses & decreases within .....

Pain is mostly in .....

Pain starts few months after menarche because initial few cycles are .....

Pain .....on its own after .....activity, ...

Mx

1. DOC- .....
2. Other drugs

↓

If no response

↓

↓

If no response

↓

3. Sx- surgical dilatation of cervix
   - Presacral nerve ablation

.....

Advice 1. Get her .....

2. Get first child .....

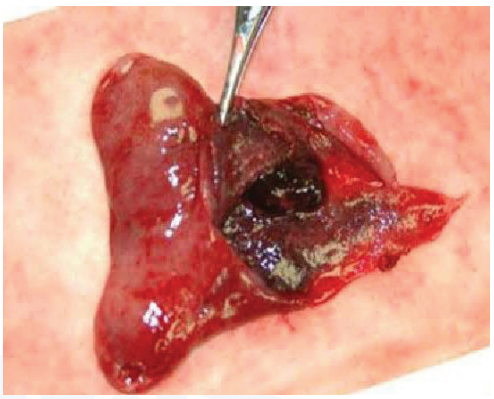
C. Membranous dysmenorrhea d/t Expulsion of .....

Active space

During menses d/t defect in

.....

.....



1. Amenorrhea: absence of menses

   A. Physiological

      a, In pregnancy - due to human chorionic gonadotropin (hCG)

      Support Pregnant .....

      b. In lactation - due to .....

Amount of prolactin is more in .....

Activity of prolactin in pregnancy is inhibited by .....

B. Pathological

   a. Primary amenorrhea- she has never menstruated before

   i. absence of menses by ..... years if secondary sexual characters are not developed, and by .... years if secondary sexual character are developed

   ii. absence of menses within .....of thelarche

   iii. absence of menses by ..... with signs of hirsutism, excessive exercise .....

Active space

b. Secondary amenorrhea – she was normally menstruating before, but now not menstruating

Absence of menses for ..... in a previously normally menstruating female

Absence of menses for .....with history of .....

Causes of primary amenorrhea

1. Endometrial level

a) Mullerian agenesis :

- Uterus
- Cervix
- Upper ..... of vagina Absent
- Fallopian tube

Secondary sexual characters.....

Tanner stage .....

Can have her own .....

FSH, LH .....

Mx ..... At the time of .....

Best .....

Active space

MRKH TYPE 1 - Only mullerian Agenesis

MRKH TYPE 2 - Triad Mullerian Agenesis + Renal Anomalies + skeletal Anomalies.

MRKH Type 2 (Mayer-Rokitansky kuster-hauser syndrome) mullerian agenesis +

..... anomalies +

..... defects

b) Testicular feminization syndrome (TFS) or androgen insensitivity syndromes (AIS)

Male→46XY→Testes→Testosterone→testosterone Receptor are poor

.....

1. Internal genitalia .....
2. External genitalia .....
3. Secondary sexual characters.....
4. Testosterone .....
5. Axillary - pubic hairs are .....

Development of axillary - pubic hairs and libido in females depend on .. ..

6. Testis ..... because final descent of testis in scrotum depends on .....
   a. MC tumor to develop.....
   b. MC malignancy to devlop .....
7. Mimic .....
8. Male pseudo .....
9. FSH, LH .....

Management of TFS - Goal - to lead .....

1. ..... At marriage best
2. Orchidectomy .....
3. .....Therapy after orchidectomy

MC cause of male pseudohermaphroditism (ambiguous genitalia in male) - .....

Active space

MC cause of female pseudohermaphroditism (ambiguous genitalia in female)-Congenital adrenal hyperplasia d/t 21-a, Hydroxylase deficiency

Q. 1 → Pri. Amen. 2. Breast 3. USG → Absent UT., Vagina

Q. 2 → Pri. Amen. 2. Breast 3. USG → Absent UT., Vagina

4. Absent axillary pubic hair

True hermaphroditism New Name .....

If ovary and testis are preset in .....

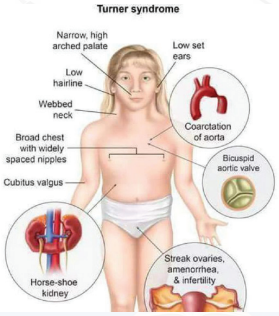
Best method to differentiate TFS & Mullerian agenesis.....

Best clinical method to differentiate- .....

2. At ovarian levels - MC - Gonadal dysgenesis - 3 types

a. Turner syndrome

Mutation in .....



45 XO

↓

Ovary are Streaky

↓

..... markedly reduced .....

Hypogonadism incrresed .....

i. Uterus is .....

ii. Endometrium .....

iii. Secondary sexual characters .....

iv Height .....

Turner mosaic may present either with primary or with secondary amenorrhea

Active space

Management

Initially ..... for approximately .....
followed by addition of ..... > .....

b. Pure gonadal dysgenesis 46 XX

some part of one X chromosome

Gene required for ovarian development is abnormal WNT-4, DAX-1

↓

reduced

i. Uterus is .....

ii. Endometrium .....

iii. Secondary sexual characters .....

iv. Height

Management

Initially ..... for approximately .....
followed by addition of ..... > .....

c. Sweyer syndrome

Male→46XY→Testis(Dysgenetic)

due to inactivation of .....

SOX-9

SF-1 (Steroidogen factor-1)

..... Sertoli cells → No .....No.....

..... Leydig cells → No .....No.....

Active space

Secondary sexual characters .....

Testis .....

FSH, LH .....

Management Goal → To lead feminine life

1. Orchidectomy

2. Estrogen +

3. At Pituitary level - Tumor (Adenoma or Prolactinoma)

- Craniopharyngeoma - MC age .....

Secretes ..... -> ..... feedback to .....

Other associated features

- ..... Spontaneous excessive secretion of .....
- .....
- .....
- .....

..... Hyperprolactinemia → M/C Presentation : Amenorrhea + Infertility.

Microadenoma .....

Macroadenoma .....

Diagnosis .....

Active space

Management

4. At Hypothalamus level - Kallmann Syndrome

Due to mutation in .....

Defective migration of ..... from .....

Estrogen, Progesterone .....

FSH, LH .....

..... Condition

Secondary sexual characters .....

Mx - GnRH agonist in .....

Causes of Secondary amenorrhea

1. Endometrial level

   - MC →.....

2. At ovarian level

   - MC →.....

   - .....

3. At Pituitary level

   a. Pituitary tumor

   b. Sheehan syndrome - postpartum necrosis of anterior

Pituitary due to excessive .....

MC Symptom .....

First hormone to decrease .....

Active space

Cause of amenorrhoea .....

Earliest feature .....

Other features.....

Management

Timely Steroids Can Reverse Features of Sheehan Syndrome To Some Extent.

4. At hypothalamus

Other systemic causes of secondary amenorrhoea

.....

.....

In Hypothyroidism .....

In Hyperthyroidism .....

.....

Severe chronic renal failure

Severe anemia

AIDS

Active space

Diagnostic protocol of secondary amenorrhoea

Ist step

↓

Urine pregnancy test (UPT)→If negative

↓

TSH, prolactin

↓

If normal

↓

..... Challenge test

If bleeding occurs (test .....) If bleeding does not occur (test .....)

↓

..... Estrogen + progesterone Combined challenge test

↓

If negative

Diagnosis .....

If positive

ovarian disorder or .....

↓

Next step is ..... (Normal .....

If ..... Disorder

If ..... Disorder

↓

Next step is .....

Diagnostic protocol of primary amenorrhoea

I. With secondary sexual characters → Next step .....

i. if uterus is present .....

ii. if uterus is absent

Active space

II. Without secondary sexual characters

i. .....syndrome

ii. .....dysgenesis

iii. .....syndrome

iv. .....syndrome

Best method to differentially diagnose various causes of primary amenorrhoea .....

Best method to differentially diagnose various causes of secondary amenorrhoea .....

MC cause of primary amenorrhoea .....

MC cause of secondary amenorrhoea .....

MC cause of pathological secondary amenorrhoea .....

Primary amenorrhoea with decreased FSH, LH, Estrogen, Progesterone .....

Primary amenorrhoea with normal FSH, LH .....

Primary amenorrhoea with increased FSH, LH & decreased Estrogen, Progesterone .....

Primary amenorrhoea with cyclical pain in abdomen every month ....

Active space

Primary amenorrhoea with normal height, anosmia, absent secondary sexual chracters .....

A 25 yr old female presenting with secondary amenorrhoea with history of dilatation and curettage. FSH levels are 6 IU/L. diagnosis is .....

MIST

POLYCYSTIC OVARIAN SYNDROME

Polycystic Ovarian Syndrome

- MC cause of.....

  MC .....disorder in female

  MC cause of.....

  onset and progress .....
- Stein-leventhal syndrome
- MC age -
- Gene mutation –

Insulin resistance

↓

Hyper glycemia/ Hyperinsulinemia

↓

Altered body function inculding liver function

↓

(Normally SHBG bind with .....Sex hormone binding globulin .....) ↓

So free androgen.....

↓

.....esterone

-ve to FSH +ve to LH

Active space

FSH..... LH.....

LH levels are raised throughout cycles & there is no .....

Smaller follicles..... Theca cells → Androgens

No. ..... aromatisation

Esterone

PCOS - Hyperandrogenic & ..... condition

If ovary produce more androgens → Toxic for follicles

..... of follicular growth

Symptoms

- ..... obesity
- Acne, ..., androgenic Alopecia→Fronto-Temporal Baldness

Hirsutism - development of excessive, coarse, thick, male pattern, sexual terminal hairs

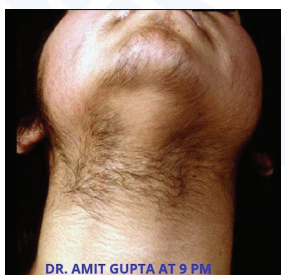
Diagnosis of hirsutism - ferriman gallwey score - if .....

- Oligomenorrhoea , Amenorrhea

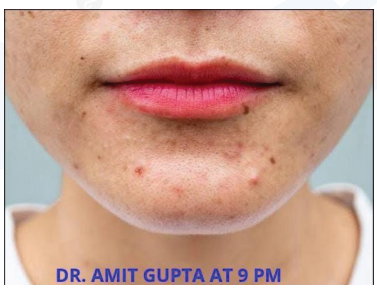
Thin PCOS Female → Does not have heavy bledding → Persistent Amenorrhoea.

Obese PCOS Female → High Estrogen → Heavy bleeding

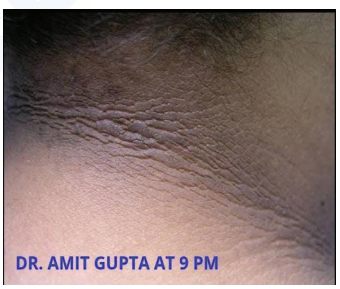
- Infertility



Hirsutism



ACNE



Acanthosis Nigricans

Long term complications

- Diabetes mellitus

Active space

- Coronary
- Cerebral } stroke

Androgens ↑ LD.L
↓ HDL

- Endometrial Hyperplasia
- Ca endometrium, breast, ovary
- Metabolic x syndrome → (Obesity + Diabetes + Dyslipidemia + Hypertension + Pcos) If any 3/5 →
- Sleep Apnea syndrome
- Psychiatric disturbance
- Non alcoholic steatohepatitis

Dx ..... criteria 3 parameters. if 2/3 present .....

1. Menstrual dysfunction
2. Clinical and biochemical evidence of ..... androgen
3. USG features
   - Size of follicles .....
   - Number of follicles ≥ .....

According to european society for human reproduction & embryology .. ..

According to PCOS & Insulin resistance society.....

- volume of ovary > .....
- Features in .....ovary are sufficient for diagnosis

- necklace sign or ..... sign .....not included in rotterdam criteria

Active space

23 year old girl with oligomenorrhoea, acne and normal appearance of ovaries on USG. what is most likely Dx .....

23 year old girl with oligomenorrhoea, acne, sudden onset or virilising features .....

MC cause of rapid onset hirsutism .....

Virilising features

- Atrophy of breast
- Clitoromegaly
- Hoarseness of voice
- Masculine body Countour

Virilising tumors - androblastoma / arrhenoblastoma, sertoli cell tumor , Leydig cell tumor, Hilus cell tumor

- Normally → androgen →
- In PCO→ androgen→

- In andogen secreting .....

Biochemical changes

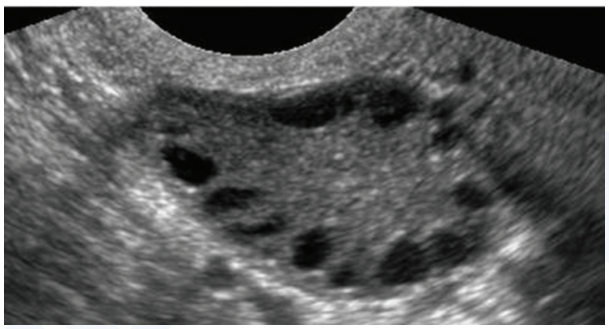
↑	↓
- Insulin	- SHBG
- Glucose	- FSH
- Free Androgens	- Progestrone
- Esterone	
- LH	
- LDL	

Active space

Feature syndrome → HAIR-AN syndrome

Hyperandrogenemia insulin resistance acanthosis nigricans

Central Thick Stroma d/t .....

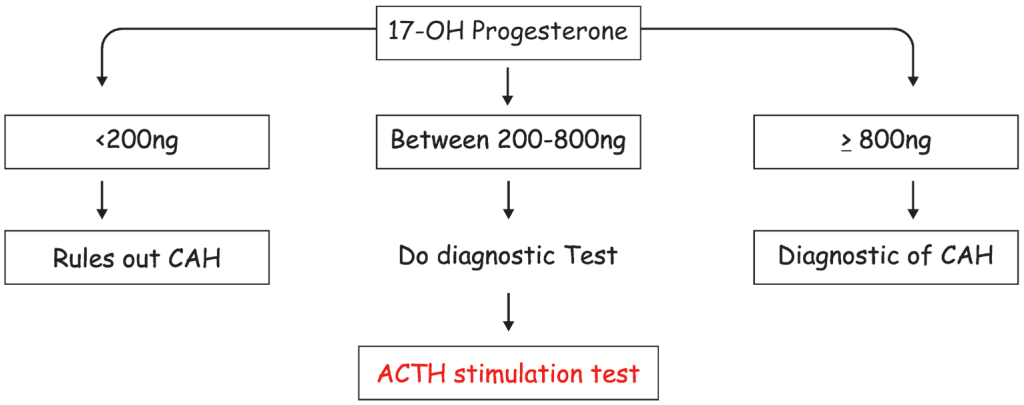


- Normal E2:E1 → .....
- PCO E2:E1 → .....
- Normal FSH:LH → .....
- PCOS- FSH:LH → .....

In every patient of Provisional diagnosis of PCOS Rule out Adult onset Congenital adrenal hyperplasia

Screening test - 17 OHP levels

Since precursor 17 hydroxy progesterone has increased due to increased ACTH, it is used as a screening test for CAH.



ACTH stimulation test - Give injection ACTH and check levels of 17-OH Progesterone. If levels are .....

Active space

Mx

1. First step - Life style modification

   A. Physical activity → Jogging for 4km / 45min. / Daily

   B. Dietary modification → To Reduce Weight

   C. Stress Relaxation Exercise

   If no Response in

   DOC for insulin resistance .....

2. To Mx Androgenic features

   - DOC -

   Ethinyl estradiol Progesterone

4. Generations of Progesterone

   1st - Norethindrone

   & 2nd Gen-Levonorgestrel

   Androgenic S/E

   3rd Gen Desogestrel, gestodene

   Norgestimate

   → Least androgenic

   4th Gen Drospirenone, Dinogest

   Cyproterone acetate) Antiandrogenic

OCPS Initially given for 6 months then we see the response. if Hirsutism Show less response then may add .....

2nd Line .....

Others

3. Oligomenorrhoea

   Progesterone withdrawal bleeding Followed by .....

Active space

4. Infertility

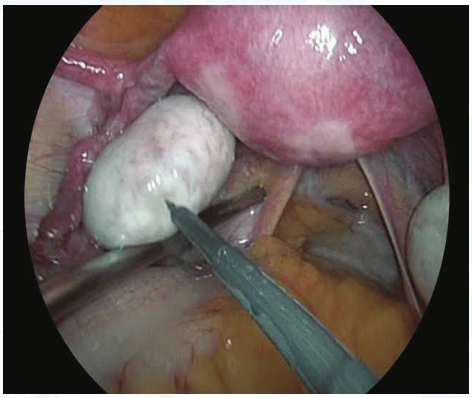
1st Line (DOC for anovulation due to PCOS.....)

(DOC for anovulation .....)

2nd Line

A. Injection human menopausal gonadotropin (.....)

B . Laparoscopic ovarian .....



3rd Line treatment :- IVF (Ovulation induction + IVF)

Side effects of ovulation inducing drugs .....

..

Risk is maximum with.....

Triggering factor .....

Active space

PELVIC INFLAMMATORY DISEASE

Pelvic Inflammatory Disease (Lower Abdominal Pain Syndrome)

PID - Pelvic Inflammatory Disease

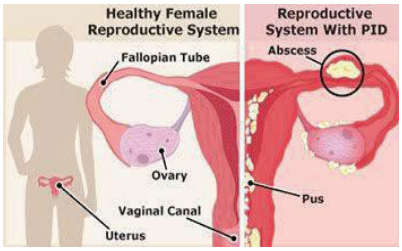
- Inflammation of uterus, fallopian tube and ovaries, Pelvic Peritoneum
- Route of spread → Ascending Infection along with .....
- MC microorganism → Polymicrobial > .....
- MC microorganism in acute case .....
- MC microorganism In IUCD User .....
- Post menopausal female :- organism = E.Coli
- MC age .....
- Risk factors

- Most common- Intercourse with.....
- Early age of intercourses
- History of instrumentation
- Douching
- .....

- Protective Factors
- Barrier contraceptive
- OCP uses → .....Cervical Mucous
- .....

Active space

Sequence of events in PID



Endometritis

↓

Endosalpingitis

↓

Oophoritis

↓

Tubo ovarian mass

↓

Tubo ovarian abscess

↓

Pyosalpinx

↓

----------PID → Hydrosalpinx

- MC symptom in PID- ...Lower Abdominal Pain - Bilateral
  Fever
  Malaise
  Pain worsens with coitus/movement

Diagnosis

1. Minimum criteria - Lower abdominal pain with any one of the following

- ..... tenderness Adnexae or Tubo - Ovarian Region
- ..... tenderness
- ..... tenderness

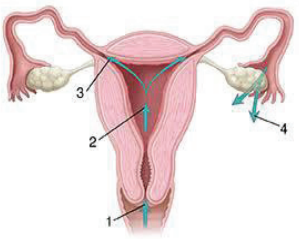
Active space

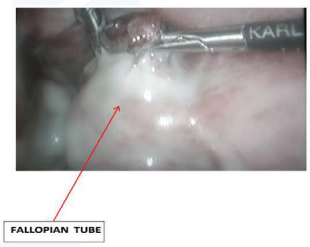
2. Additional criteria

- Temperature >.....
- ↑WBC
- ↑CRP
- Mucopurulent discharge
- Laboratory demonstration of .....

3. Definitive criteria

- Endometrial Biopsy
- USG- .....
- Laparoscopy (Gold standard) Most Conclusive → Trickling of... via Fimbrial ends of fallopian tube





Syndromic Management –

With Yellow Kit → Kit (6)

(1) Tab. ..... 100mg BD (For Chlamydia)

(2) Tab. ..... (For Various Anaerobic Infections)

(3) Tab. ..... (For Gonorrhoea)

Male Partner is also treated

with Grey Kit → { Kit -1}

(1) Tab. .....

(2) Tab. .....

Active space

→ Prognosis Scoring (Prognosis for conception) "Boer Meisel"

Chronic PID

Symptom .....on movements due to fibrous bands dut to frozen pelvis (pelvis with fibrous bands)

Sometimes pain in .....

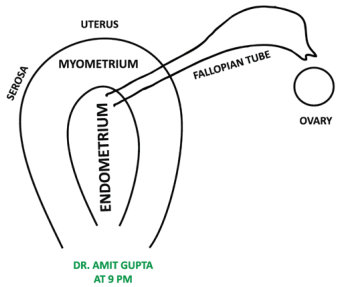
A lady presents with history of lower abdominal pain 2 years back. She was treated with yellow kit. Now she persents with restricting pain on movements. Likely diagnosis is .....

Complications

MC Complication .....

a. Complication If tube is completely blocked-

b. Complication If tube is partially blocked-



Investigations -

A. USG

1. Bead on string sign
2. Waist Sign
3. Cog - Wheel Sign
4. Hydrosalpinx

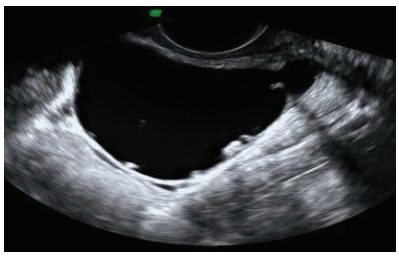
.....

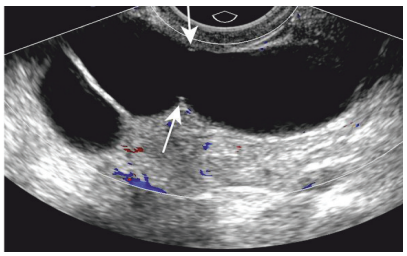
B. Laparoscopy

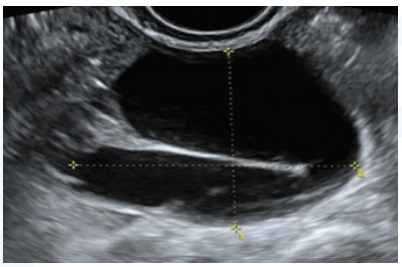
Violin - String Sign

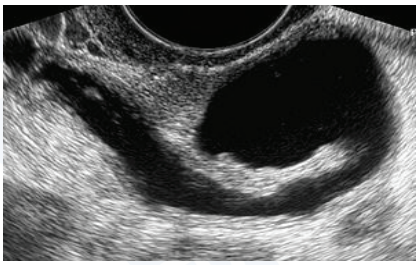
1. MC Site of tubal block due to chlamydia & gonorrhoea → .....
.....
2. MC Site of tubal block due to TB → .....

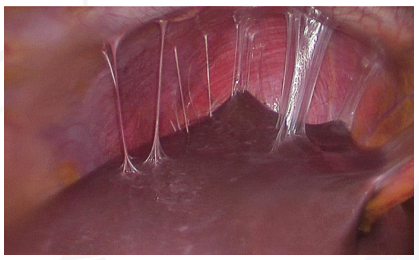
Active space







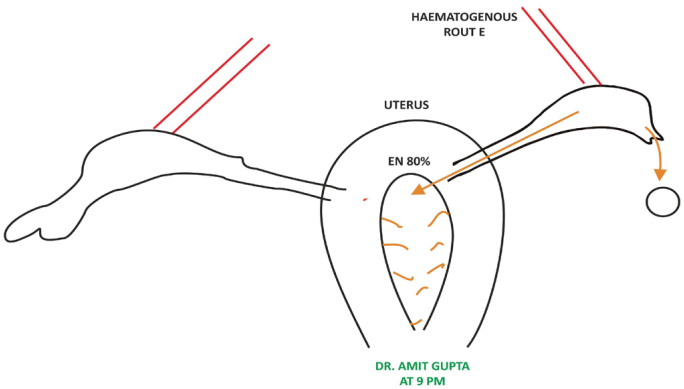




Active space

GENITAL TUBERCULOSIS

Genital Tuberculosis



Almost invariably .....

Can occur in .....

Usvally presents with .....

Genital TB

- .....infection
- MC primary site .....
- MC route of spread.....
- MC route of spread of endometrial TB →
- Earliest / Most common site of genital TB .....

Endometrium→Asherman syndrome, Pyometra (pus in uterine cavity)

Ovary→Menstrual symptom.....

- MC symptom → .....
- MC menstrual symptom :-

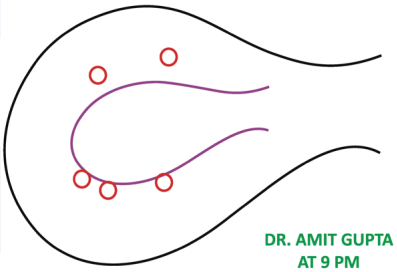
Active space

MIST

P/V : M/C finding = Normal pelvic examination
Some tenderness
Rarely Bilateral adnexal mass.

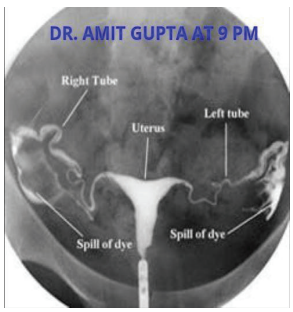
- Diagnosis →Endometrial biopsy on
.....
before menses > ..... of menstural fluid > .....
.....
within first ..... of cycle

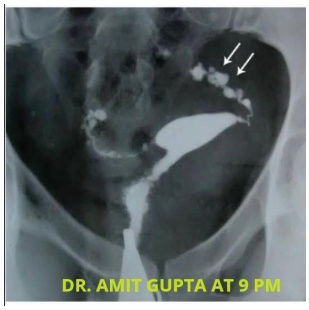
- Endometrial biopsy - sample sent for ..... Examinations.
1 Histopathology
2 PCR
3 Culture (L-J Medium)

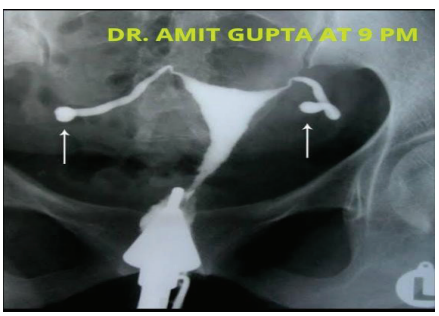


HSG is contraindicated

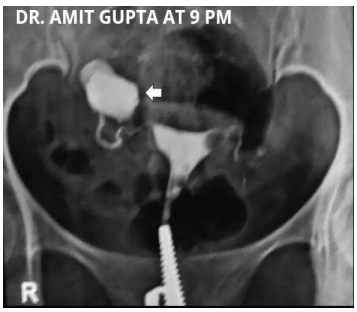
If done (1) Beaded ..... Appearance
(2) Golf - Stick ..... Appearance
(3) Tobacco - Pouch ..... Appearance
(4) Lead - Pipe or Pipe - Stem ..... Appearance
(5) Bilateral cornual block
(6) Cotton wool appearance
(7) Bilateral hydosalpinx (Rare)

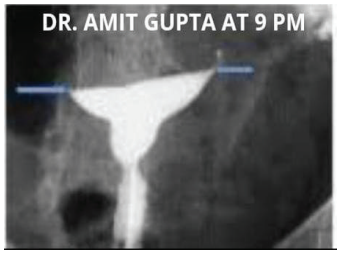


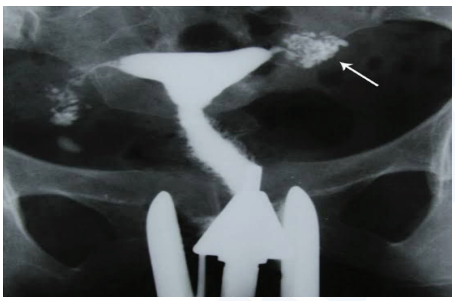


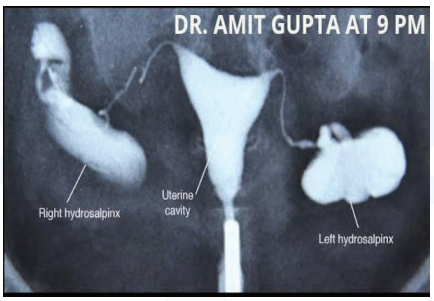


Active space









Mx→Antitubercular therapy (ATT) .....

First 2 months

a. Rifampicin
b. Isoniazide
c. Ethambutol
d. Pyrizinamide

Next 4 months

a. Rifampicin
b. Isoniazide
c. Ethambutol

Active space

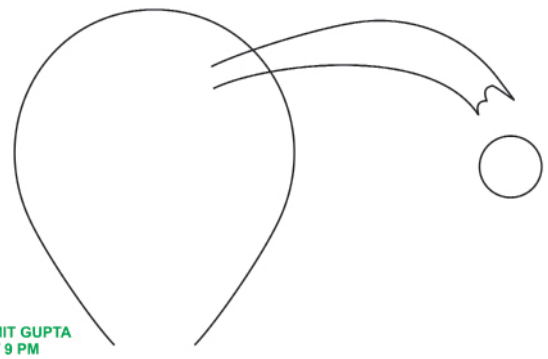
ENDOMETRIOSIS AND ADENOMYOSIS

Endometriosis

Presence & persistence of ..... outside uterus

- Risk factors

1. Genetics→
2. ..... Estrogen
3. .....parity
4. .....menarche/ .....menopause
5. Obesity
6. Age .....
7. Scar areas
8. Socioeconomic class .....
9. Menstrual Bleeding .....
10.



DR. AMIT GUPTA
AT 9 PM

Active space

- Protective factors
    1. Physical exercise
    2. ..... Inhibits .....
    3. OCP - A. inhibits ..... delay .....
       B. .....
    4.

- Most accepted theory..... of retrograde menstruation or .....

- Other theories
    1. Meyer & Ivanoff
    2. Halban - Metastasis via blood vesseles & lymphatics - liver lungs brain, eyes, nose, chest - wall etc. {rarest site - ..... }

- Vicarious menstruation - bleeding from ..... during .....
- Catamanial .....
  Bleeding in ..... during .....

3. Immune theory
   Female with endometriosis have deficient cell mediated & humoral immunity

- 1st organ involved.....
- Finally settle in .....

- Symptoms → Clinical Triad
    1. MC - Dysmenorrhea ....
       .....
       Sometimes chronic pelvic pain

Active space

pain is .....

pain progressively .....

pain is directly proportional to .....

2. Dyspareuria - pain during intercourse

3. Infertility -

A. Mild - due to .....

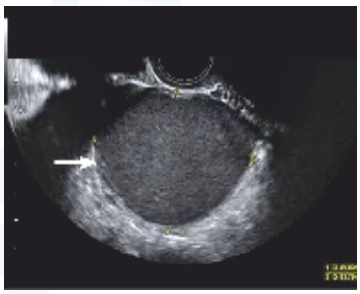
B. Severe - due to .....

Endometriosis may sometimes be associated with ..... and ..... (usvally if associated with adenomyosis)

- P/V exam -

.....

- USG .....



Management for mild pain in endometriosis

1st Line

If she wants to conceive .....

Active space

2nd Line

Anti ..... effect

3rd Line

GnRH Agonist in ..... fashion

Eg. → Leuprolide

or

GnRH Antagonist → Eg. - Elagolix

4th Line

.....

Management for severe endometriosis

1st Line

GnRH Agonist in ..... fashion

Eg. → Leuprolide

or

GnRH Antagonist → Eg. - Elagolix

2nd Line

Laparoscopy

Other drugs

Danazol - Side effect .....

Drug of last resort .....

Drug for post-op follow-up .....

Active space

Mx For mild infertility

- .....

Mx for severe infertility

.....

- IOC - .....

Laparoscopy findings -

1 Endometriotic deposit in peritoneal cavity→

On -

peritoneum

Intestine

Omentum

Urinary bladder

Rectum

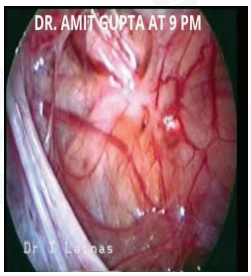
POD

Uterosacral ligament

Broad ligament

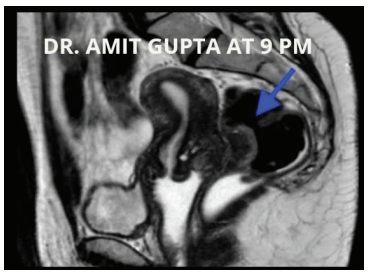
a) Early lesion→ .....

b) Late lesion→.....



MRI (for rectosigmoid endometriosis) .....

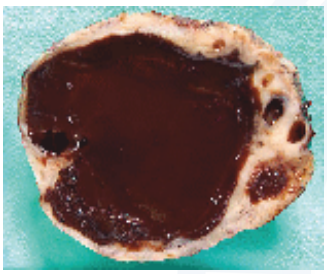
Active space

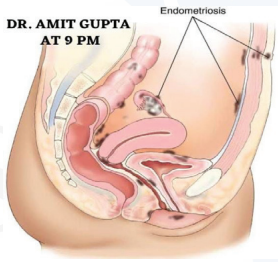


2 Ovarian cyst

..... cyst

True Cyst lined by .....





Manegement of chocolate cyst .....

Chocolate cyst + infertility = IVF

Routine removal of chocolate Cyst before IVF is not done

3 Adhesion

- Gold standard- .....

Severe Disease - Any of the following

- Adhesions
- Chocolate cyst
- Distorted pelvic anatomy

Active space

Laparoscopic Management

1. Ovarian cyst

a) If size < 5 cm→ Follow-up with .....

b) If size ≥ 5 cm →

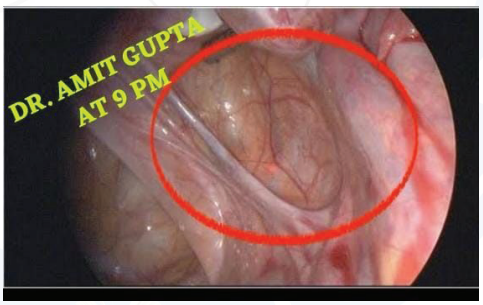
To prevent recurrence .....

2. For Endometriotic deposit → .....

3. For Adhesions→.....

LUNA (laparoscopic uterosacral nerve ablation)
If laparoscopic Mx fails → Last resort
If family is complete - do hysterectomy

If there is break in ligaments – ..... syndrome



Adenomyosis

Endometrium in .....

Old Name .....

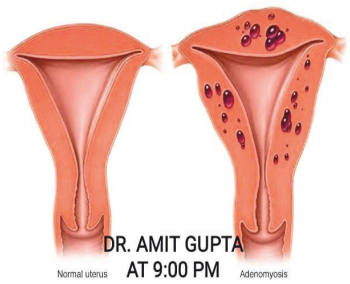
MC Age .....

More in .....

symp. MC .....

P/V Exam .....

Active space



Uterine enlargement may be upto .....

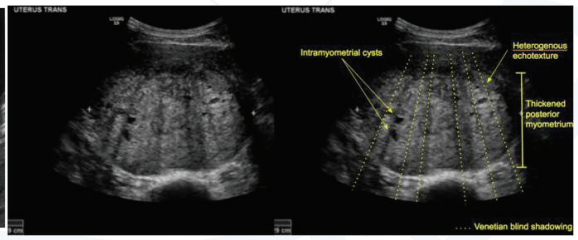
- USG →TVS

1 Salt-pepper appearance

2 Venetian blind (alternate light & dark bands) appearance or fan shaped shadowing

3. Myometrial cyst

4. Loss of endo-myometrial junctional Hypoechoic area - blurring of junction



Salt - Pepper Appearance Myometrial Cyst Fan - Shaped Shadowing

- IOC→ MRI shows increse in thickness of endomyometrial junctional zone .....

- Gold standard - Hysteroscopic curettage from..... and if endometrium is seen .....

Active space

Management

1. If multiparae - .....

   If unwilling for hysterectomy and does not want to conceive

   A. Uterine artery .....

   B. High intensity focussed .....

   C. Mr guided high intensity focussed .....

2. If wants to conceive

   A. Mirena IUCD

   B. GnRH Agonist in .....

MEDICAL INSTITUTE FOR SCREENING TEST MIST

Active space

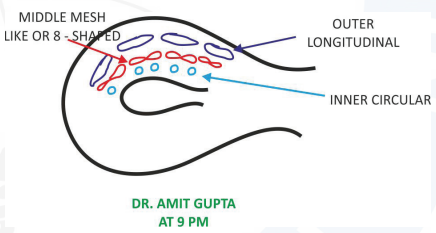
Fibroids

Fibroids cause ..... enlargement of uterus

Uterus = Irregular ..... Nontender

Uterine enlargement may be upto .....

- ..... Have no effect on size of fibroids.
- Originate in .....
- MC Solid benign pelvic tumor in female
- Risk factors :-



1. ..... Estrogen
2. Genetics - Mutation in .....

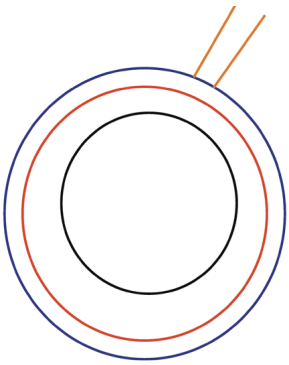
   Translocation between chromosome .....

   Upreguation of collagen. ..... in Angiogenic Growth factors.

3. Nulliparity
4. ..... menarche- ..... menopause
5. Obesity
6. Reproductive age group
7. High socio economic class
8. Decreased vitamin .....

Active space

- Protective factors
    1. Physical exercise
    2. .....
- Gross examination
    1. Whorled
    2. Rubbery consistency (firm), pinkish - white
    3. .....capsule
    4. ..... vascularity
    5. Originate from ..... base



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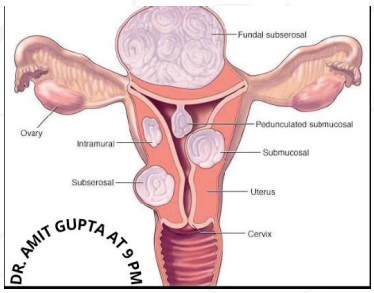
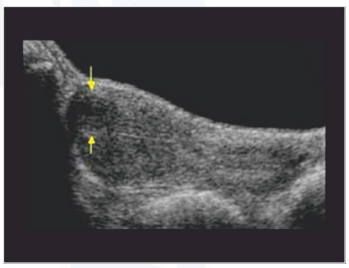
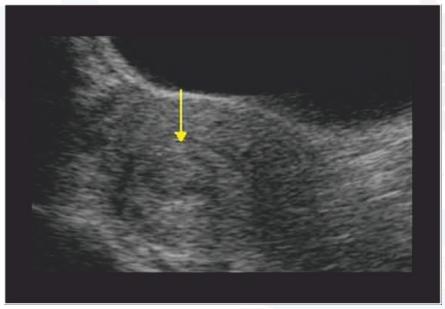
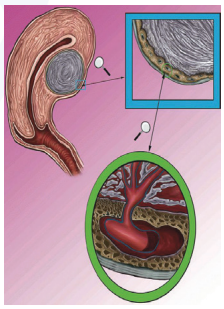




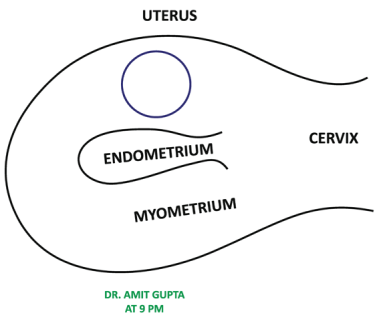




Types of fibroid

I. Uterine fibroid

II. Extrauterine fibroid



Types of uterine fibroid

a) MC→----------

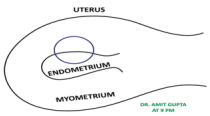
Confined to .....

Active space

MIST

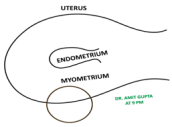
b) Submucous

Protrude towards .....



c) Subserous

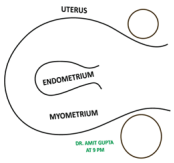
Protrude towards .....



II. Extrauterine Fibroid

a) Cervical fibroid

b) Broad ligament fibroids



1. Cervical fibroid

a) Anterior Wall cervical fibroid .....

b) Posterior wall cervical fibroid .....

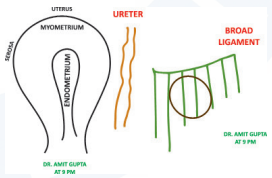
Post wall fibroid enlarge → POD → Push Rest of uterus forward → Obstruction of outflow tract

b) Lateral wall cervical fibroid .....

2. Broad ligament Fibroids

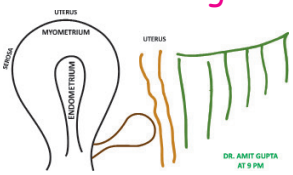
a) True broad ligament fibroid

- Originate in broad ligament
- Ureter is ..... to it



b) Pseudo broad ligament fibroid

- Originate from elsewhere but encroach toward broad ligament
- Ureter is .....to it



MC fibroid causing symptoms .....

MC fibroid causing heavy menstrual bleeding .....

MC fibroid causing malignancy .....

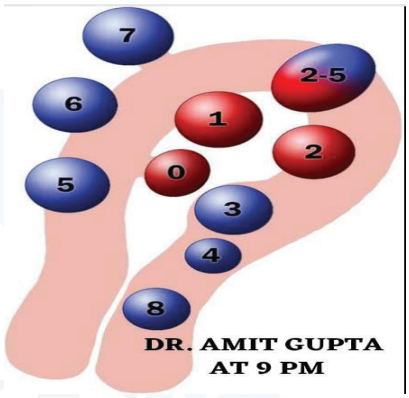
MC fibroid causing pain (pressure symptom) .....

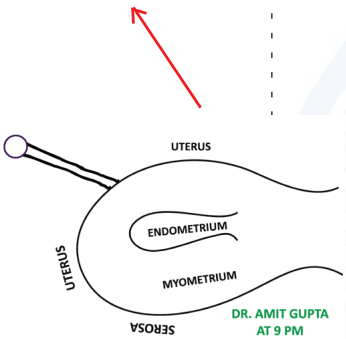
MC fibroid undergoing calcification

Active space

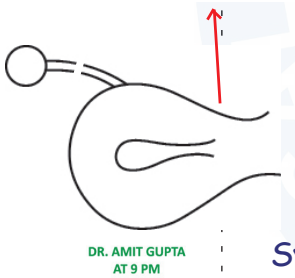
Classification (FIGO)

- 0 (Submucous)→Completely in endometrial cavity
- 1 (Submucous)→..... in endometrial cavity
- 2 (Submucous)→..... in endometrial cavity
- 3 (Intramural)
- 4 (Intramural)
- 5 (Subserous)
- 6 (Subserous)
- 7 (Subserous)
- 8 (Extrauterine)





- Pedunculated fibroid - fibroid with .....



- Wandering fibroid

Pedunculated subserous fibroid whose pedicle .....

Symptoms

1. MC presentation.....
2. MC Symptoms.....

Symptoms specific to fibroid

a. Submucous fibroid

Which fibroid can cause irregular bleeding .....

b. Subserous fibroid

- IOC .....
- Best .....

Active space

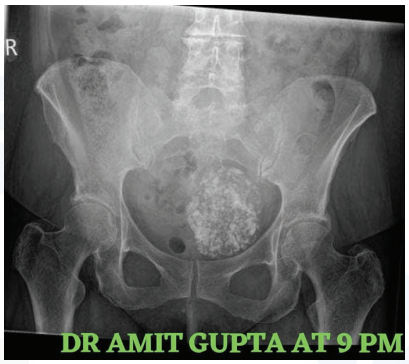
MIST

- For follow up---------

Complication - Fibroid - Degeneration from centre

Except Calcification - start from periphery

- MC.....



Other complications

- Fatty(Myxomatous) degeneration
- Cystic degeneration
- Calcareous degeneration ($CaCo_3$)

↓

- .....

- Red degeneration
  - Atrophy-after .....

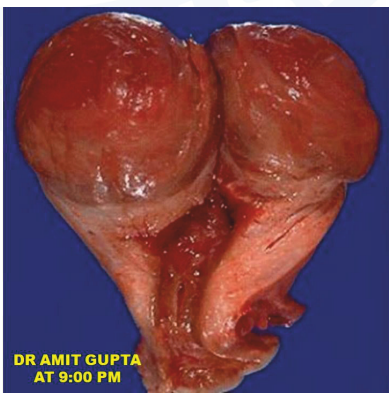
  Rarest - Myoma or Leiomyoma →Risk of Malignancy.....

  Leiomyosarcoma

Red degeneration - Due to thrombosis of the vessel supplying fibroid

MC time - In ..... of pregnancy

Red degeneration of fibroid may present with fever, acute pain in abdomen and mild leucocytosis.
If no fever and leucocytosis - may be due to ..... of fibroid



MX:-

Indicators of malignancy

Active space

- Sudden Increase in size
- Loss of whorled pattern
- Softer
- Pain

Specific Investigations for submucous fibroid

- Sonohysterography or saline infusion sonography
- Hysteroscopy
- Hysterosalpingo-graphy (HSG) .....

Pregnancy with fibroid - Complications

1) Pre term labour - IUGR
2) Abruption
3) Malpresentation
4) PPH

Endocrine effects may be -

1) Polycythemia
2) Hypercalcemia (PTH)
3) Hyperprolactinemia

Management

Asymptomatic fibroid .....

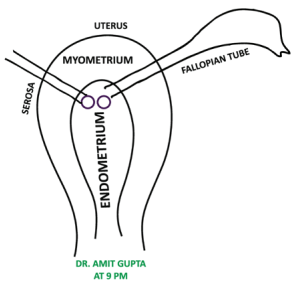
Perimenopausal / asymptomatic .....

wants to conceive / asymptomatic .....

exception :

a submucous fibroid.....

b bilateral cornual intramural fibroid .....



Active space

Follow up in asymptomatic fibroids in >= 40 years age lady is done if size is less than 12 weeks.

Symptomatic fibroid .....

1. Heavy menstrual bleeding

Type 0,1 fibroid

.....

Other fibroid - medical management

1st line

- .....
- Tranexemic acid
- Mirena IUCD

2nd line

GnRH agonist in

.....

or GnRH .....

A. Other drugs decreasing estrogen - danazol, letrozole

B. Other drugs decreasing progesterone - mifepristone, ulipristal

3rd line - If unwilling to conceive

a. Uterine artery .....

2. pain/pressure

1st line

GnRH agonist in .....

..... or GnRH .....

A. Other drugs decreasing estrogen - danazol, letrozole

B. Other drugs decreasing progesterone - mifepristone, ulipristal

2nd line (If unwilling to conceive)

a. Uterine artery.....

b. High intensity focussed .....

.....

c. MRI guided high intensity focussed .....

If no response

1. If wants to conceive - .....

.....

2. If does'nt want to conceive - .....

Active space

b. High intensity focussed .....

c. MRI guided high intensity focussed .....

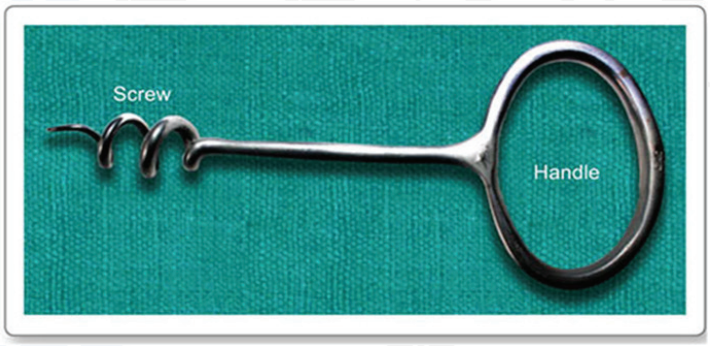
If no response

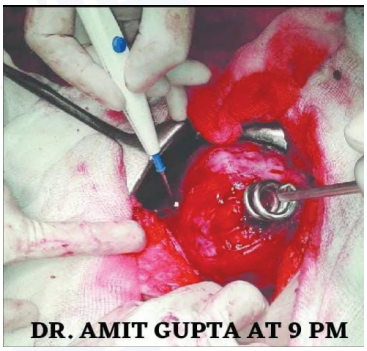
1. If wants to conceive - .....

2. If does'nt want to conceive - .....

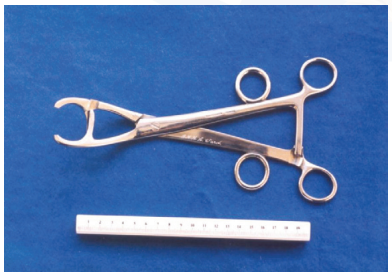
Fibroid + heavy menstrual bleeding + want to concive .....

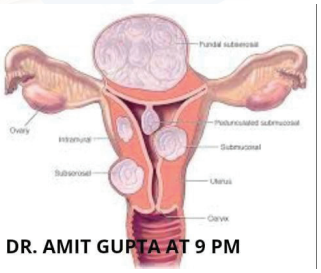
- Bonney's myoma screw- to hold.....

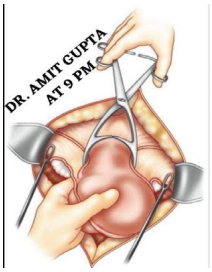




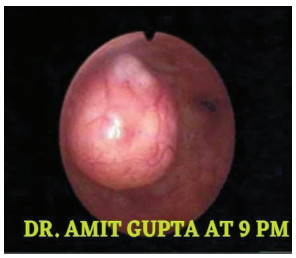
- Bonney's myoma clamp- to intermittently relase blood supply



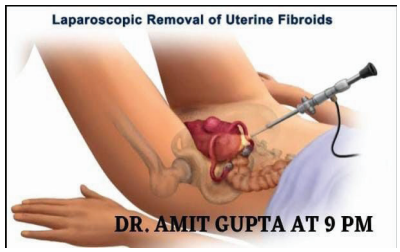




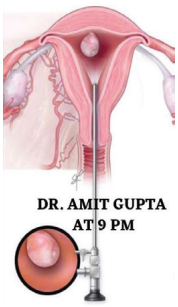
Active space



Hysteroscopy



Laparoscopy



Hysteroscopy

HSG →

Hysterectomied Specimen

Laparoscopic removal - done for .....

fibroid if size .....

Hysteroscopic removal - done for .....

fibroid if size .....

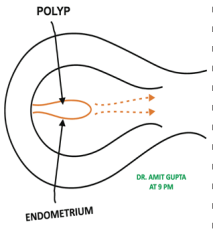
During caesarean section - ..... fibroids may be removed easily

Disadvantage of Laproscopic hysterctomy - unable to know extent of .....

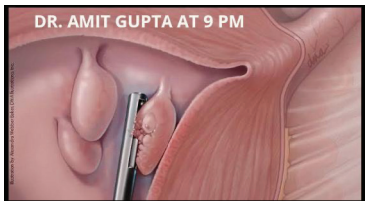
Complication of hysteroscopy .....

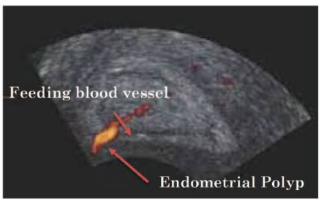
normal intrauterine pressure in hysteroscopy .....

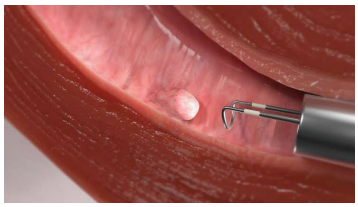
Active space



Endometrial Polyp







Color Doppler Hysteroscopy

....

Is a .....

polyp originate from ..... base

MC age .....

Mc symptom .....

P/V finding - Uterus .....

.....

First investigation .....

Color doppler USG finding .....

IOC .....

Complications .....

Mx of endometrial polyp .....

Mx of endocervical polyp - remove with .....

Active space

Uterocervical Prolapse and Vaginal Wall Prolapse

Most important support of uterus .....

Most important ligamentous support of uterus .....

Ligament which maintains anteversion of uterus .....

Ligament which prevents retroversion of uterus .....

- De lancey supports

- Uterocervical prolapse
- Due to laxity of support
- MC----------

  Abnormal conduct of Labour
  Mismanaged second stage of Labour
  Precipitate Labour
  Prolonged Labour
  Premature bearing down

- Others

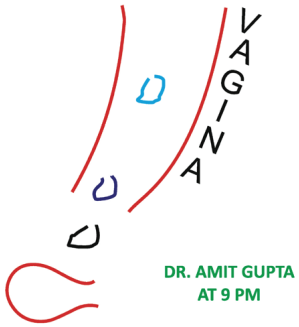
  Multiparity

  Menopause

  COPD, Obesity, severe ascites, smoking

Classification→ (old →SHAW's classification)

- 1°→Prolapsing part is in vagina
- 2°→At vaginal introitus

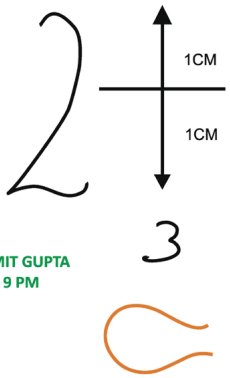


Active space

- 3°→out side introitus
- 4°→Whole uterus outside or procidentia

Universal Classification of Prolapse →

New –Based on distance from hymen. Pelvic organ prolapse Quantification system. POP-Q classification



Reference Point .....

- 1→If prolapsing part is >1 cm above hymen
- 2→Anywhere between 1 cm above hymen to 1 cm below hymen
- 3→ >1 cm below hymen
- 4→ 3 + complete vaginal eversion

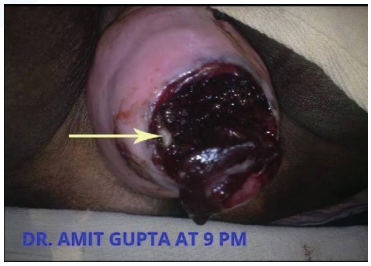
Elongation of supravaginal part of cervix occur in .....

And does note occur in .....

First step in prolapse .....

Symptoms

- Something is coming out / Dragging sensation / feeling of insecurity in vagina
- Pain - due to stretching of ligaments, mainly of .....



- Decubitus ulcer

  Due to .....

Active space

Management for Ulcers

..... with glycerine (.....) & acriflavine (.....)

Congenital prolapse

- Causes –Spina bifida
  ..... Syndrome
  ..... Imperfecta

For treatment of 1° prolapse

1. ..... - Voluntry tightening of perineal muscles .....

2. .....

Management for 2° & 3° prolapse → Depends on parity & age

1. If nulliparous / congenital prolapse

Abdominal sling operation or ..... Pexy - Fixation or Solidity

- Sling Surgery can be done only when uterocervical length .....
- IF uterocervical length is more & patient doesn't want Hysterectomy - Manchester Repair or Fothergills Repair

.....

....

Right Sacral promontory Left Rectus sheath

Mersilene tape

.....

Abdominal sling operation

Purandare

- Attached to rectus sheath
- Dynamic sling, anterior sling, provide.....

Active space

Shirodkar's

- Attached to sacral promontory
- Static sling, posterior sling
- More complication specially on LT. side
- Provide.....

Virkud's

- Lt. side- Purandare
- Rt. Side- Shirodkar

2. If multiparous - Ward-Mayo vaginal hysterectomy

Ward-Mayo Surgery - Vaginal Hysterectomy + Pelvic floor repair

3. If multiparous but unwilling for hysterectomy- <40yrs
   - Doesn't want child birth -Forthegill operation or Machester operation
   - Shirodkar's modification of forthegrill's - no cervical amputation done

4. If her age is around 60 - 65 years with some contraindications to surgery eg. diabetes, hypertension etc.
   a) Le- forte colpoclesis
   b) Daniel stiching

5. Ring pessary .....

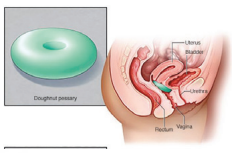
Indications -

- Pregnancy
- Young girl who is unwilling or C/I for surgery
- Puerperium

Active space

- for temporary relief
- If her age is around 60 - 65 years with co-morbidities and absolute contraindication for surgical procedure
- Put at the level of.....
- Removed/ changed after.....

Pessary - made of ..... - space occupying device put in vagina.



A 26 year old woman presented in ER with fully outside cervix and a cystocele. Next step in management is .....

6. For vault prolapse-
   - Vault-vaginal stump left after hysterectomy
   - SacraL colpopexy → Best results but difficult to perform
   - Sacrospinous fixation
   - MC Performed → Uterosacral Suspension

For Prevention of prolapse .....

Vaginal wall prolapse

- Due to ..... defect

MC Prolapse .....

1. (MC) Cystocele→prolapse of upper .....of anterior wall of vagina + Urinary bladder

   Cystocele Advances :- Progress to difficulty in passing urine & retention of urine. (Kinking of urethra)

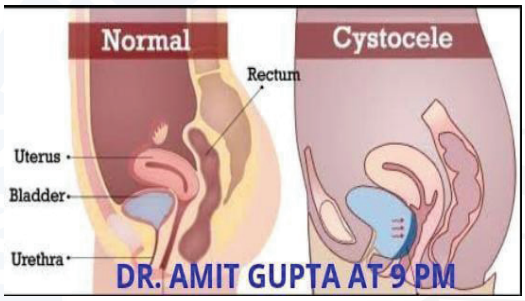
Active space

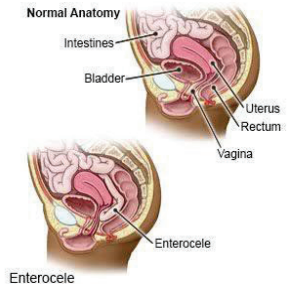
2. Urethrocele→ Lower ..... of anterior vaginal wall + urethra

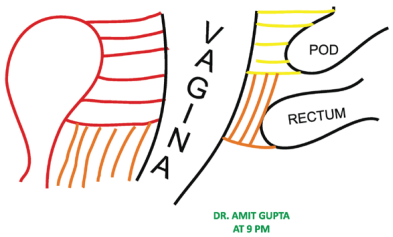
3. Enterocele →If upper ..... of posterior vaginal wall + pouch of douglas

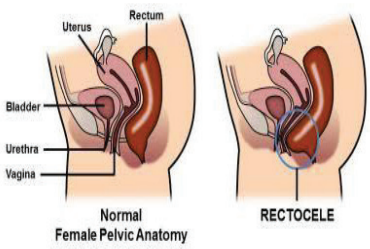
4. Rectocele→Middle ..... of posterior vaginal wall+ Rectum

5. Lax perineum→Isolated prolapse of lower..... of posterior vaginal wall









Rrhaphy → Repair

Mx

- For Cystocele & Urethrocele - ..... colporrhaphy
- For rectocele & Lax perineum ..... colpoperineorrhaphy
- For enterocele – Best .....
  – Moschcowitz repair, Halban Repair

Stress Urinary incontinence - Involuntry escape of urine during .....
.....

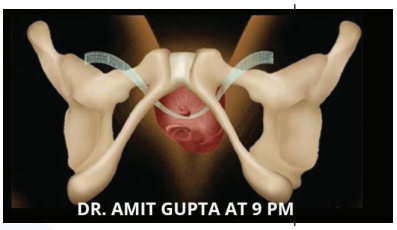
Active space

MC cause - .....

Normally bladder neck compress .....

When abdominal pressure .....

Incontinence in Reproductive Age female M/C - .....



Management

- Goal - to lift .....

Best procedure .....

- Middle part of urethra is supported.

Best surgery .....

- proximal part of urethra is supported.
  Support is Given by cooper's ligament

Urge incontinence

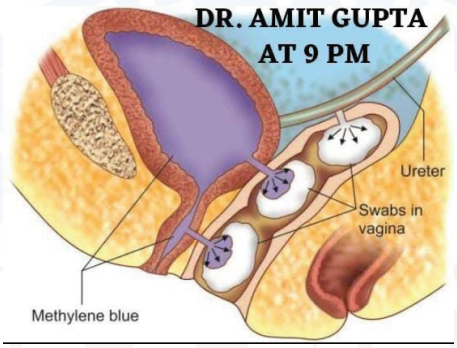
MC cause -

Mx-

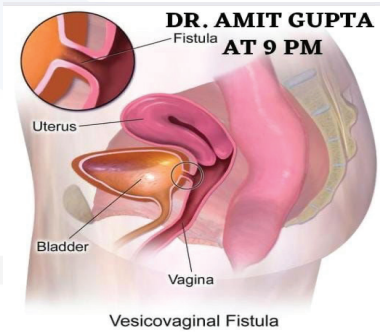
Advice -

Active space

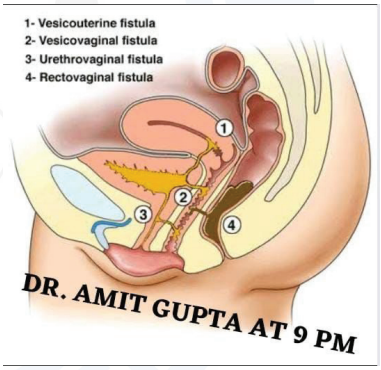
Urinary Fistula



Methelene blue '3' swab test



Vesicovaginal Fistula



1. Ureterovaginal fistula ..... dribbling of urine through vagina and .....micturition

2. Vesicovaginal fistula .....dribbling of urine through vagina .....micturition

3. Urethrovaginal fistula - dribbling of urine through vagina .....

Active space

- Screening test..... test

1. If top most swab become wet but not blue→

   → Later on all swabs affcted

2. If middle swab wet/blue →

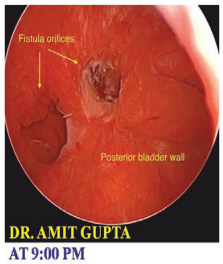
   → Later on Lower Swab also affected

3. If lowermost swab turns wet/ blue →

Gold standard

For Vesicovaginal fistula and Urethrovaginal fistula.....

For Ureterovaginal fistula .....



- MCfistula
- MC cause - Developing countries- .....

  Developed countries - Post Surgical - as complication

Mx

1. Pre operative Mx

   Most important aspect →.....

   To choose most appropriate
2. Operative Mx
3. Post operative Mx -
   - Most important aspect.....
   - Antibiotics for .....
   - No per-vaginal, per-speculum exam, intercourse for ....
   - No pregnancy allowed .....
   - If become pregnant →

Active space

Timing of repair

If diagnosed within ..... Immediate repair

If diagnosed after ..... After .....

Fistula formation occurs after ..... in obstructed labor

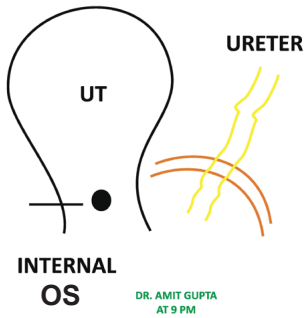
1. Urethrovaginal fistula
   - Very very rare
   - Cause - Fall from height on a penetrating object over urethra
   - Mx- Repair

2. Ureterovaginal fistula
   - MC cause - ..... hysterectomy

     Highest risk of ureteric .....

     MC site of ureteric injury A. at pelvic .....

     B. Place where ..... is crossed by .....
     ..... lateral to internal os.



Highest Risk of injury to urinary tract in Hysterectomy :
Radical Hyst. > laparoscopic Hyst. > Abd. Hyst. > Vaginal Hyst.

   - Mx - .....

Timing of repair.....

Menouria / Mensouria - hematuria at menses

   - ..... fistula (Youssef syndrome)

     .....

Active space

Rectovaginal fistula

- Symptoms- passage of faeces through vagina
- Most common cause .....

↓

- MC cause.....
- Mx- Initial colostomy followed by repair.....

Types of Perineal tear

1° Tear - Tear of Skin+Fascia

2° Tear - Tear of Skin+Fascia+muscles

3° Tear - Tear of Skin+Fascia+muscles+Anal Sphincter

4° Tear - Tear of Skin+Fascia+muscles+Anal Sphincter+.....

Another classification based on distance of fistula from external uriinary meatus

Types I > 3.5 cm

II 2.5-3.5 cm

III 1.5 to < 2.5 cm

IV < 1.5 cm

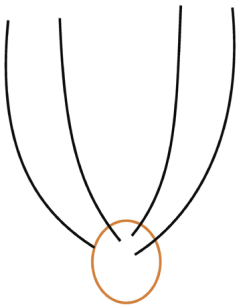
Depending on diameter of fistula -

a < 1.5 cm

b 1.5-3 cm

c > 3 cm

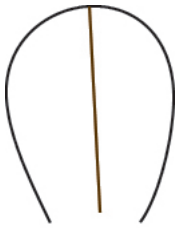
Active space



DR. AMIT GUPTA
AT 9 PM

CONGENITAL MALFORMATION OF UTERUS

Congenital Malformation of Uterus



DR. AMIT GUPTA
AT 9 PM

- 2 Mullerian duct, start developing at -------weeks of IUL in above - below fashion
- They fuse in below-Upwards fashion at -------- wks, but their inner walls don't disappear
- Inner walls disappear at --------- weeks in ....fashion

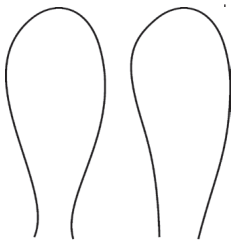
Normal uterine cavity is formed at .....

DR. AMIT GUPTA
AT 9 PM

Mullerian duct- Uterus, cervix, upper ..... of vagina, Fallopian tube

Urogenital sinus.....

Abnormalities

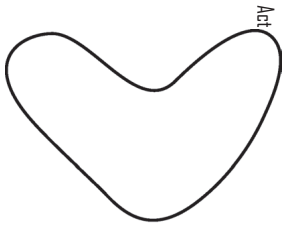


DR. AMIT GUPTA
AT 9 PM

1. If Müllerian duct don't fuse at all -

   Uterine ..... or ..... uterus

   Bicornis bicollis is also categorised as .....



DR. AMIT GUPTA
AT 9 PM

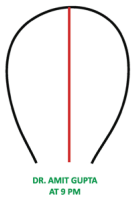
2. If mullerian duct fuse partially

   ..... Uterus (..... shaped uterus)

   2 types - bicornis .....> bicornis .....

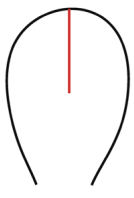
   may be associated with .....

3. If mullerian duct fuse but their inner wall don't disappear even after 20 weeks –



4. If mullerian duct fuse but their inner wall disappear partially even after 20 weeks –

Septum is present in ..... of uterus



5. If only one mullerian duct develop

..... uterus or ..... or ..... uterus

may be associated with .....

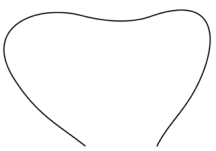
2nd MC Malformation associated with .....

MC Malformation associated with renal anomalies .....

Associated with .....

6. Arcuate uterus .....

Dome of uterus has .....



7. T-shaped uterus .....

Due to use of ..... (synthetic estrogen) in -utero or intrauterine life (IUL)

most specific malformation .....

MC malformation .....

MC malignancy .....

Active space

Trasverse lie will not occur in uterine didelphys and unicornuate uterus

MC congenital malformation .....

- MC complication -
- Other complication -
- MC malformation responsible for abortion (Worst reproductive outcome) .....
- Best reproductive outcome -

Diagnosis

- Most commonly used USG → TVS > TAS TVS is C/I in virgin girl
- Gold standard.....
- Other investigation HSG hystero-Salpingo graphy

Which Investigation should be done in women with mullerian anomalies - Assessment of .....

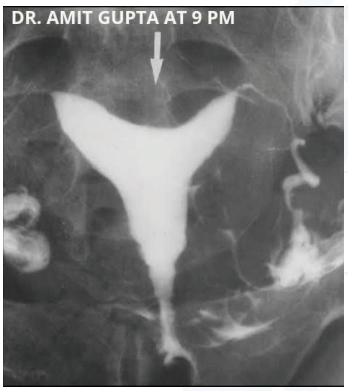
Indication of management

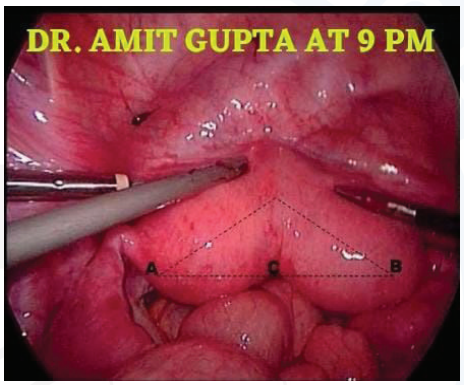
- If H/O recurrent abortion

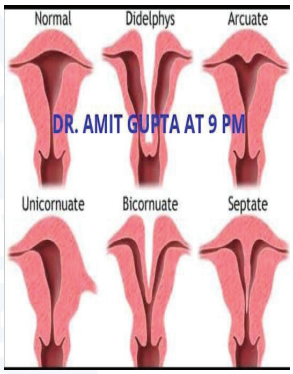
1. For septate uterus.....
   Hysteroscopic septoplasty or .....
2. If uterine didelphys - repair .....

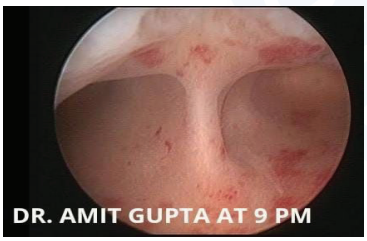
Active space

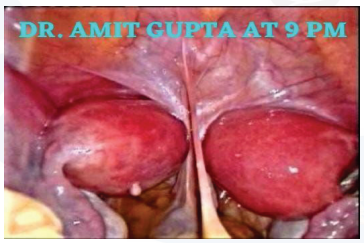
3. If Bicornuate - repair .....

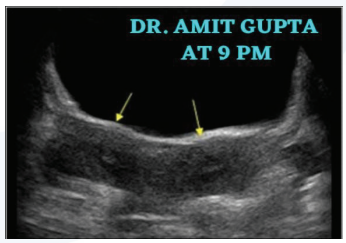












Classification : American fertility society classification

Class - 1 Mullerian Agenesis

Class - 2 Unicornuate Uterus

Active space

MIST

Class - 3 Double Uterus

Class - 4 Bicornuate Uterus

Class - 5 Septate Uterus

Class - 6 Arcuate Uterus

Class - 7 T - Shaped Uterus

HSG - Iodinated water soluble dye (..... Dye) is instilled via .....

Contraindications - a. Pregnancy, b. Active Genital TB, c. Active PID

Bicornuate Uterus Septate Uterus

1. Distance between cornuae

2. Angle between cornuae

Active space

Infertility

- Failure to conceive even after ..... of unprotected and regular intercourse Couple - 8mth. Back

Primary infertility - She never conceived before Conception - Fertilization

Secondary infertility - She conceived before but now .....

Sterility - Absolute inability to conceive due to permanent damage to .....

Fecundability - probability to ..... in one cycle

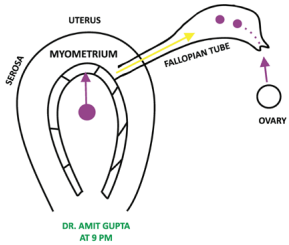
Fecundity - probability to ..... in one cycle

approx.

- Male-35%, Female- 45%, Combined- 10%, Unexplained-10%

Female factor approx.

- MC- Ovarian disorder-45 %, Tubal disorder-35%
- Uterine/cervical-10%, Unexplained infertility -10%



Management of unexplained infertility

→ Clomiphene citrate f/b intrauterine insemination for .....

Mineral required for fertility is .....

Ovarian disorder

- MC cause, most easily diagnosed cause, most easily treatable so most reversible cause of infertility .....

Causes of anovulation (WHO Classification)

Class 1 H-P-O axis failure-10%

- ..... syndrome

Class 2 H-P-O Axis dysfunction-85%

- MC - .....

Class 3 Premature ovarian failure (old name) or
Primary ovarian insufficiency (new name)
If ovaries fail before 40yrs.

Class 4 .....

Diagnosis of anovulation

1. Maturation index -on day .....
   If ovulation occurred-Intermediate cells max.- 0/70/30
2. Cervical mucus - on day ..... →Thick, Scanty, Non elastic, Tack phenomenon
3. Endometrial Biopsy- on day .....→ vacuolation(earliest)
   cork- screw glands or sawtooth glands
   If ovaries fail before 40yrs.
4. Easiest/Best- ----------on day ..... → > 3ng/ml
5. MC used - serial .....(follicular monitoring or ovulation studies) Evidence of ovulation

Active space

6. Basal Body Temperature→Progesterone ..... temperature by 0.5-0.8^0 F

Predictor of ovulation - .....kits if +ve .....

LH kits detect .....

LH surge in urine occurs .....

Mx of anovulation

for class 1 - If H-P-O axis failure - GnRH agonist in .....

for class 2 - If H.P.O Axis dysfunction - .....

for class 3 - If primary ovarian insufficiency .....

for class 4 - Hyperprolactinemia .....

First line drug for infertility management

DOC to treat anovulation - .....

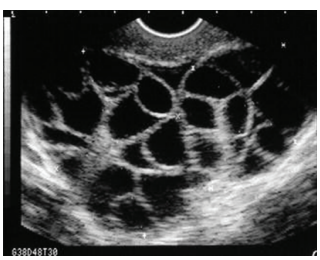
DOC to treat anovulation due to PCOS .....

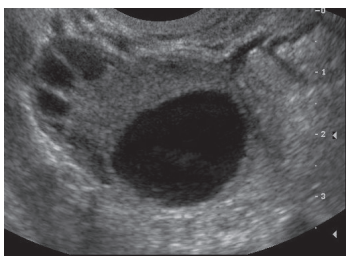
DOC to treat anovulation in pituitary ablation .....

Various S/E of Clomiphene Citrate

S/E : Hot flushes, breast tenderness, headache, twin pregnancy, OHSS, Endometrial thinning, Visual symptoms

Indication to stop clomiphene .....





Normal Single Mature Follicle

Active space

Tubal disorder

- MC → ..... block
- MC cause → PID> Endometriosis
- ..... syndrome

MC microorganism causing cornual block .....

MC microorganism causing fimbrial block .....

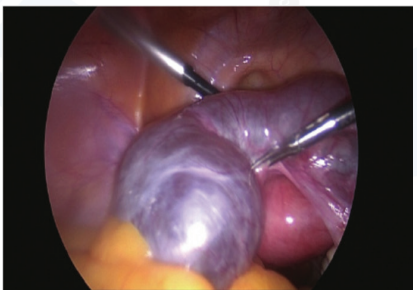
Diagnosis of tubal patency test-

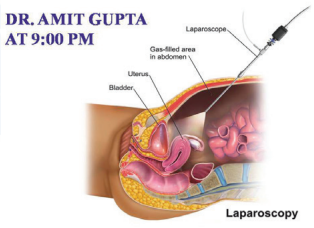
1. Hysterosalpingograpy (HSG) → X-ray
2. Sonosalpingography (SSG) → By USG
3. BEST Laparoscopic Chromopertutation - .....
4. Hycosy - Hysterosalpingo - contrast-sonography - look for tubal patency.

   Contrast : microbubble contrast or agitated saline

   Advantage over HSG : ..... is also seen

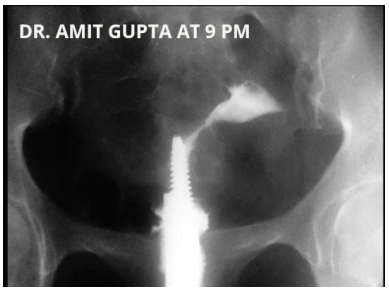
Test for tubal patency -Done on D 10





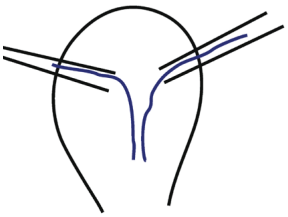
Mx-Depend on site of block

1. MC Cause of bilateral Cornual block
   - .....



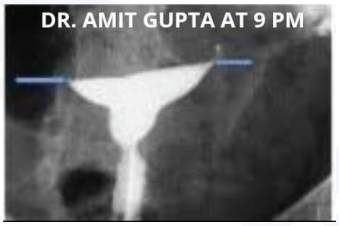
Active space

- Hysteroscopic tubal .....
  - If able to pass .....
  - Followed by Laparoscopic Chromopertubation

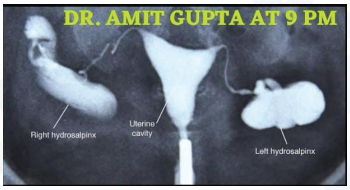


If not able to pass cannula .....

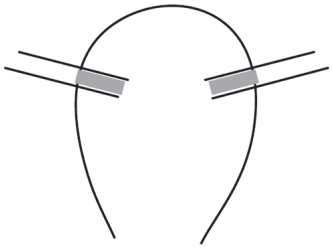
2. MC Pathological cause of bilateral cornual block.....
   Mx- IVF



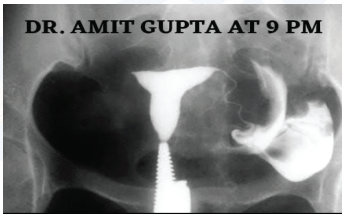
3. Bilateral fimbrial block-
   Mx For Mild .....
   Mx For Severe .....
   Best Mx .....



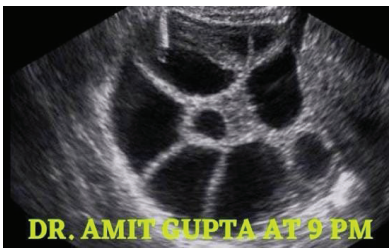
4. Mid segmental block
   - MC cause - H/O tubal ligation
   - Mx - Recanalisation

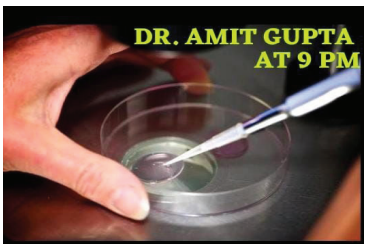


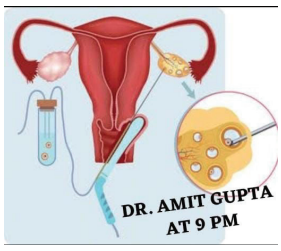
Prerequisite



Best Mx for tubal block- IVF.....







Active space

DOC for ovarian stimulatⁿ for IVF .....

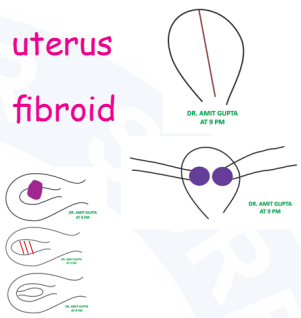
..... Sperms are dropped /oocyte

..... is essential for IVF

..... Transfer is done - ..... CM Below
(Embryo-Transfer) Uterine Fundus

Uterine factors

1. ..... uterus
2. ..... fibroid
3. ..... Fibroid
4. ..... Syndrome
5. Endometrial Polyp



Gold standard investigation for uterine factor infertility

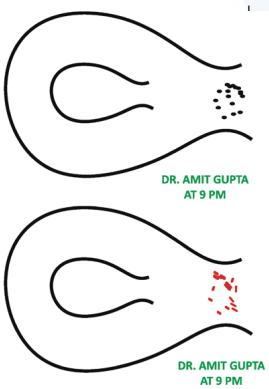
.....

Cervical factor

1. Poor cervical mucus- impermeable for sperm

   Dx- Insler score → normal

2. Anti sperm antibody.....



Diagnosis

- Post coital test or..... test
- 2 slides on Day.....
- ..... intercourse

a. From cervical canal - Movement of sperms .....

b. From posterior fornix - Movement of sperms .....

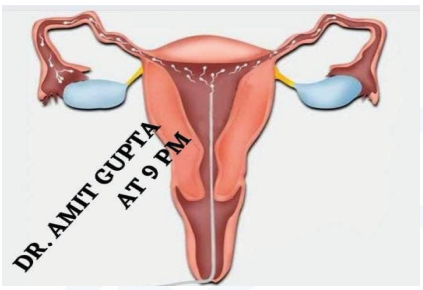
Active space

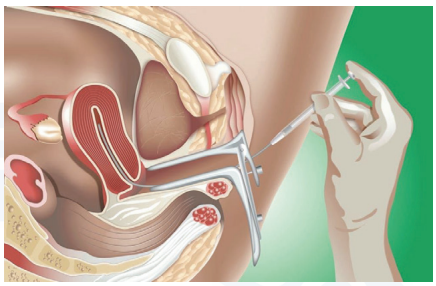
Mx - Intrauterine .....

..... ml of ..... semen is inseminate near

..... via .....

source of semen .....





Best method for semen washing .....

Advantage of semen washing

A. Removal of .....

B. Sperm concentration is .....

Prerequisite for IUI Sperm Conc. ≥

Other indications of IUI

- Impotence
- Vaginismus- Involuntary contraction of vagina at intercourse due to anxiety
- Mild Oligospermia
- Epispadias, hypospadias, Retrograde ejaculation
- Poor cervical mucous
- Immunological .....

1st investigation advised in infertile couple .....

If unable to conceive even after six months of marriage

.....and follow up after six months.

If married for more than one year - .....

Active space

Min. 2 Days
Max. 7 Days

Seminogram - based on ..... reference range

Done on Liquefied sample

Prerequisite - abstinence from .....
..... for at least .....

most important criteria

- sperm Morphology >

sample should reach lab within .....

liquefaction time .....

if semen analysis report is azoospermia - repeat after .....

if the report is same - get done .....

Seminogram - WHO - Based on lower reference range

1. Amount ≥.....ml
2. Sperm count ≥.....Million/ml
3. Total sperm count ≥ ..... Million / Ejaculate
4. Morphologically normal sperms ≥ .....
5. Progressively motile sperms ≥ .....
6. Total motility ≥ .....
7. Viable forms ≥ .....
8. Leucocytes < ..... Million/ml
9. Round cells < ..... Million/ml.....
10. pH.....

Abnormalities

1. If semen .....ML →Hypospermia
2. If semen .....ML →Hyperspermia
3. If semen..... - Aspermia
4. If sperm ..... Azoospermia
5. If sperm count .....Million/ML - Oligospermia

Active space

a) If ..... Mild oligospermia

b) If ..... - Moderate oligospermia

c) If .....Severe oligospermia

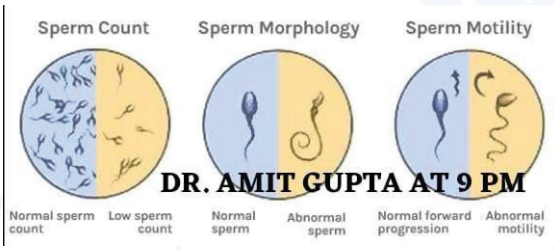
6. If Morphologically normal sperms ..... Terato spermia
7. If Progressively motile sperms ..... Asthenospermia
8. If Viable sperms ..... Necrozoospermia
9. If Round headed sperm ..... Globozoospermia

(Acrosomal cap is absent)

Male factors

I. Pre testicular

- Hypothalamus
- Pituitary



II. MC Testicular → MC → Idiopathic → Defective .....

Other Causes

- Accident
- H/o radiotherapy
- Kleinfeilter syndrome - 47 XXY, small hypoplastic testis, Gynaecomastia, tall stature (Long Legs), mentally retarded, IQ ↓, life span ↓ false positive baar body etc.
- Refeinstein syndrome
- H/o mumps etc.

III. Post testicular

- Obstructive - Obstruction in .....

Seminal vesicle & vas deferens may be bilaterally absent in .....

MC cause of male infertility .....

MC cause of reversible male infertility .....

Active space

Diagnosis by FSH, LH, Testosterone. Best marker .....

If pretesticular

- FSH, LH, Testosterone

If testicular

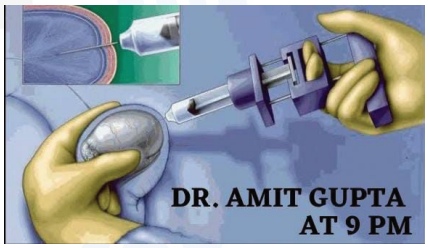
- Testosterone
- FSH, LH

If post testicular

- FSH, LH, Testosterone

In all male partners when the count <10 million /ml - Do hormonal analysis

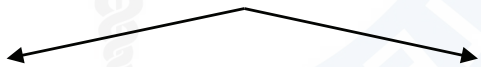
If count : <5 million / ml - Hormonal + Genetic Analysis



Mx

- If pre testicular .....
- If testicular problem - Accordingly
- If post testicular problem-

Azoospermia with normal FSH, LH, Testosterone



next step

Testicular sperm .....

.....

- OPD procedure
- L.A. - .....

Best Mx

- Microsurgical ..... sperm .....
- G.A.
- IPD
- From epididymis

A. Sperm attain .....

B. Sperm are in .....

C. Can correct .....

Active space

TESA/MESA - f/b .....

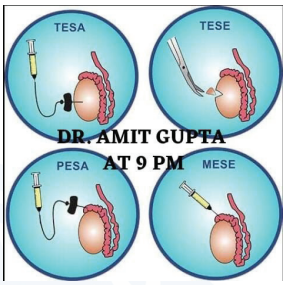
Azoospermia, normal FSH, LH, Testosterone, Failed MESA

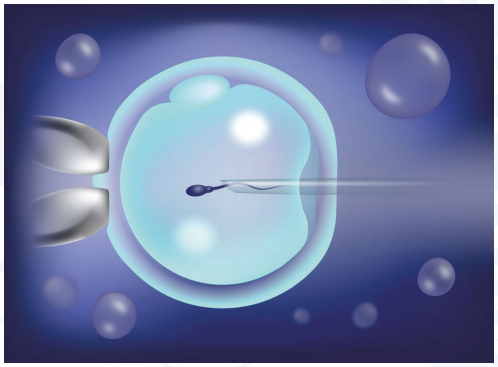
↓

Next step

Therapeutic Donor insemination

(TDI or IUI-D)





ICSI.....

Indications

- ..... spermia
- ..... spermia
- Obstructive Azoospermia

ART (Assisted reproductive technique)

- Any technique involving .....
- IVF
- ICSI
- ZIFT (..... Intrafallopian Transfer)
- GIFT (..... Intrafallopian Transfer)

- IUI .....

Active space

MENOPAUSE

Menopause

- Cessation of menses for at least .....In menopausal age
- Menopausal age.....
- In Developing countries.....
- Perimenopausal period - 4-5 years period of transition
- Premature menopause Before .....
  New name - Primary ovarian insufficiency
- Delayed menopausal-Not even till.....yrs

Maximum no of follicles are at .....

Maximum no of follicles at birth .....

Maximum no of follicles at puberty .....

Ouvlate - .....

every month .....undergo atresia

Event at menopause -Decline in activity, ultimately leading to cessation of H-P-O axis

Estrogen, Progesterone .....

So FSH, LH.....

Uterine size.....

Endometrial thickness.....

ACOG.....

FIGO.....

HRT is given upto .....

HRT is given upto .....

HRT decreases risk of colorectal cancer

Active space

Principle of Hormone replacement therapy (HRT)

- If Uterus is intact→
- If Hysterectomy done →

Progesterone is added for at least

Estrogen used in HRT .....

Progesterone used in HRT .....

Symptoms

MC , Earliest, Most Characteristic, symptom for which maximum treatment is required

1. .....sweating (accompanied by tachycardia/Palpitations)

Feeling of warmth due to dysregulation of thermodynamic centre (thermostat) due to imbalance between .....

.....Due to Estrogen

Sweating palpitation, nervousness .....

Mostly at .....

one episode lasts for.....

correspond to .....

Mx- DOC.....

Other drugs

SSRI .....

For acute symptom .....

Gabapentin, isoflavones .....

2. .....

- DOC .....
- Side effects.....
- Precaution.....
- Other drugs

Active space

Bazedoxifene, Teriparatide, salcalcitonin, denosumab, Strontium ranelate

For prevention - .....

Most of drugs inhibit osteoclast. only 2 drugs act on osteoblast namely Strontium, Teriparatide

3..... Vaginitis - Vaginal Itching Due to .....

Urogenital Syndrome = Vulvovaginal Atrophy
atrophic vaginitis

Urinary Symptoms Genital Symptoms

Mx - First step A. - Vaginal lubricants / moisturisers
B. - Local Esrogen cream
C. - .....

4..... Libido

Mx - .....

Other symptoms

5. Sagging of skin

6. Psychiatric disturbances

7. Risk of cardiac disease → Refer to cardiologist

Cardio-Protective
Estrogen ↑ HDL ↓ LDL

Maturation Index .....

Maximum estrogen in menopausal lady .....

C/I of HRT :-

1. CAD (Coronary Artery Disease)

2. VTE, 3. Smoking, 4. TIA, 5. Breast Ca

6. Active liver disease, 7. High risk of Endometrial Ca

Active space

Premature menopause (Primary .....

- MCC -
- H/o radiotherapy
- Accidental removal of ovaries
- Diagnosis - Best→.....

Normal →

In Pcos → AMH ↑

Secreted from ..... of ovary (reserve pool)

AMH prevents .....selection of follicle

AMH decreases sensitivity of preantral follicles to FSH, it corelates with pregnancy rate of IVF cycle, it is best tested in fasting stage on day 2 of menstrual cycle

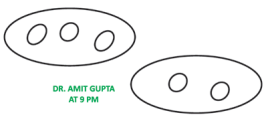
- Can be done on any day

Ovarian failure if .....

E_2 in menopausal lady .....

Others

- S. FSH on D3 →.....→ Ovarian failure
- Antral follicular count.....→ ..... Ovarian failure
- Inhibin -B if.....→Ovarian failure



In low ovarian Reserve, AMH Level will be < 1 NG / ML

Best S. ..... - (Anti mullerian Hormone)

MC Used S. .....- (done on D3 (D_2-D_4)

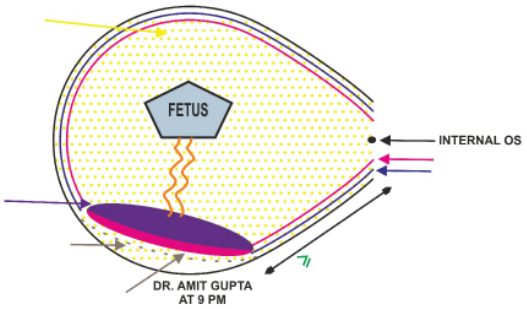
Active space

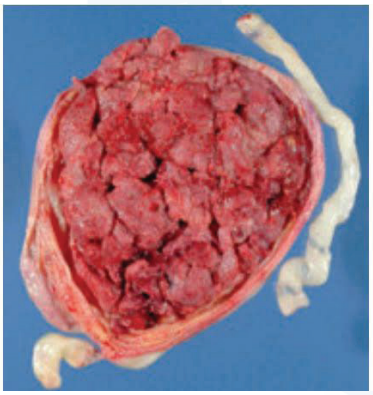
II
OBSTETRICS

Active space

PLACENTA AND UMBILICAL CORD

Placenta and Umbilical Cord





Placenta- formed by

Maternal component

- .....
- Form.....part of placenta
- Formed by.....
- ..... Plate
- 15-20 lobes present
  Each lobe contains
  Lobules or .....

Fetal component

- .....
- Form.....5 Part
- By.....
- .....Plate
- .....is attached

Active space

- Functional unit of placenta→.....

- Starts forming at .....wks.
- Anatomical development completes at ..... wks.
- Functional maturation of placenta occurs .....

Decidua - formed by ..... . Rich in secretions.

1. Uterine blood flow at term .....
2. Uteroplacental circulation
   - Established by- .....
   - At term- .....
3. Fetoplacental circulation
   - Established by .....
   - at term .....
4. Fetal circulation is established by .....
5. Fetal blood flow at term .....
6. Rate of oxygen supply to fetus .....

Shape of placenta

- Disc shaped discoid
- Circumference .....
- Thickness ..... correspond to gestational age in .....
- At 40 weeks ..... 35wk → 3.5cm
- Average thickness ..... 36wk → 3.6cm
- It increases by .....
- Placentomegaly → > ..... cm
- Deciduate – sheds off after delivery
- Haemochorial – contact with blood
- Weight of placenta .....
- Volume of placenta .....

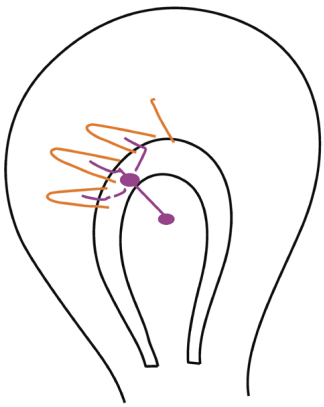
Active space

- Villi .....
- Inter villous space .....
- Weight of Placenta is equal to Fetal weight at .....
- At term ratio of placental weight : fetal weight .....

Normal Location → Mc → .....part of uterus

because MC site of interstitial inplantation is .....
part of uteru

Hartman sign - spotting p/v at the time of .....



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AT 9 PM

Various Hormones are secreted by placenta

- hCG (Human chorionic Gonadotropin)

secretion begins on day ..... post-fertilization
secreted by .....
it doubles every .....
peak values are at .....
minimum levels at .....
disappear from circulation .....After delivery
half life .....
2 subunits .....
.....non specific ..... similar to .....

..... specific to .....
Maximally responsible for .....

Active space

Luteo - placental shift - Function of corpus luteum is taken over by .....
.....

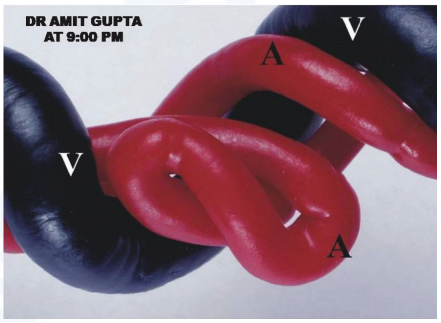
Occur at approx. .....

Life span of pregnant corpus luteum .....

Life span of non-pregnant corpus luteum .....

- Estrogen
- Progesterone
- HPL (human placental lactogen) New name :- HCS (human chorionic somatomammotropin)
- hCT (human chorionic thyrotrophin)
- PAPP-A (pregnancy associated plasma protein-A)

Distance between lower edge of placenta & internal OS should be .....
.....

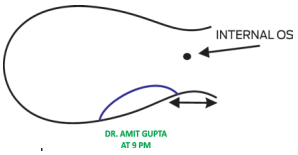


Umbilical cord (Funis) - Inflammation of cord .....

- Length .....Mean Length.....
- Connects fetus with placenta
- Develop from connecting stalk
- Coils/ Folds/ loops →Folds of hoboken
- Connective tissue of umbilical cord - Wharton's Jelly
- It is attached on central part of placenta

Active space

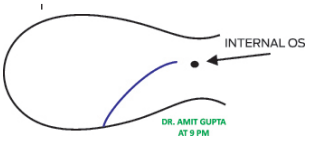
- Initially has 2 arteries/2 veins
- later on -Right umbilical vein disappears
- finally ..... umblical artery and ..... umblical vein.....

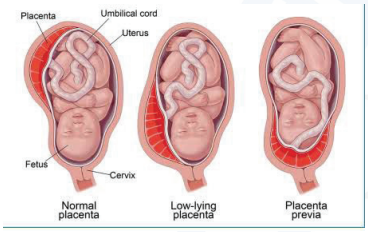


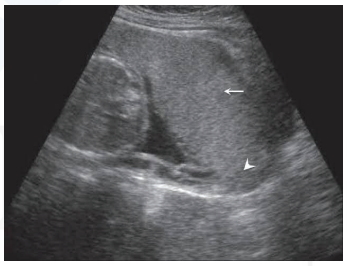
Abnormalities

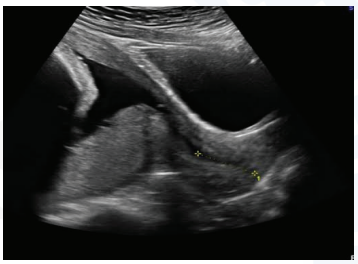
1. A. Low lying placenta –if Lower edge..... cm from internal OS

B. Placenta Praevia - If lower edge of placenta .....



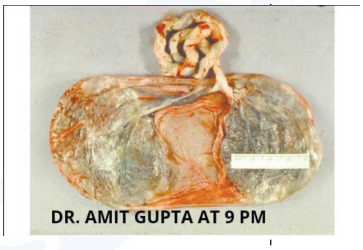






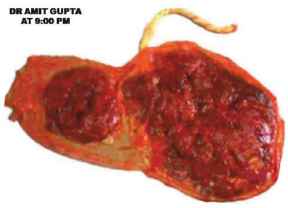
Complication

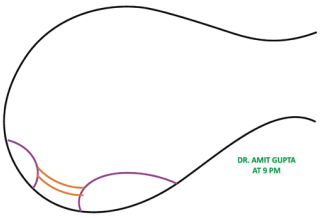
2. Accessory lobe



1. Placenta bilobata or duplex or bipartite
   - ..... sized lobe
   - Connected by .....

   Umbillical cord attached to both the lobes



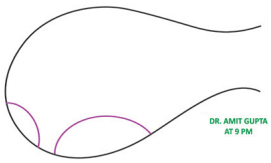


2. Succenturiate lobe
   - ..... accessory lobe
   - Connected by blood vessels
   - Umbillical cord is attached to main lobe & vessels connect main lobe to Accessory lobe.

MIST

3. Placenta spuria –

- ..... accessory lobe
- ..... vascular connection

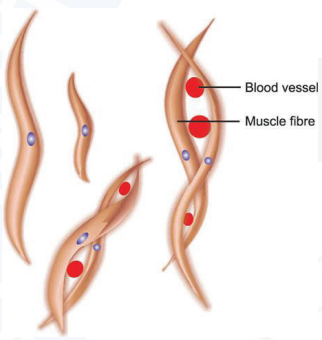


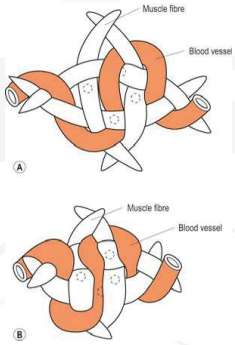
Placenta multilobata .....

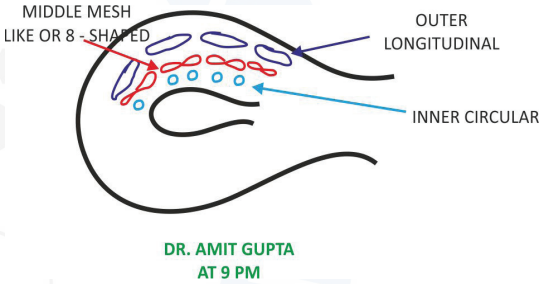
Complication -.....

Living ligature of uterus → present in .....

Advantage it Prevents .....





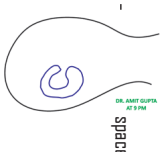


Normally fetal surface=maternal surface

3. Extrachorial placenta - Maternal surface ..... than fetal surface – C shaped

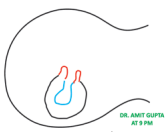
Two Types A. Circummarginate

B. Circumvallate



SPACE

A. Circummarginate - Maternal Surgace continues.....



B. Circumvallate placenta

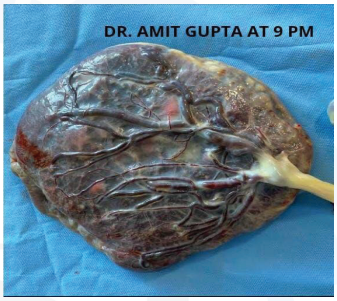
- ..... is present between maternal and Fetal surface

MIST

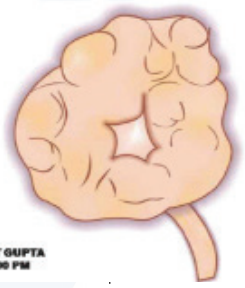
Complication

1. .....
2. .....
3. .....

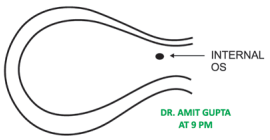
4. Battledore placenta
   - Umbilical cord is attached at placental .....
     Complication .....

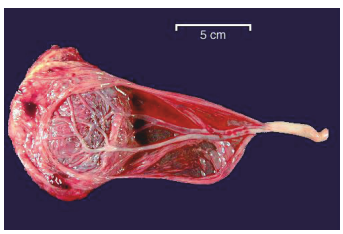


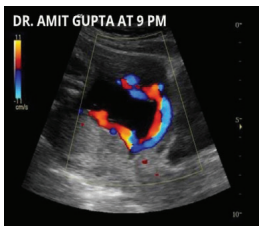
5. Fenestrated placenta - if .... of chorionic surface is absent

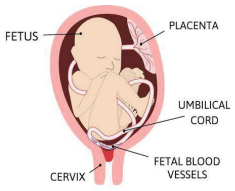


6. Placenta membranacea-
   - Thin membranous placenta occuplying whole uterus
   - Associated with
     a. Placenta Praevia
     b. Morbidly adhered placenta
   - Complication
     a. Antenatal bleeding P/V
     b. PPH

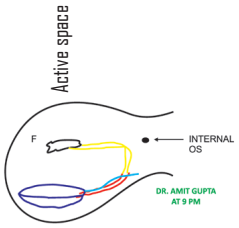








VASA PREVIA



MIST

7. Velamentous insertion of cord

- Umbilical cord is inserted .....

- Complication- .....- Bleeding p/v.....

- Diagnosis of vasa praevia-.....

Differentiation between maternal and fetal blood→Singer's Alkali Denaturation test or .....

- Reagent .....
- Principle.....
- So fetal blood .....
- Maternal blood .....
- Mx .....

Short umbilical cord - If length is .....

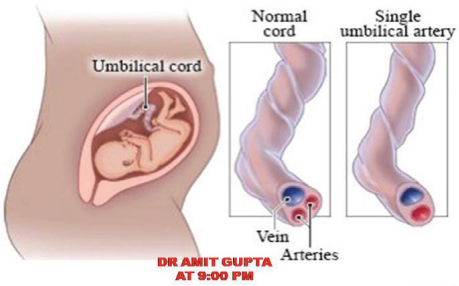
Complications

Active space

Long umbilical cord - If length is .....

Complications

Single umbilical artery



Incidence .....

MC Vascular Anomaly of cord

MC type .....

Associated with

A. Congenital malformation .....

B. Aneuploidy - MC-.....

C. Associated with Diabetes, Twins, Black african population

Morbidly adherent Placenta

Cause .....

.....

-Risk Factors - Same as that of placenta previa

- Underlying decidua is absent.

Highest Risk : Previous H/O Caesarean Section + Placenta Previa

Since there is no plane of seperation of placenta

Placenta enters .....

Histopathological Examination - Absent Nitabuch's Membrance.

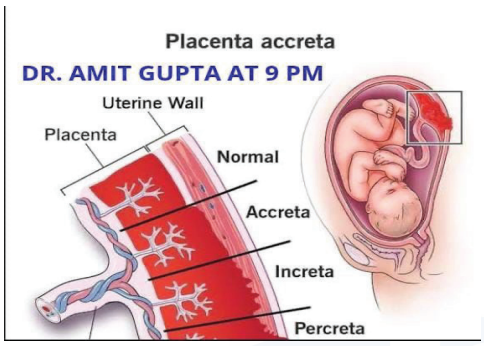
(Layer of fibrinoid degeneration - Normally present between trophoblast & decidua)

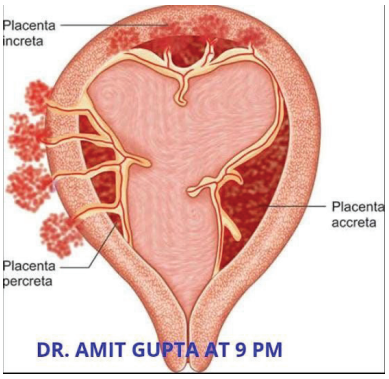
Types (Grades) - FIGO

Grade 1 - Placenta .....

Grade 2 - Placenta .....

Active space





Grade 3 - Placenta .....

3 a - Reaches Upto - .....

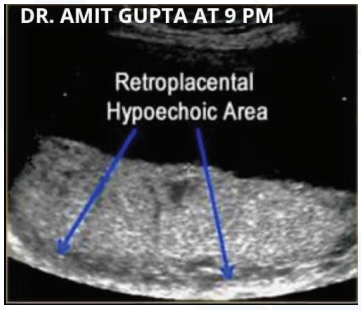
3 b - Reaches Upto - .....

3 c - Reaches Upto - .....

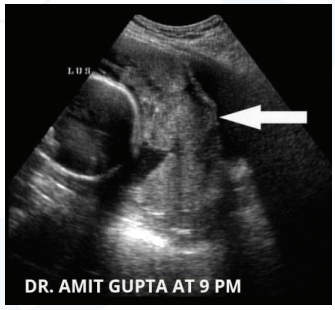
IOC - .....

Findings in color doppler:- 1) Abnormal vessels going from Placenta to myometrium

Gold Standard .....



Normal



Morbidly Adhered placenta

Complications .....

Management in diagnosed case -

A. If Multiparae .....

B. If Primiparae .....

Management in undiagnosed case -

If she presents with severe intractable post-partum haemorrhage .....

.....

Active space

AMNIOTIC FLUID

Amniotic Fluid (Liquor Amnii)

Composition 99% Water .....

1 % solids - Na^+, k^+, cl^-,

- Urea, Uric Acid, creatinine

Functions of amniotic fluid No .....

1. Prevent fetal compression by uterine wall
2. Shock absorber
3. Temperature maintenance
4. Bacteriostatic property

Hypotonic to .....

Isotonic to .....

Source

- In Early pregnancy - Diffusion or Ultra-filtrate from maternal plasma
- 12 to 20 week: diffusion from fetal skin
- After 20 week - .....

Major contributor .....

Fetal kidneys become functional at .....

Entire Amniotic Fluid is replaced every 24 hours.

- pH .....
- Specific gravity- .....
- Turn over - .....

Active space

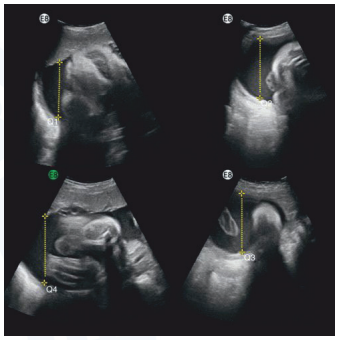
- Turn over - Maintained by fetal swallowing
- Osmolality .....
- Assessment By .....

2 Parameters

a. Single Deepest Pocket or largest vertical pocket.....

Normal .....

b. Amniotic fluid index (AFI) Summation of ..... pockets of amniotic fluid



Normal .....

c. Amount .....

If less – Oligohydramnios

If excessive – Polyhydramnios

Oligohydramnios:

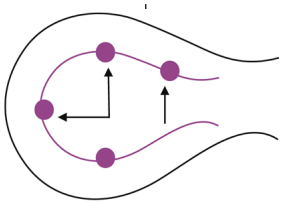
- SDP<.....
- AFI<.....
- Amount: <.....

Causes of Oligohydramnios:

- MCC of mild oligohydramnios – idiopathic
- MCC of severe oligohydramnios –
  1. .....
  2. Fetal renal problems
     a. Bilateral renal agenesis
     b. Poly cystic kidney disease
     c.

Active space

Other causes



3 Prelabor rupture of membranes (PROM)

4 Post term pregnancy d/t calcification in placenta

5 Indomethacin

6 Amnion nodusum nodules on amnion- absorb amniotic fluid

7 Triploidy

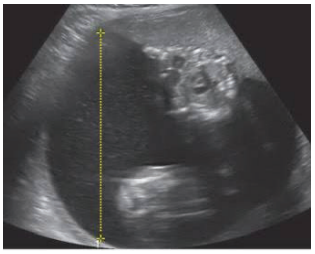
Complication

1. of Early Oligohydramnios –upto 30 weeks
   - .....
   - .Amniotic band syndrome or Constriction band synd or Streeter syndrome
2. of Late Oligohydramnios(>30week)
   - Meconium aspiration syndrome
   - Congenital talipes equinovarus {CTEV}

Potter's sequence- Severe oligohydramnios - facial features and pulmonary-Hypoplasia

Management

- Accordingly
- but if USG show severe compression of fetus or .... deceleration in CTG →Amnioinfusion (Normal saline at body temperature)



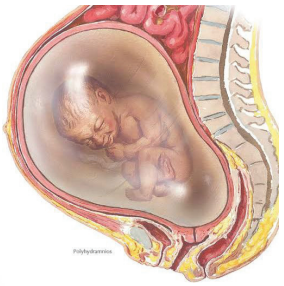
Polyhydramnios

- SDP > ..... cms

Active space

- AFI > ..... cms
- Amount: > ..... ml,
  - MCC of mild polyhydramnios –idiopathic
  - MCC of severe polyhydramnios – GI Defect

    2nd M/C open neural tube defect



GI defects

- Cleft lip /cleft palate
- Esophageal atresia - Intestinal obstruction
- Duodenal atresia
- Annular pancreas - Congenital diaphragmatic hernia

Open neural tube defect - Anencephaly >Spina bifida aperta

Other Causes

- Multifetal pregnancy
- Trisomy 21,18
- Chorioangioma of placenta
- Rh incompatibility
- Diabetes

Couses of both Polyhydramnios and Oligohydramnios

Complications of polyhydramnios

- Respiratory discomfort
- Preterm labour

Active space

- PROM
- Malpresentation
- Cord prolapse
- Abruptio placentae
- Post partum hemorrhage

Features

1. Excesive abdominal distention
2. Fetal parts are not well felt
3. Fetal heart sound may be .....
4. Symphysiofundal height is .....
5. Abdomen tense but not .....

Normally symphysio - fundal height beteen ..... wks. correspond to gestational age in weeks.

Mx

- Accordingly but if severe respiratory discomfort (if SDP >= ....., AFI >=

A.→Amniocentesis

B. Indomethacin →should not be used beyond 32 weeks otherwise it may cause premature closure of ductus arteriosus

Indomethacin Stimulates fetal production of Arginine, vasopressin

Colour-Colorless, ...

- At term – straw coloured
- Rh Incompatibility-Golden yellow

Active space

- Listeria infection green
  Breech green
  Transverse presentations green
- Abruptio placentae- Dark coloured

- Intrauterine Fetal demise – Tobacco colour (blackish brown) d/t thromboplastin
- Post term pregnancy- Greenish Yellow (Saffron Colour)

Amount of amniotic fluid

- At 12 weeks-50 ml
- 40 week -800ml
- 42 week - 200- 400ml

Maximum Amount of amniotic fluid

- Between - 32-34 Weeks

Active space

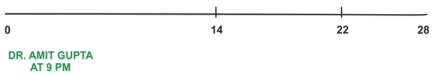
Pregnancy

- Period of amenorrhoea or gestational age- (POA OR POG)
- 40 weeks
- 280 days
- 9 Months+7 days
- Calculated from ..... day of LMP

Conception - Fertilization

Growth period or True conceptional age (TCA)

280 - ..... =



DR. AMIT GUPTA
AT 9 PM

- Calculated from .....
- Ovular period- .....
- Embryonal period- .....
- Fetal period - .....

Cardiac activity by TVS can be seen earliest on day..... post-fertilization means ..... weeks gestation (TCA.....)

Trimester

- 1st trimester- upto 14 weeks
- 2nd trimester -14-28week
- 3rd trimester -29 week to 42 weeks
- 4th trimester- upto 12 weeks from date of delivery

Active space

- Gravida – No. of .....
- Parity – No. of delivery after .....

present pregnancy is not included

No. of fetuses are of no concern in gravida/ parity scoring Eg. - G_XP_y

Old Method

A lady delivered twins 3 yrs. back. Now she is pregnant. What is her Obstetric score .....

Most accepted scoring system (Universal formula) - Provide more information

GTPAL Scoring system

- G- No. of times of pregnancy
- T- No. of term deliveries
- P- No. of preterm deliveries
- A- No. of abortion
- L- No. of living children
- A 30 year old lady came with twin pregnancy at 10 weeks of gestation. She has a 3 year old female child who was born at term and had 2 first trimester abortions previously. Her obstetric score is .....

- Nulligravida –Never got pregnant
- Primigravida –Pregnant for 1st time
- Multigravida –Pregnancy for at least 2 times
- Nulliparae –Her pregnancy never delivered beyond age of viability
- Primiparae– ..... time her pregnancy delivered beyond age of viability
- Multiparae– ..... times her pregnancy delivered beyond age of viability
- Grand multiparae– .....times her pregnancy delivered beyond age of viability

Active space

- Great grand multiparae - times her pregnancy delivered beyond age of viability
- Puerpera- In puerperium (Within 6 wks after delivery)
- Parturient – In labour. Parturition - Uterine Contraction

Early term - ..... weeks -..... weeks 6 days

Term -..... week-..... weeks ..... days

Late term > ..... W - ..... W ..... D

Post term >= ..... weeks

Preterm-Before ..... weeks

Calculation of expected date of delivery (EDD)-By NAEGLE'S formula

- Add ..... to days
- Subtract ..... months

LMP = 15/11/2024

In 30 day cycle →

EDD – ..... In 25 day cycle →

But if month is changed while adding days- .....

LMP = 28/11/2024

EDD – .....

Applicable for regular 28 days cycle

But If prolonged cycle - ..... additional days in EDD

If cycle is shorter - ..... shorter days

Active space

In IVF pregnancy - add ..... from day of .....

Best assessment of gestational age is by .....

Best clinical assessment of gestational age is by .....

Symptoms

1st trimester symptoms

- Earliest symptom-.....
- Others symptoms
- Morning sickness - nausea, vomiting due to .....

Mx - dietary modification - meals .....

if no response - tab. .....

if still no response - tab. .....

last resort.....

- Increase in frequency of micturition

.....

- Lethargy
- Breast discomfort due to .....

2nd trimester symptoms

- ..... - Perception of first fetal movement

Multiparae .....

Primiparae .....

3rd trimester Symptoms

- .....
- Reappearance of increased frequency of micturition

M/C cause of UTI in Pregnancy .....

UTI in pregnancy is growth of .....in voided urine sample

Active space

MIST

Antibiotic of choice for Urinary tract infections in pregnancy.....
In non - pregnant .....
C/I In .....

Signs
Presumptive Signs - Signs that are felt by Patient herself.
- Amenorrhea
- Nausea, vomiting Morning Sickness
- Urinary feequency
- Breast changes
- Ed pigmentation
- Quickening

Probable signs - signs seen by doctor.
Observed on bimanual Examination
1. Hegar Sign : Soflening of uterine isthmus.

Positive signs
when it is sure that pt. pregnant.
1. USG findings in early Pregnancy. (Gestational sac with embryo, yolksac
2. Fetal cardiac activity
3. Palpate fetal parts
4. Auscultate fetal heart sound.

1st trimester
- Earliest sign -.....

Softening of cervix due to .....
.....

Active space

Other signs -At 8 weeks
- Jacquimer/Chadwick sign -bluish violet discoloration of cervix/ vagina/d/t venous congestion
- Osiander sign → palpation of pulsation of ..... on p/v exam

- Piskacek's sign - ..... enlargement of uterus
- Palmar sign -sense some tone d/t utrine contraction of ..... intensity

  Uterine contraction Palpable .....

  Uterine contraction Pain .....

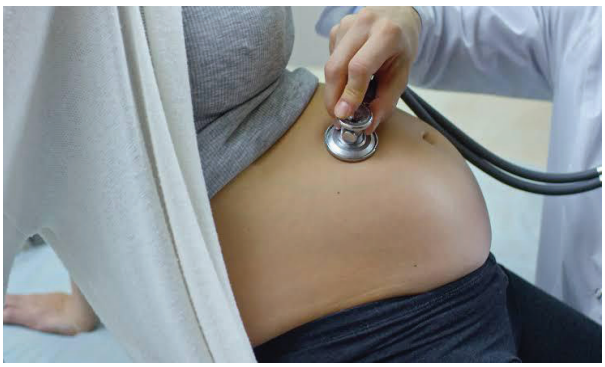
  Cervical dilatation at .....
- Hegar's sign (6-10 weeks)- ..... sign abdominal & vaginal fingers seem to approximate each other
- Von - Braun - fernwald sign .....
- Landin sign .....
- Mc Donald Sign .....

2nd trimester sign

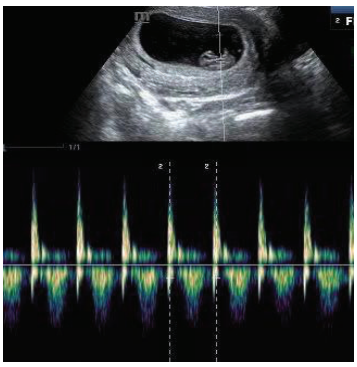
- Palpation of fetal parts - can be done .....
- Auscultation of fetal heart sound (stethoscope) .....

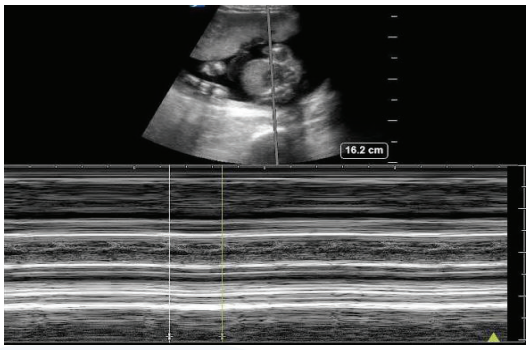
  by fetal doppler .....

  by USG .....



Active space





3rd trimester sign

- Lightening or ..... sign

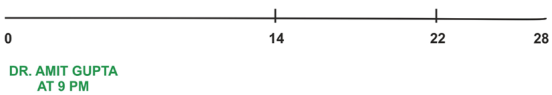
Size of Uterus -

At 6 weeks pregnancy uterine size - Egg of hen
At 8 weeks pregnancy uterine size - Cricket ball
At 12 weeks pregnancy uterine size - Fetal head

6-8 wk - Pear
8-10 wk - orange
10-12 wk - Grape fruit
at 12 wk - suprapubic area
abdominal organ after 12wk

Investigations

- Earliest - ..... → Early morning sample → it is positive only when she is at least .....
  Minimum hCG required for UPT to be + ve .....
  .....test

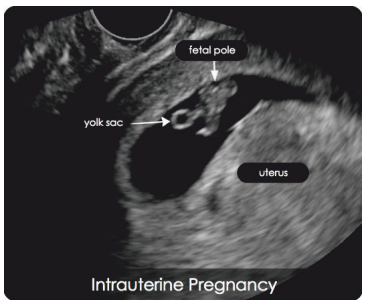


- To know before overdue/best test → ..... → diagnosis
  May be made ..... Before overdue

Fluoroimmuno assay (0.001miu/ml)>Radioimmunoassay (0.002miu/ml)

Active space

Ultrasound



Seen Earliest By TVS

1st Gestational sac 4.5 - 5wks (4wk & 3days)

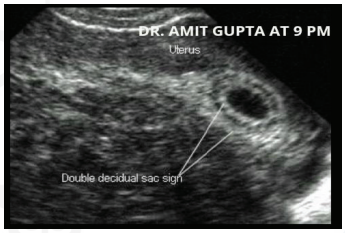
2nd yolk sac 5-5.5 wks

3rd embryo/cardiac activity 5.5-6 wks

1st sign - ..... sign

2nd sign - ..... sign

3rd sign - ..... sign



Double Decidual Sac Sign

Absolute /positive sign - Confirmatory signs of pregnancy

- Auscultation of fetal heart sound
- Palpation of fetal parts
- USG- Fetus → fetal cardiac activity
- X-ray -Fetal skeleton (>16 week)

- Best clinical signs -.....
- Best sign -.....

Active space

Pseudocyesis- pseudo (false) pregnancy

She thinks that she is pregnant but actually not pregnant

- Absolute sign are ..... in pseudociesis

Maxium permissible radiation in pregnancy is upto .....

if > .....

if x-ray is required in pregnancy .....

because if 1 or 2 x-rays done, radiation is .....

CT scan if required .....

Active space

LEOPOLD MANEUVRE

Leopold Maneuver or Obstetric Grips

Leopold maneuvres (pelvic grips) – 4

Ist – ..... grip – lie, presentation

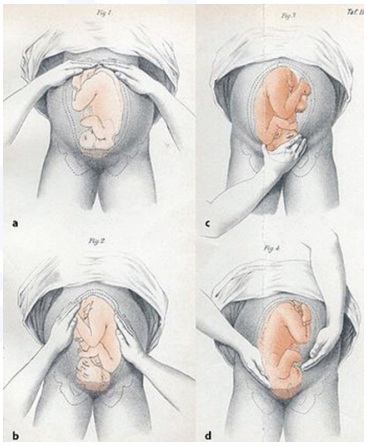
Head .....

Buttocks .....

2nd – ..... grip. either side of maternal abdomen – position and back of fetus

Spine .....

Limbs .....



3rd –.....

..... – head engaged or not

If head moves .....

If head does not move .....

4th – ..... – head engaged or not (.....comfortable for patient)

If able to converge fingers below fetal head .....

Leopold 1 , 2 , 3 grips are done facing ..... of female. leopold 4 is done facing ..... of female.

Active space

CHANGES IN PREGNANCY

Changes in Pregnancy

Face - Estrogen stimulates melanocyte

- Blackish-brown Pigmentation on cheeks, nose, fore head d/t melanin
- Cloasma gravidarum or melasma or Pregnancy mask

Breast

- Increase in size (mammogenesis)

  Ductal growth is by .....

  Alveolar growth is by .....
- Secondary areola - Another dark area around primary areola due

Active space

to .....

2nd trimester onwards usvally after .....

- Montgomery tubercles -Hypertrophoid sebaceous glands at areolar margin
- Colostrum- secretion starts > ..... week

(Lactogenesis – synthesis of milk by .....)

(Galactokinesis – ejection of milk by .....)

Cramping pain during lactation are due to .....

Abdominal Pain .....

Levels of prolacting are maximum in .....

Because of Progesterone - it..... receptors of prolaction on breast tissue.

1st Stimulus to initiate lactation after birth - withdrawal of .....

Abdomen

a. Linea nigra – .....-black line between xiphisternum and pubic symphysis due to .....
b. Striae gravidum –
   - Curvilinear
   - More in lower abdomen/ upper thigh
   - d/t Dilatation of abdominal wall .....
c. After delivery of baby- striae albicans (d/t obliteration of capillaries)

.....erythema

Reddism palm due to .....

.....nevi

Reddish areas under skin due to .....

Vagina

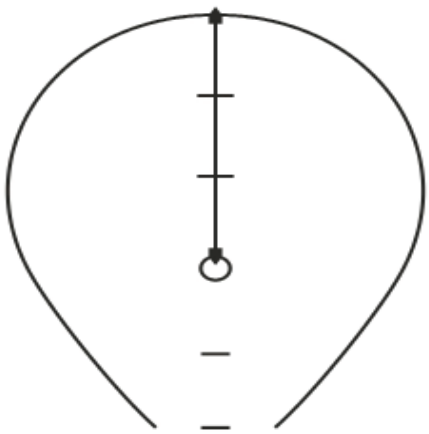
- Color -.....
- Increase in no. of doderlein bacilli or lactobacillus acidophilus - converts glycogen to lactic acid - pH of vagina ..... prevent entry of other .....
- pH - most acidic 3.5

No. is governed by.....

Weight gain

- Wide range 10-12kg (mean 11 kg) 11 > 12
- 1 trimester 1kg
- 2 trimester 5-6 kg
- 3 trimester 5-6 kg

- Net weight gain 5-6 kg



Total water retention approx 6.5 litre

BMR increase 10-20%

Increased Risk of thrombosis

Risk of Diabetes d/t HPL

Edema- Physiological due to A. Water retention,

B. Compression of iliac veins C. Decreased albumin.

If disappears after 12 hours of

rest- .....

Fasting : Hypoglycemia

Post prandial : Hyperghycenia

Best clinical assessment of gestational age-

by .....

Active space

Best assessment by .....

Best time for assessment by usg

.....

Best USG Parameter

A lady is on OCP - Best method to assess gestational age is ..... by USG

Fundal height

- At ..... weeks - Pubic symphysis UT. Just palpable become abdominal organ
- At ..... weeks - Midway between pubic symphysis & umbilicus
- At ..... weeks - Lower border of umbilicus
- At ..... weeks - Upper border of umbilicus
- At ..... weeks - Junction of..... from umbilicus & ..... from xiphisternum
- At ..... weeks - ..... from umbilicus & ..... from xiphisternum
- At ..... weeks- Xiphisternum
- At ..... weeks (head ..... palpable)
- At ..... weeks. Level

Umbilicus

.....

Fundal height > Period of amenorrhea

1. Wrong dates
2. Full bladder
3. Multifetal Pregnancy
4. Molar Paregnancy
5. Polyhydroamnios
6. Placental Abruption- Concealed
7. Macrosomia
8. Fibroid uterus (Pelvic Tumor)

Active space

Fundal height < Period of amenorrhea

1. Wrong dates
2. Missed Abortion
3. Oligohydroamnios
4. IUGR
5. Transverse lie
6. Intrauterine fetal Demise

Conditions in which hCG ↑

1. Molar pregnancy
2. Choriocarcinoma
3. Twins
4. Down syndrome
5. Rh negative pregnancy

Conditions in which hCG ↓

1. Ectopic pregnancy
2. Abortion
3. Any trisomy other than down syndrome

Changes in Hematological system

- Plasma volume increase→starts increasing at 5th week of IUL→Max. between 28-32 weeks by 40%
- RBC increased by 20 %
- Blood volume 30% ..... – physiological anemia

  Total Hb .....

Increased blood volume at term is due to increased fluid retention, action of aldosterone & estrogen.

(Hb% - Not less than .....)

- WBC↑
- Platelets-unaffected but due to hemodilution, concentration decresed → benign gestational thrombocytopenia)
- TLC -15000
- Post partum 25000

Active space

- s. Iron ↓
- S. Ferritin ↓
- Reticulocyte ↓
- Total iron binding capacity ↑
- Transferrin ↑
- All clotting factors ↑-Except XI and XIII
- A:G 1:1 pregnant
- A:G 1.7:1 Non pregnant
- Albumin ↓
- Globulin ↑
- Serum urea, Serum uric acid, Serum creatinine ↓
- More Na+ retention but Na+ conc. ↓
- ESR, CRP ↑
- Tissue demand of O_2 ↑ by = 20% in pregnancy
- Humoral immunity .....
  T Helper cell 2 .....
  interleukin 4 .....
  Cell mediated immunity .....
  T Helper Cell 1 .....
  interleukin 2 .....

- BT-CT – normal
  B Cell
  Anti thrombin time
  CD4 : CD8 Ratio

Active space

CVS

- Heart rotates upward & outward
- Apex beat → In 4th ICS, 2.5 cm left lateral to mid clavicular line

- ECG – left axis deviation echocardiography
- Pulse rate/H.R ..... BPM ↑ HR
- Increase in cardiac output- ↑ →starts increasing at 5th week→ max. between ..... wk

  ↑ By ..... at labour

  ↑ ..... immediately after delivery → max risk of .....

BP ↓ =

- Systolic, - Diastolic ↓, - Mean , Maximum- ↓ DBP

  BP → Reaches minimum value at ..... wks. After this BP begins to increase & in ..... trimester it comes to pre-pregnant value.

- S1 –loud with wide splitting
- S2- Normal
- S3 easily heard
- Ejection systolic murmur – < .....
- Echocardiography- Mild increase in cardiac silhouette

Respiratory system

- Progesterone ..... Sensitivity of Respiratory centre to CO2

  PCO_2 ..... (..... ed wash out)

  PO_2 ..... Compensated Respiratory .....
- Total lung capacity ↓
- Residual volume ↓
- Vital capacity, - Respiratory rate, - Inspiratory reserve volume unaffected
- Inspiratory capicity ↑, Tidal volume ↑, Minute volume ↑
- Respiratory alkalosis .....

Active space

- Upliftment of Diaphragm →.....
- Transverse diameter of chest -.....
- Chest circumference-.....
- Subcostal angle ..... in pregnant
- Non Pregnancy-.....

Renal system

- Renal Blood flow increase by.....
- ↑ GFR .....
- Hydronephrosis and hydroureter .....
- Renal size .....
- Bladder Pressure .....
- Stress urinary incontinence (SUI) .....
- Frequency of micturition .....
- Glycosuria due to decrease in renal threshold from 195 to 155. HCO3 excretion increased

Lipid metabolism

- LDL .....
- HDL .....
- Cholesterol .....
- Triglycerides .....
- Lipoprotein .....
- Apo lipoprotein .....

GI system

- ↑ GI Reflux due to relaxation of .....
- Heart burn (Pyrosis)
- Constipation-
- Risk of Gall bladder stones (Cholesterol stones)
- Pruritus due to ..... biliary stasis (..... gravidarum)

Active space

- Mx -DOC -.....Ursodeoxycholic Acid

Endocrine system

- GH .....
- ACTH .....
- Prolactin .....
- FSH and LH .....
- TSH = ...
- T3,T4 increases, thyroxine binding globulin .....
- FT3, FT4- .....

Adrenals

- Aldosterone .....
- DHEA,DHEA-S .....

MC Cause of hypothyroidism in pregnancy- .....

MC cause of acute viral hepatitis : Hepatitis .....
Highest mortality in pregnancy : Hepatitis .....

DOC for Hyperthyroidism in pregnancy .....

MC Anomaly seen in fetus whose mother is on lithium during pregnancy .....
High ..... inhibits milk production in pregnancy

A 26 Year old primigravida with 12 weeks pregnancy present with history of lower quadrant pain and fever. Her usg shows appendix 10 mm thick and WBC count 18000. Manangement is .....

Appendicitis - USG - Appendix Thickness .....

Appendicitis - USG - Appendix Thickness .....

Active space

Other changes in pregnancy -

1. Stress Urinary Incontinence :
   Mx ..... Exercises (Pelic floor muscle exercise)
   Can be started anytime in pregnancy safe to do.

2. Varicose vein : Due to ..... venous pressure ..... blood volume.
Mx - 1) Avoid long standing
     2) Exercises
     3) Leg elevation
     4) .....

3. Vulval Varices :
Mx - ..... support, avoid long standing.

4. Leg cramps
Mx - 1) Improve hydration status
     2) At the time of cramp : a) Hot massage
                               b) Knee ..... & ankle .....

5. Bell's Palsy - (Due to fluid relention)
Cause : Perineural edema/Hypercoagability Immunosuppression.
Mx - Short term therapy with .....

6. Dizziness - due to fall in .....
   Advise -
     1) Avoid Rapid postural changes
     2) avoid ..... position
     3) stay hydrated.

Active space

7. Bleeding Gums
   Reddish purple papule with granulation tissue
   (Pyogenic granuloma or granuloma gravidarum)
Mx - Conservative, Don't Excise

High Risk Preganancy -

1. Systemic illness
2. Severe Anemia
3. Pre-eclampsia or GDM.
4. Extremes of Age (< 20 yr or > 35 yr)
5. HIV / Hepatitis .....
6. ..... Pregnancy
7. Previous .....Section
8. Previous bad obstetric history
   (Abortion/Still birth / Pre-term)
9. Placenta .....
10. Other Pregnancy Complications.

Active space

Ectopic Pregnancy

- Implantation of fertilized ovum outside uterus

- MC Risk Factor - .....
- 2nd MC .....

MC cause

- PID
- Others
  - Early age of intercourse
  - Multiple partners
  - Low socioeconomic class
  - H/o instrumentation
  - Endometriosis
  - Contraceptive failure

    Caesarean Section

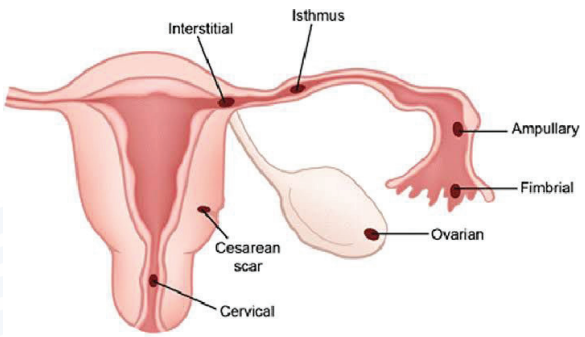
    Tubal ligation failure > IUCD failure (progesterone > mirena >cu-T) > Progesterone only pill failure
  - Cornual Fibroid
  - Smoking

Active space

- MC site - ..... of fallopian tube because it is MC site of fertilization

MC Nontubal site .....

1. Interstitium - Narrowest
2. Isthmus - 2nd Narrowest
3. Ampulla
   - Widest and longest
   - MC site of fertilization
   - MC ectopic pregnancy
   - Earliest / MC genital TB
4. Infundibulum



MC site - Ampulla >Isthmus >Infundibulum>Interstitium

Earliest site of rupture

- ..... at 6 weeks
- Last site- .....

Most dangerous tubal ectopic .....

Overall longest survival is in .....

Least common type of ectopic pregnancy:-
C-Section scar ectopic > cervical ectopic > Abdominal ectopic

Symptoms

- MC .....
- Others – amenorrhea, mild spotting

Sign

- P/V exam- Empty uterus and Adnexal mass
- USG- to confirm it. If Push/pull probe→

feels pain (..... movement tenderness)

Active space

Discriminatory titre or critical titre of hcg - Gestational sac should be visible by USG at this hcg titre.

>= ..... IU/L {TVS}

>= ..... IU/L {TAS}

If >=..... IU/L {TVS} - .....

If < .....

repeat after 48 hours because-

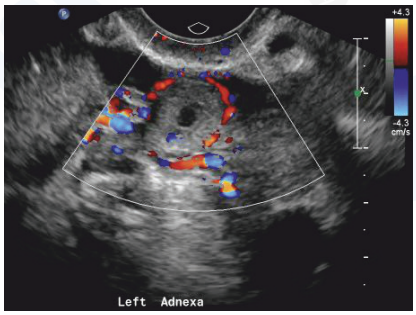
1. If β-HCG is doubled – .....

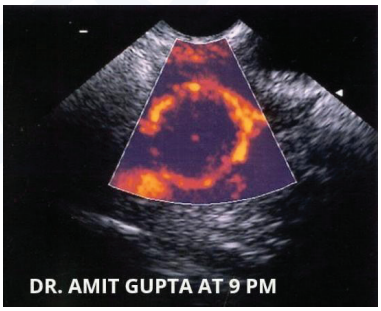
2. If β -HCG ↓ - .....

3. If β-HCG ↑ but doesn't double – .....

Other USG signs

- Bagel sign- Visualisation of gestational sac or yolk sac or embryo with or without .....in adnexae .





Active space

- Color Doppler - .....
- IOC – .....
- Gold standard –.....
- Blood doesn't clot on keeping (If ruptured ectopic)

S. Progesterone - Suggestive of live intrauterine .....
Suggestive of Ectopic/abortion - .....

Management

1. Expectant mx

- waiting for spontaneous resolution
- if β-hCG< ..... IU
- Best if < .....
- if Ectopic mass size .....

Follow up with S. Beta hCG till ..... consecutive reports are .....

2. Medical mx

Indications

- Serum beta hCG.....
- Size of mass .....
- Cardiac activity preferably is .....
- DOC -.....
- If jaundice – .....

Protocal

Day 1 - S.Beta hCG, LFT, KFT, CBC,

Day 2 - give methotrexate

Day 4 - S. Beta hCG

Day 7 - beta hCG - If fall in S. Beta hCG on day D7 is >15% → job is done → monitor → -hCG ..... consecutive reports

Active space

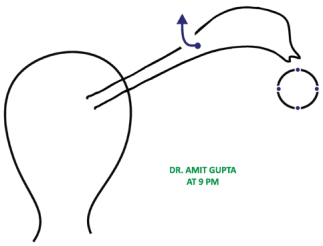
are negative But If fall in β - hCG is ..... →Repeat methotrexate

→Max ..... doses of Methotrexate can be given

If no response even after 3 doses →

3. Surgical mx

Indications



- Serum beta hCG.....
- Size of mass .....
- Cardiac activity is .....
- no response to .....
- Preferred – .....

  but can only be done if she is hemodynamically stable

  A. If primi - linear ..... {don't suture incision site}

  B. If multi - unilateral .....

- If she is haemodynamically unstable or BP 80/60 or Shock or ruptured ectopic pregnancy - Ruptured ectopic pregnancy

Mx - ..... - Unilateral .....

- In ruptured ectopic → pain in shoulder → .....

  (referred pain due to irritation of diaphragm)

Urge for ..... due to pooling of blood in pouch of douglas

Indications of Salpingectomy -
Ruptured Ectopic
IVF
Complete family
Sac Size 5 cm
To maintain hemostasis

Active space

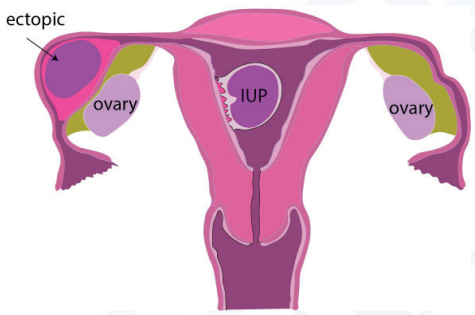
- Heterotopic pregnancy-Intrauterine +Extrauterine pregnancy

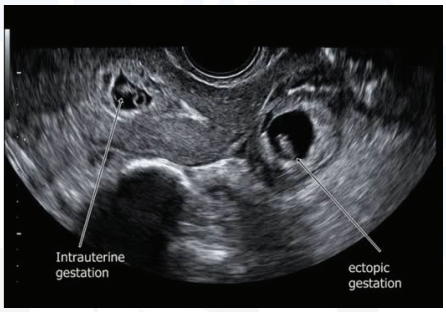
Incidence-1 in 3000

Risk → due to use of ovulation inducing drugs

If no response even after 3 doses -

Management - May continue with .....





- Studiford criteria -For primary..... pregnancy
- Spigelberg criteria- Primary .....pregnancy
- Rubin/Palman - For Primary.....pregnancy

Active space

MOLAR PREGNANCY

Molar Pregnancy

Cause - due to abnormal fertilization

Definition - Abnormal proliferation and degeneration of chorionic tissue (syncytiotrophoblast and cytotrophoblast)

Risk -It is benign but carries malignant potential

Types of Molar Pregnancy (Gestational Trophoblastic Disease)

1. Complete mole or vesicular mole .....

2. Partial Mole .....

Risk factors:-

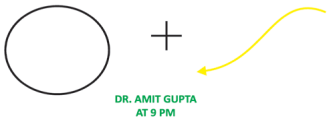
Most imp,.....

- Age>(35-40yrs) 40 > 35
- Deficiency of beta carotene /Vitamin A/Folic acid
- OCP
- Female- A and male - O Blood Group
- AB Blood Group
- Smoking
- Developing countries
- Maximum -Philippines

Active space

Complete mole:

If empty ova is fertilized by sperm .



Empty Ova + sperm = zygote. .....

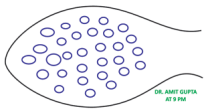
Androgenesis duplication

.....

↓

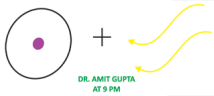
Complete abnormal proliferation and degeneration of chorionic tissue

→ Uterus is filled with grape like vesicles



Partial mole :-

If a normal ova is fertilized by 2 > 3 sperms



Partial abnormal proliferation & degeneration of chorionic tissue

Fetus develops but usvally dies around .....

Abnormal hydropic .....

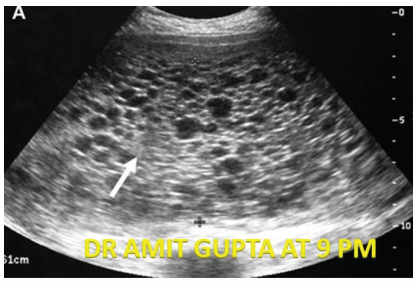
Symptoms:-

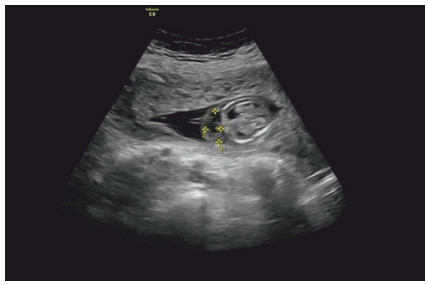
- MC-.....

  With specific to complete mole .....

- Amenorrhea
- Hypermesis gravidarum
- Sometimes thyrotoxicosis

Active space





IOC → USG

- For complete mole → .....
- For partial mole -.....
- Gold standard :.....

	Complete mole.	Partial mole
- Theca Leutein cyst - Bilateral &	- Beta hCG > 10^5 IU/L	- < 10^5 IU/L
	- Size of theca lutein cyst-	- .....
Other causes of theca leutein cyst :-	- Corpus Luteum Cyst	- .....
1) Multifetal pregnancy	- Thyrotoxicosis -	- .....
2) Hydrops fetalis	- Hyperemesis -	- .....
3) Ovulation induction	- PIH -	- .....
4) Renal failure	- Malignant- potentials -	- .....
5) Long Standing Severe hypothyroidism	- Trophoblastic hyperplasia-	- .....
	- Hydropic degeneration	- .....
	- Villi are avascular	- Villi have vascularity
		- Scalloping
		- Stromal inclusion bodies
		- Immuno staining with .....

Mx

- If multiparous - .....
- If nulliparous- ..... f/b check curettage with

MIST

Active space

sharp curette

OCP for 6 Months

Monitor serum beta HCG

1st test is done after .....

Monitor serum beta HCG weekly till 3 consecutive report are negative then monitor beta HCG monthly for 6 months

Surveillance in molar pregancy is done .....

Repeated assessment of hcg levels are done in .....

What we can do to prevent pulmonary embolism - begin oxytocin after initiating .....

When is the risk of thyroid storm highest- during .....
when is the risk of pulmonary embolism highest- during .....
.....

Most important advise during survillence:- .....

hCG levels usvally become normal in complete mole .....

hCG levels usvally become normal in partial mole .....

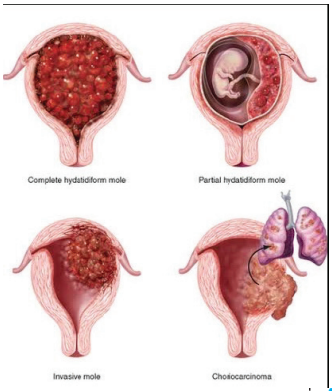
Management for theca lutein cysts .....

High risk patients receive prophylactic chemotheraphy after S & E.
DOC:- Methotrexate or Actinomycin -D

Complication

H. Mole→ MC-..... > Most.....
Dangerous.....> Placental site

trophoblastic tumor > Epithelial trophoblastic tumor



MIST

Gestational Trophoblastic neoplasia (GTN)

1. Invasive Mole
2. Choriocarcinoma
3. Placental site trophoblastic tumor
4. Epitheloid trophoblastic tumor

MC GTN after full term pregnancy:- .....

Choriocarcinoma MC happens after which type of pregnancy :- .....

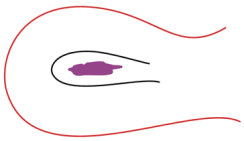
.....

Chorionic Villi are seen in .....

1. Complete Mole
2. Partial Mole
3. Invasive Mole

Invasive mole or chorioadenoma .....

- Invasion in myometrium (locally invasive)
- Histopathological examination- Chorionic villi seen-Invasion in myometrium
- Mx hysterectomy



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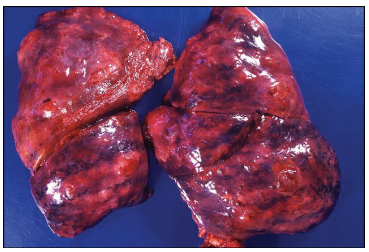
Choriocarcinoma - gross examination .....

- Rare but most dangerous
- Suspicion



1. Persistent Irregular bleeding P/V despite of complete evacuation even after

.....

Active space

2. 4 consecutive plateu levels of S. Beta HCG or rising again
3. 3 consecutive rising hCG levels
4. Histopathology Choriocarcinoma
5. Persistent theca leutein cyst, Normally after S & E, Theca leutein cyst disappear in 2-4 months.

- IOC - beta HCG bcz it is highly malignant & metastasize via hematogenous route

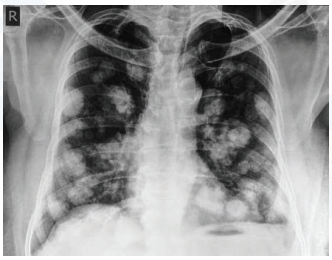
FIGO Staging:-

- Stage I→ uterus
- Stage II →f.t / ovaries/ vagina
- Stage III →..... (most common site of metastasis)
- Stage IV → brain / liver

MC site of metastasis .....

Lungs -X-ray chest

- MC appearance of metastasis of lung - ..... appearance > Snow storm appearance



- Low risk Choriocarcinoma

1. Age .....
2. Duration of previous pregnancy to diagnosis of GTN .....
3. Develops after .....
4. S beta hCG .....

- High risk Choriocarcinoma

1. Age .....
2. Duration of previous pregnancy to diagnosis of GTN .....
3. Develops after .....
4. S beta hCG .....

Active space

5. No. of Metastasis .....
6. Metastasis .....
7. Size .....
8. After which chemotherapy recurrence occurred .....
Score .....

5. No. of Metastasis .....
6. Metastasis .....
7. Size .....
8. After which chemotherapy recurrence occurred .....
Score .....

Mx

- For low risk -.....
  On Day
  Alternating with injection

- For high risk - Multi - .....

Bagshaw Regime (EMACO Therapy)

- E- Etopocide
- M- Methotrexate
- A- Actinomycin -D
- C - Cyclophosphamide
- O - Oncovin

Follow up

.....

Indication of radiotherapy with chemotherapy .....

Indication of total abdominal hysterectomy

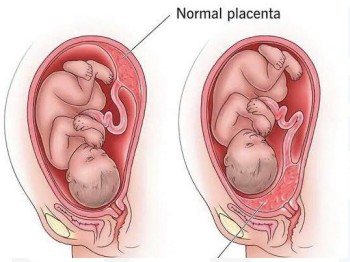
1. Multiparous with complete mole
2. Invasive mole
3 Excessive, uncontrollable bleeding with suction evacuation

Active space

ANTEPARTUM HAEMORRHAGE

Antepartum Haemorrhage

- Bleeding from/into genital tract after age of viability .....
  Delivery of baby

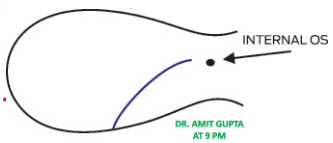


- MC Cause - 1. Abruptio placentae
- Other causes

2. Placenta praevia
3. Circumvallate, circum marginate placenta - Abruption
4. Vasa praevia
5. Local tear
6. Preterm Labor
7. Uterine Rupture

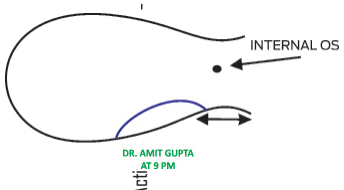
New Classification

Placenta praevia - if lower edge of placenta is .....



Low lying Placenta

- If lower edge of pacenta is within ..... from internal OS



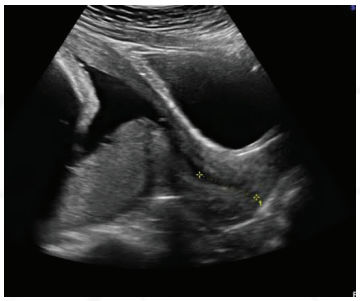
- MC Risk factor
- -H/O previous placenta praevia
- 2nd MC - H/O previous lower segment C.S

-Other risk factors -

- Twins
- Smoking
- H/O Curettage
- Age >=35 year
- Multiparity

Placenta previa has negative association with preeclampsia

- Incidence - .....
- Recurrence rate - .....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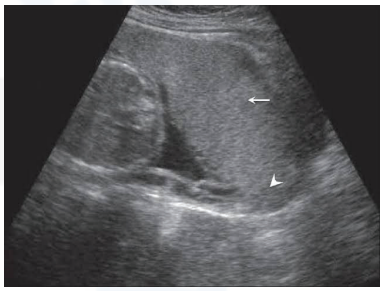
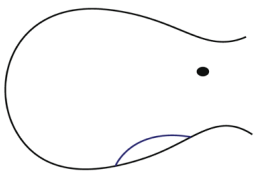


Trophotropism - Normally placenta migrates up with increasing gestational age due to .....

Lower segment of uterus .....

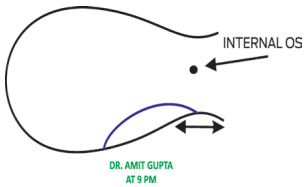
Many low lying placenta become normal with ..... pregnancy.

Diagnosis of placenta praevia by USG is confirmed at .....



Old Classification by - Browne

Type 1 (Lateral) - If lower edge of placenta is within .....from internal os but away from os.



Type 2 (Marginal) - If lower edge is reaching upto ....of internal os

Type 3 (incomplete or partially central) lower edge covers internal os when os is .....but unable to cover os when os is .....

Type 4 (Complete central) Lower edge covers internal os even when os is .....

Minor degree

Type 1

Type 2 Anterior Mx → Vaginal Delivery

Major degree

Type 2 Posterior

Type 3 Mx → C. S.

Type 4

Stalworthy sign - seen in

- ..... onwards
- If fetal head is pressed in pelvis - FHR → Fetal Heart Rate
- When pressure is released - FHR .....

Dangerous placenta praevia →.....

Symptoms – Bleeding p/v

1 Recurrent
2 Massive
3 Painless
4 Causeless
5 Bright red
6 Usually preceded by small warning haemorrhage

Active space

Cause of bleeding
uterine contractions or cervical changes
apply shearing forces on placenta - bleeding
(maternal blood coming from intervillous space)

Sign

P/A Exam - abdomen/uterus .....

fetal parts .....

fetal heart sound .....

sometimes .....

MC malpresentation in placenta praevia .....

Screening test for APH (TAS) Trans - Abdominal ultrasound

IOC for placenta praevia (TVS) Trans - Vaginal ultrasound

- Ultrasound 100% diagnostic (TVS)

P/s & P/V Examination shoud not done until placenta previa is ruled out

- P/V exam is C/I - if mandatory done in Double set - up operation theatre

Mx

I. If she is haemodynamically unstable / BP 80/60 / Excessing uncontrolled continuous bleeding p/v /fetal distress / fetal death / gross congenital malformation .....
.....

Active space

II. If she is haemodynamically stable .....

a. if gestational age is > .....

b. if gestational age is < ..... and no excessing bleeding p/v / no fetal distress / no fetal death / no gross congenital

malformation .....

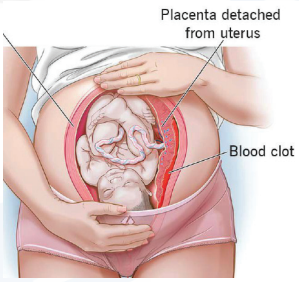
By Expectant Mx - Macafee and Johnson regime

- Bed rest
- Blood grouping/ matching
- Steroids
- Tocolytics if required
- Anti-D- If Rh negative
- Mg So4 ..... (To Prevent Cerebral Palsy) → For Neuroprotection
- Fetal monitoring by Non - stress test (NST), Bio physical profile score (BPS), Ultrasound (USG) .....
- Maternal monitoring for Vitals, Bleeding, all Baseline investigations and arrange blood.

Management of low lying placenta -
a. if within 1 cm.....
b. if between 1-2 cm. .....

Abruptio placenta -
- Premature separation of placenta

Normally placenta separates from uterine wall after .....
incidence .....
recurrence rate .....



MC Risk factor
- H/O previous abruption

Others
- Hypertension

Trauma
- Polyhydraminos - Prelabor rupture of membranes
- Retroplacental Fibroid
- Multifetal pregnancy

Active space

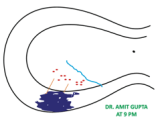
- ↓Folic acid
- Smoking
- Thrombophilia
- Short umbilical cord
- Age >=35 years
- Multiparty

Types - 3

1. Concealed
2. Revealed
3. MC - Mixed

1. Concealed

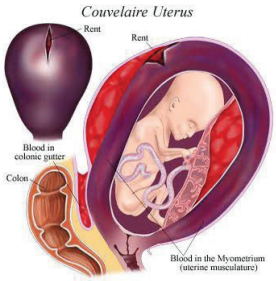
- Lower edge of placenta is intact



Symptoms-

- Abdominal pain

Sometimes- Retroplacental blood may reach up to serosa→ .....uterus→..... .....or uteroplacental .....



Risk

2. Revealed

- Lower edge of placenta is separated

Active space

Symptoms-

1. Abdominal pain
2. Bleeding p/v
   a. Continuous
   b. Dull red/Dark red
   c. Painful
   d. With cause

Cause of bleeding :
Rupture of blood vessel in D. Basalis
Decidua releases : tissue thromboplastin, a potent uterotonic which initiates DIC - matrix metalloprotinase - tissue necrosis

Thromboplastin - Abdominal .....
- Uterine .....

Signs-

P/A Exam - abdomen/uterus .....

sometimes fetal parts .....

sometimes fetal heart sound .....

in concealed variety - fundal height .....

most dangerous abruption .....

3. Mixed → Concealed + Revealed

Investigation

- Clinical corelation > USG

USG signs

Active space

What finding on TAS makes diagnosis of abruption
Retroplacental clot (RPC) (clot b/w placenta & decidua)
.....Thickened placenta that shimmers with maternal movement

Abruption - women can undergo precipitate labour due to tissue thromboplastin

MX -

I. If she is haemodynamically unstable or BP 80/60 / fetal distress / fetal death / excessive bleeding p/v .....

II. If she is haemodynamically stable .....

a. if gestational age is > .....

b. if gestational age is < .....

Goal - to continue till ..... wks.

1. Steroids
2. Tocolytics if required
3. ..... if gestational Age < 32 wks
4. Fetal monitoring by Non - stress test (NST), Bio physical profile score (BPS), Ultrasound (USG) .....

Active space

5. Maternal monitoring due to risk of .....

If DIC Developed → .....

Classification- PAGE classification

Obstetric causes of DIC -

MC .....

Others Amniotic Fluid Embolism .

Severe Sepsis

Retained Dead fetus

preeclampsia

Hellp syndrome

eclampsia

Acute fatty liver of pregnancy

Active space

ABORTION AND MTP

Abortion - Miscarriage - Pregnancy loss

- Death of fetus before age of viability < .....
- Death >= ..... gm intrauterine fetal demise (IUFD) or still birth

WHO definition

Abortion - death of fetus before fetal weight is <500gms

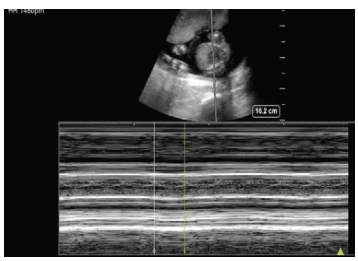
intrauterine fetal demise (IUFD) or still birth - death of fetus once fetal weight is .....

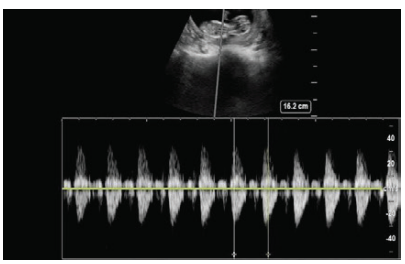
Maximum abortions accour before .....

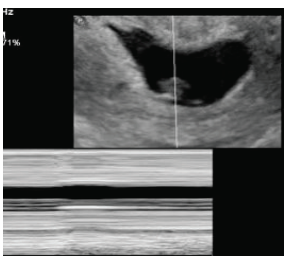
Early pregnancy loss - death of fetus before .....

live fetus shows cardiac activity

dead fetus ..... cardiac activity







Active space

Most important cause of bleeding in early pregnancy - .....

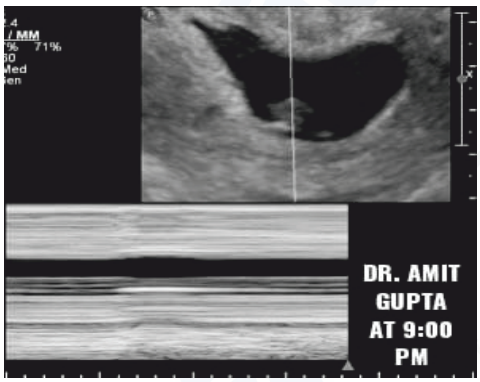
MC cause of abortion .....

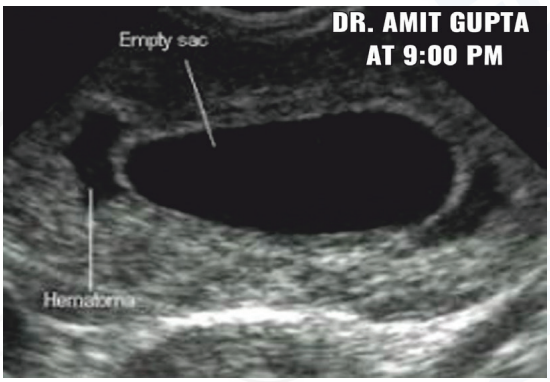
MC time of abortion .....

Classification

USG based

1. Blighted ovum or Anembryonic Pregnancy
   - Only gestational sac, no embryo
   - Mean sac diameter is >..... & no embryo seen





2. Missed abortion
   - Embryo without cardiac activity if CRL .....

2nd classification – episode based

Isolated abortion and recurrent abortion

1. Isolated abortion → More Common in

- Mc cause- .....
- MC - ..... – MC - .....MC trisomy .....
  .....

  Monosomy x (20%) > Trisomy 16 (16%)

Active space

- Diagnosis - karyotyping

MC cause of isolated abortion

A. Chromosomal, B. Aneuploidy, C. Trisomy, D. Trisomy 16,

E. Monosomy X

Single mcc of abortions - ..... (20% of all abortionas)

Most viable trisomy - Trisomy .....

M/C sex chromosomal abnormality - ..... syndrome

2nd MC - ..... causes

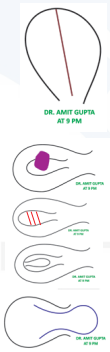
a. ..... Uterus

b. ..... Fibroid

c. ..... syndrome

d. .....

e. Cervical .....



3rd Trauma - Direct impact on uterus

4th Infection → TORCH

→ Chorio Amnionitis

5th Thrombophilia

Kassowitz Law -

Kassowitz rule is related to congenital syphilis & chances of infection of fetus in later pregnancies become less

2. Recurrent abortion or Habitual abortion or recurrent .....

- If H/O abortion

More in ..... Single MC cause →

MC Cause .....

MC group of cause .....

Active space

a. uncontrolled .....
b. thyroid.....
c. hyperprolactinemia
d. luteal phase defect
e. PCOS

2nd MC group of cause .....
MC-a. Cervical incompetence
b. ..... Uterus
c. ..... Fibroid
d. ..... syndrome
e. .....

3rd .....

Anti phospholipid antibody syndrome (APLA)→Causes thrombosis-IUGR, Abortion, Early onset preeclampsia, early onset preterm labor

- Lupus anticoagulant
- Anticardiolipin
- B_2 Microglobulin Antibody

If any 1/3

- Dx by- Modified sapporo criteria

Present

- Mx –low dose aspirin

Low molecular weight heparin .....

4. Chromosomal - MC .....

Toxins - Lead, Arsenic

Cervical incompetence

Active space

Causes

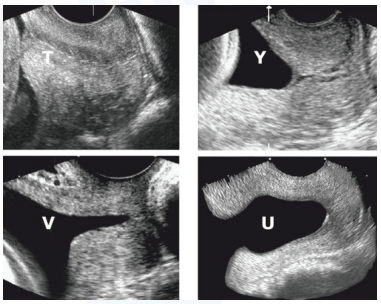
- Congenitally short cervix
- Conization
- Amputation
- Cryocauterisation
- LEEP
- Cervical Trauma During Labor

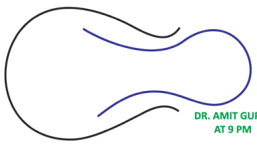
Max. Abortions occur in 2nd trimester .....
painless dilation f/b expulsion of fetus

Diagnosis of incompetent cervix in nonpregnant female .....

Incompetence

a. Hysterocervicography- .....of dye
b. Hegar dilator no. 8 - can be .....
c. Foley's Catheter no. 16 - can be .....





Diagnosis- In pregnancy →USG→

- Cervix length < .....
- Internal OS > .....
- U shaped

Mx –Cervical cerclage - MC - ..... at .....
junction in .....
applied at ..... but can be applied upto .....
removed at .....

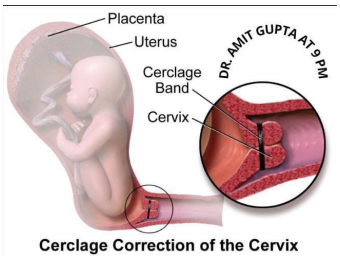
Along with cerclage, Progesterone is given

Contraindications of cervical cerclage .....

.....

.....

.....



Cerclage Correction of the Cervix

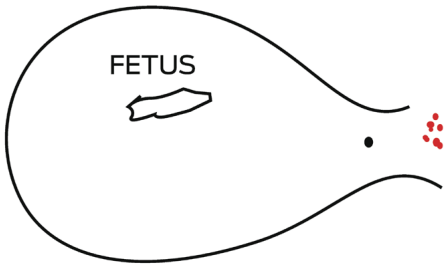
MC Uterine cause of recurrent abortion / (in second trimester) .....
.....

Active space

MC Uterine cause of recurrent abortion in first trimester .....
.....

3rd classification – clinical presentation based

1. Threatened abortion – reversible
   - C/o → .....
   - P/A → Fundal Height
   - P/V - internal OS –.....
   - Mx .....
   - USG –live fetus - not a part of protocol



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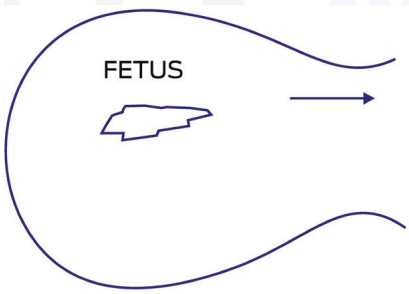
2. Inevitable (Unavoidable) abortion-irreversible int. os is ..... & fetus is in

   process of expulsion
   - Symptoms
   - Bleeding
   - P/A exam → Fundal Height
   - P/v exam-internal OS —.....
   - Fetus is in process of expulsion
   - MX - .....
   - If G.A.> 12 week - add .....



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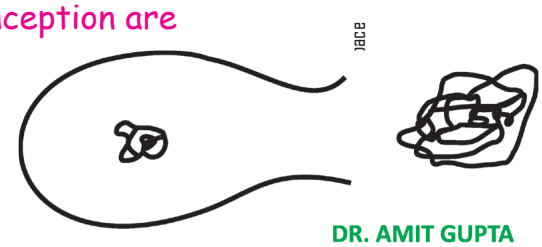
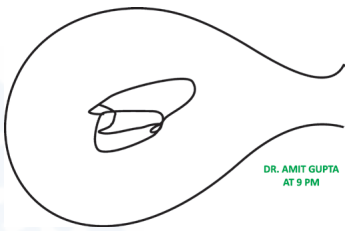
3. Incomplete abortion
   - Abortion occurred but some products of conception are
     .....
   - Symptoms
   - Bleeding P/v
   - H/O.....



DR. AMIT GUPTA
AT 9 PM

- P/A →Fundal Height
- P/v -internal OS —.....
- Mx – .....

4. Missed abortion – Abortion occurred but remained unnoticed
   - Mostly Asymptomatic sometimes- dirty brown discharge p/v
   - P/A → Fundal Height
   - P/V internal OS - .....
   - Mx - .....



5. Complete abortion

   Abortion occurred, all products of conception expelled out & uterine cavity is .....

Mx

Toxins : - Ionizing Radiation Lead/Arsenic.

Medical termination of pregnancy (MTP)

Indication

1. Contraceptive failure - upto ..... weeks pregnancy only

Upto ..... weeks pregnancy

2. Congenital malformation incompatible with continuation of pregnancy
3. Maternal disease incompatible with continuation of pregnancy
   A. Heart disease clark's class III
   B. Malignant hypertension
   C. Severe chronic renal ds
   D. Schizophrenia

Active space

4. H/o rubella infection
5. Histroy of taking teratogenic drugs
6. Rape, incest, minor, mentally unstable
7. H/o radiation exposure

OLD ACT

- It was passed in august 1971 . -Effective in april 1972
- MTP can be done legally up to 20 week
- Upto 12 wk opinion of one doctor is enough
- After 12 week-opinion of two doctors is required but can be done by one doctor

- Registration (License)
  1 Both place and person are registered
  2 It is given by chief medical officer of district (CMO)

- Consent
  1 If age< 18 years: guardian
  2 If age≥ 18 years: herself
  3 If mentally unstble - consent of .....
  4 Consent of husband is not required

- Submission of records of MTP of Every month to .....
- Keep MTP Records of last .....

Eligibity

- Can be done by Gynaecologist – M.S./D.G.O. or Residency in OBG
  For 6 months or Registered medical practitioner M.B.,B.S. with experience certificate of assisting ..... cases, out of which ..... cases should be performed independently

Active space

Amendments in MTP - 2021

Was passed in Lok Sabha 17/03/2020, Rajya Sabha 16/03/2021

Effective .....

State Medical Board
- Radiologist
- Gynaecologist
- Paediatrician

.....

- MTP can be done upto ..... weeks
- Upto ..... – opinion of one doctor is enough
- Between ..... weeks- opinion of two doctor required
- For gross congenital malformation eg. Anencephaly- no age limits but on recommendation of .....

Methods

- First trimester
  - MC → medical abortion
  - Best → suction evacuation
  - Other – Menstrual regulation
  - Manual vaccum aspiration

  ] Old, Obsolete

Medical abortion

May be done upto .....

Protocol -

on day 1 - Tablet Mifepristone .....(Anti .....

on day 3 (After 48 hrs.) - Tablet Mistoprostol (PGE1)

- Uterine .....
- Cervical .....

- Contraindicated in .....
- Side effect.....

Active space

- Abortion occurs in .....
- Review After .....

  If incomplete .....

Suction evacuation (7-15 week) Best up to .....

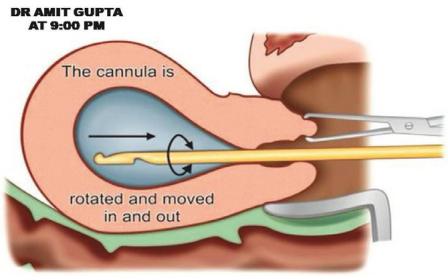
May be done upto .....

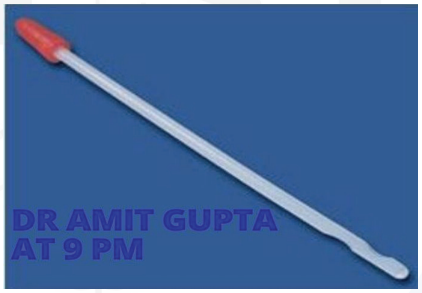
- Number of Karman cannula → diameter of cannula in .....
  gestational age in .....

Suction pressure required – ..... mmHg

Eg. for 10 weeks MTP - no. of canula required .....

diameter of canula .....





Indication of emptying

- Griping of cannula
- Air bubbles in cannula
- Grating

Complication

- MC – .....
- Rare –uterine perforation

Mx

- If perforation occurs while cervical dilation – Nothing

Active space

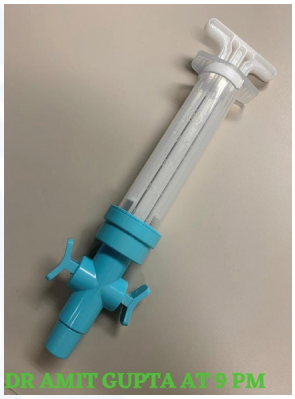
- If perforation occurs during suction –stop procedure, do not take out cannula, do immediate laparotomy

- In rural area, where electricity is not available

Mestrual regulation (oldest)

- By MR syringe.....
- Upto ..... week pregnancy
- ..... Syringe is used

Manual vacuum aspiration



- by MVA Syringe.....
- .....mmHg
- Done upto ..... week pregnancy

2nd trimester MTP

- Latest method →.....

First step - Tab. .....

Sencond step - Dilaation & Evacuation by .....

Followed by check curettage with .....

Sequence of events in dilatation and curettage are

A. Determining the size of uterus by uterine .....

B. Serial .....

C. Curettage

Active space

Others (Old, Obsolete)

- Intra amniotic instillation of hypertonic solution like saline / urea/ mannitol
- Extra amniotic instillation of ethacrydine lactate 0.1%

13 year old girl presents with UPT positive.

next step is .....

Minor girl found pregnant by doctor

- Immediate next step is - .....

.....

Active space

PREGNANCY INDUCED HYPERTENSION

Pregnancy Induced Hypertension

PIH - Gestational hypertension - only hypertension

- Preeclampsia - hypertension + ...

better reliable marker is .....

Gestational hypertension - Increase in BP after ..... pregnancy which become normal within 12 week of delivery on two occasions ..... apart No proteinuria or end organ damage.

Preeclampsia-Increase in BP & Proteinuria and / or organ damage

Preeclampsia is most commonly diagnosed in pregnancy in ..... trimester

A lady presents with blood pressure 150/90 mm HG at 34 weeks gestation. Next step is .....

Best next step

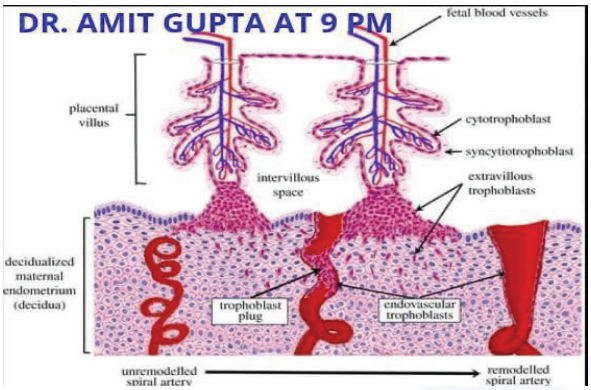
Chronic HTN or overt HTN - Increase in BP unrelated to pregnancy

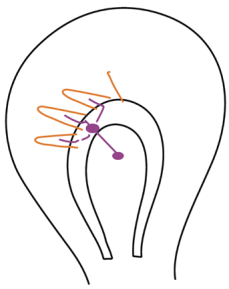
Eg. - At .....weeks

Chronic HT^{N} with superimposed PE :- Any of the following :-

1) New onset proteinuria >20 wk
2) New onset end organ damage >20 wk
3) if she develops uncontrolled HT^{N} >20 wk

Active space





Normally extravillous cytotrophoblast specially .....
..replace lining of spiral vessel → elasticity of vessel ..... →
..... blood flow to fetus

Why vascular remodelling is absent ? Altered immune response
Towards paternal antigens

Cause : ..... trophoblastic invasion → vesseles remain
..... → blood flow ..... → placental
..... → .....mediators → ..... in
capillary permeability & generalised vasoconstriction → Edema and
Haemoconcentration → ..... stasis → platelet .....
..... End organ damage

Placental ischemia (small, pale placenta)
1. ..... Release of anti-angiogenic factors
2. ..... Release of angiogenic factors
3. ..... Inflammatory mediators

Risk factors

1 Primi

2 New paternity

3 Twin

4 Rh incompatibility

5 Metabolic x syndrome

6 APLA syndrome - Early onset preeclampsia

Active space

7. Molar Pregnancy

8. H/o Chronic Hypertension

9. Collagen vascular disease

Smoking is protective for preeclampsia

Preeclampsia - a. Eary onset .....

b. Preterm or late onset .....

c. Term onset .....

Sequence of events -

..... Preeclamsia → ..... Preeclamsia → .....

..... syndrome → .....

Types of Preeclampsia

Mild Preeclampsia (.....severe features)

- SBP>=140 mmHg<160
- DBP >=90mmHg <110
- Proteinuria .....
- On two occasions .....
- Dipstick .....
- Urine protein : creatinine ratio .....

- No. Signs of end organ damage
- No. Signs of impending eclampsia

Severe Preeclampsia (.....severe features)

- SBP>=160 mmHg — Signs of .....
- DBP >=110 mmHg — Signs of .....
- Proteinuria .....
- Dipstick .....
- On two occasions .....because we want ot start anti-hypertensive drug within .....

34 wk pregnant female
BP - 150/100 mmHg. Proteinuria +
Preeclampsia with severe features.

Active space

and / or End organ damage

Severe features

- Headache (not relieved by .....
- ..... symptoms
- Pain in ..... or right hypochondrium
- Convulsions
- Signs of .....

.

- Oliguria
- IUGR
- Proteinuria quatification

→ usvally present but not A criteria for Differentiation between mild & severe preeclampsia

Other features - Oligohydramnios, S. Urea uric acid, creatinine.....

Signs of end organ damage - (any one of the following)

- Platelets .....
- Liver enzymes .....
- Serum creatinine .....
- Pulmonary .....
- Cerebral / visual .....

PIH - ..... Disorder

MC organ / First organ involved .....

MC Histopathological finding .....

First clinical manifestation in PIH .....

First haematological manifestation in PIH .....

Active space

Target BP : Systolic 120-130mm Hg - Diastolic 80-90 mm Hg

Mx

- Mild Preeclampsia - monitoring till .....
- Severe Preeclampsia –DOC ..... + Steroid (.....) + Monitor till. .....
- Definitive Management for preeclampsia .....

HELLP syndrome

- H -hemolysis M/C in 3rd Trimester
- EL- elevated liver enzyme
- LP-low platelets

MC Presentation Epigastric pain nausea/vomiting : - Because of stretching of liver capsule. (subcapsular Hematoma) .....

..... criteria

All 3 Should be Present

- If platelets .....
- If LDH>
- Liver enzymes >.....

Hemolysis :- Any 2 of the following

a. Sr. Billirubin ≥ 1.2 (unconjugated)

b. Peripheral smear : Schistocytes / helmet cells

c. LDH & low Haptoglobins

d. severe anemia not explained by blood loss.

Mx of HELLP syndrome – .....

If gestational age ≥ 34 weeks .....

Active space

If gestational age <34 wks pregnancy + no complications

(Fetal distress / abruption / pulmonary/acute kidney injury)

Mx of of Impending Eclampsia .....

- Severe Preeclampsia + convulsions due to severe cerebral hypoxia due to loss of ..... cause of seizure : cerebral Hypoxia & Cerebral edema.

  Seizures - Tonic - Clonic Sezures

  Headache, Blurred Vision, Epigastric Pain, Persistent Pedal Edema Suspect Imminent Eclampsia

  Types

- Antepartum eclampsia- MC & worst prognosis
- Intrapartum
- Postpartum

Management protocal -

First step

↓

Maintain airway

↓

DOC to prevent & treat eclampsia MgSO4

↓

Labetalol

↓

Immediate induction of labour irrespective of gestational age

MC Regime .....

Active space

MgSo4 .....Therapeutic index.....

Mechanism of action of MgSo4 1. ACTS on NMDA Receptors

2. Block .....

3. .....

End point of giving $MgSO_4$ -24 hrs after delivery or 24 hrs after last seiure whichever is later

Toxicity of MgSO4

First sign of Toxicity Loss of patellar reflex (At 7-10meg)

- At this range - Diaphoresis - Feeling of warmth
- Slurring of speech can occur

At 12 meq - Respiratory arrest

At 15 meq - Cardiac arrhythmias

At 25 meq - Cardiac arrest

Antidote-.....

Termination of pregnancy in-

1. Impending Eclampsia
2. Eclampsia
3. Fetal distress/abruption - M/c
   Complication of PE with severe features
4. Uncontrolled BP.
5. Acute kidney injury (Rising creatinine value)
6. Pulmonary edema
7. Coagulopathy.

Active space

- Doc to treat preeclampsia .....
- Doc To treat Preeclampsia in asthmatic lady .....
- Safest drug .....

- DOC to treat emergency hypertension in pregnancy- .....

- DOC for chronic hypertension in pregnancy - Oral.....

- DOC for refractory hypertension .....

- DOC for status eclampticus .....

- Prevention of PIH –.....

Predictor of PIH – Old – ..... Roll over test

Normally when she turns from lateral position to supine position, her blood pressure .....

(Supine .....syndrome)

Predictor of PIH - Supine .....

This is why preganent lady is advised to lie in .....
.....

Latest - .....

Early uterine artery color doppler may detect .....

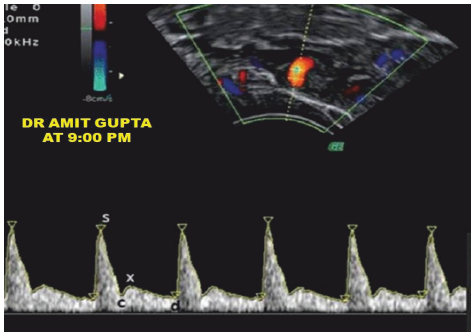
Normally diastolic notch in uterine artery disappears .....

Predictor of PIH - Persistence of ..... In uterine artery even after.....

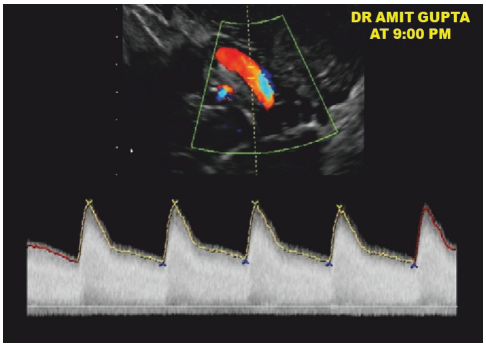
Diastolic Notch

Notch Disappeared

Active space

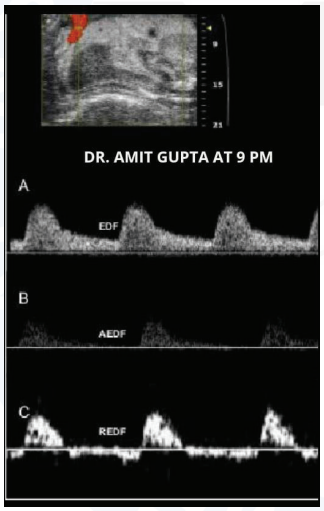


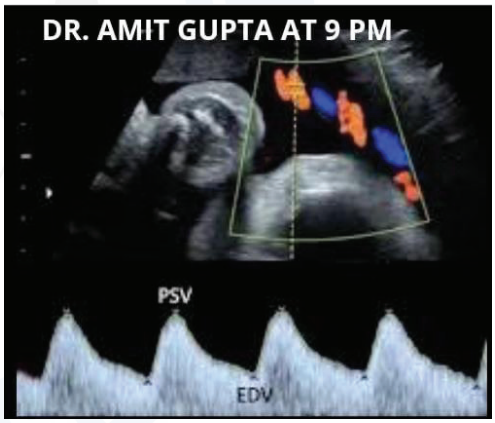
Uterine artery in non-pregnant & pregnant lady bebore 22 weeks



Uterine artery in pregnant lady >= 22 weeks

Prognosis of PIH - ..... artery Color Doppler





Active space

MULTIFETAL PREGNANCY

Multifetal Pregnancy

MC pregnancy .....

- MC multifetal pregnancy -Twins

Hellin's rule (for incidence of multifetal pregnancy)

- Twins 1 in 80
- Triplets 1 in 80^2
- Quadruplets - 1 in 80^3

Max incidence – in Nigeria → 1 in 20

Min incidence – in Japan → 1 in 200

– in India → 1 in 80

Additional 350 kcal from singleton pregnancy Additional 60 mg Iron (Additional to Singleton) 0.5 mg Folic acid (Additional to Singleton) required

Types

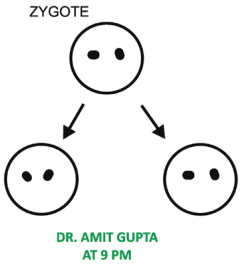
Monozygotic

- From 1 zygote
- Identical twins

Phenotype same

Blood group same

Same sex



Dizygotic

From 2 zygote

- Nonidentical twins
- Differs

Active space

Except – finger prints different

- May be monochorionic or Dichorionic (depends on timing of division)
- Incidence –1 in 250

- Always dichorionic

Risk factors -

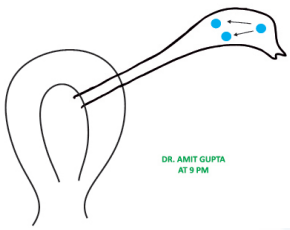
MC - Increasing due to increased use of ovulation inducing drugs

Other risk factors-

- Age >=35 years
- Multiparity
- Family H/O twins

Amnion is formed on ..... post fertilization

Chorion is formed on ..... post fertilization



Type of monozygotic twins - depends on timing of division

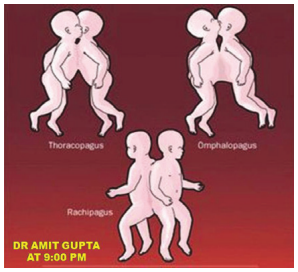
1. If division occurs within 3 days (72hrs)- .....
2. 4 to 8 days - .....
3. 9 to 12 days - .....
4. After 12 days - Partial splitting .....

MC conjoined twins → ..... > Thoracopagus

Rarest conjoined twins →.....> Craniopagus

Active space

MIST

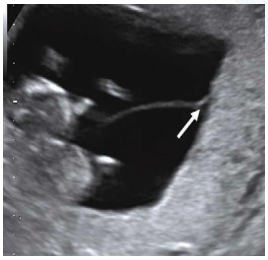


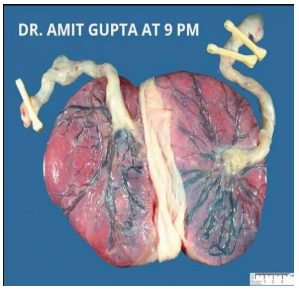
Prognosis in twins – depends on .....

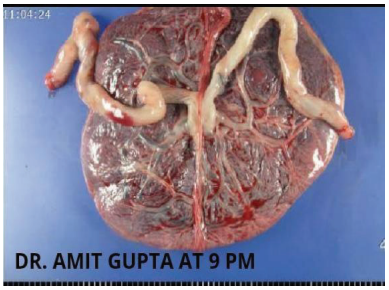
- Better in Dichorionic> Monochorionic
- Diagnosis – .....

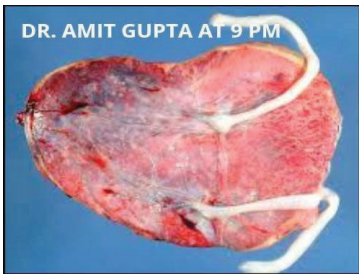
Aneuploidy more in .....

Monochorionic	Dichorionic
1. Dividing membrane- (intertwin membrane)	1. Dividing membrane - 4 layer 2 amnion 2 chorion
2. (2 layer) - 2 Amnion	2. Thickness > .....
3. Thickness	3. .....
4.	







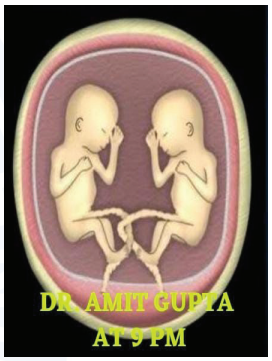


Dichorionic..... Monochorionic ..... Monochorionic .....

Active space

Overall Complications of twins - MC complications .....

- More anemia
- Hyperemesis gravidarum
- Polyhydramnios
- PROM
- Cord prolapse
- ..... MC neonatal complication
- ..... – MC maternal complication
- Placenta praevia
- Diabetes
- Pregnancy induced hypertension

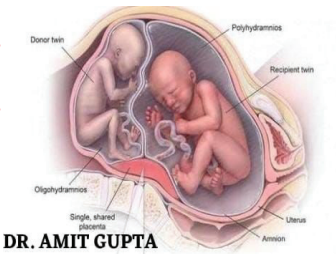


Complications Specific to monochorionic monoamniotic

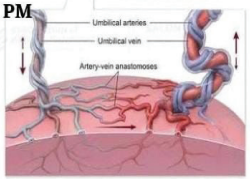
- .....

Complications Specific to Monochorionic diamniotic- Twin twin transfusion syndrome d/t abnormal deep arterio- venous anastomosis

Donor fetus	Recipient fetus
- Smaller	- Larger
- Anemia	- polycythemia
- Oligohydramnios	- Polyhydraminos
- ..... failure	- .....
- ..... prognosis	- .....



DR. AMIT GUPTA
AT 9 PM



Mx - If <28wks. ,,,,,,,,,,..... of abnormal anastomosis

If ≥ 28 weeks

Active space

.....

..... Staging

Twin anemia polycythemia sequence

Mostly after treatment of TTTS.

- Due to micro defect of .....
- ..... of one fetus enter other fetus
- Donor ....., recipient .....

V.V. Rare - Twin reversed arterial perfusion

- Due to abnormal arterio .....
- Seen if one fetus is .....
- Umbilical ..... from normal fetus carry deoxygenated blood to .....

Vanishing Twin - If one fetus .....

More Common in .....

Mostly diagnosed .....

If diagnosed antenatally .....

Active space

Concordant twins- If difference in fetal weight < 20%

Discordant twins - If difference in fetal weight >=20%

Superfoetation –

- V.V.V. Rare
- fertilization of two ova in ..... cycle

Superfoecundation –

- V.V Rare
- Fertilization of two ova in ..... cycle by different act of intercourse

Timing of delivery :

Dichorionic : 38 weeks

Monochorionic : 36 weeks

MonoAmniotic : 34 weeks

Mx - Depend on Presentation

- Rule - If first fetus is non cephalic – C.S.
  If first fetus is cephalic - vaginal delivery may be tried

1. MC Presentation - both cephalic
   - Mx -Vaginal delivery

Active space

2. 2nd MC - First cephalic and second breech
   - First cephalic – vaginal delivery
   - Second breech → assisted breech delivery

Interlocking of twins - seen with

First twin - .....

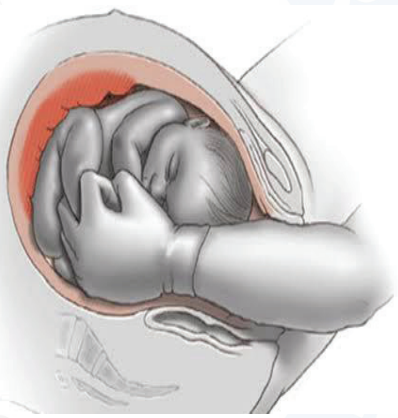
Second twin - .....

3. First cephalic and second transverse

- First cephalic – vaginal delivery
- Second transverse – ..... podalic version or Breech .....

Done in Operation theatre in .....

Complication .....



Active space

Abnormal Labour— Preterm Labour, Prelabor Rupture of Membranes, Post-term Pregnancy , Induction of Labour , Obstructed Labour

Preterm labour – if labour begins before 37 weeks

Early preterm labor .....

Late preterm labor .....

- Most imp. Risk factor- H/o previous preterm labour
- MC cause -Idiopathic
- Second most common cause of preterm labour –infection

Gardnerella
Chlamydia
Listeria
Streptococci
Covid -19
Asymptomatic Bacteriuria
Periodontal Infection

Chorioamnionitis
↓
Prostaglandins

Premature activation of fetal .....

Stess -Maternal - depression, Anxiety
- Fetal - UPI

Stress - ..... Release of Hypothalamic CRH

Other risk factors

- Polyhydraminos
- Macrosomia

Active space

- Multifetal pregnancy

Lead to overstretching of uterus - .....

.....

- Cervical Incompetence
- Anemia
- Smokings
- Antepartum hemorrhage
  Infections
  PROM - Release PGS - Preterm labour
  Short cervix
  Smoking
  Extremes of age
  Low BMI
  Under nutrition
  Uterine anomalies

- Diagnosis of preterm labour
  - Most important - Uterine contractions

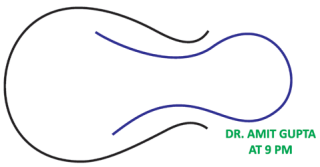
with any one of following

1 Cervical length.....

2 Internal OS diameter .....

3 ..... membranes (ballooning/funneling/herniation of

4 Fetal fibronectin -if .....



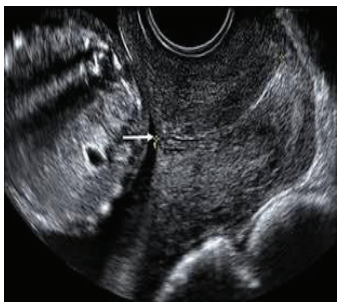
Fibronectin → Fetal glycoprotein Present between fetal & maternal membranes.

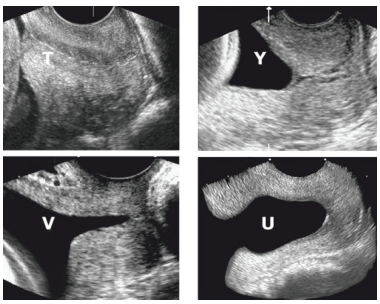
Sequence of change in shape of cervix with cervical dilatation.....

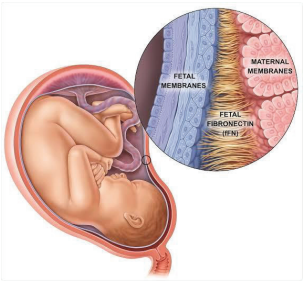
IOC to Diagonose preterm labor .....

Cervical length to predict preterm labor .....

Active space







Mx -

I. If Gestational age is < 34 week (early preterm labor)

1. Steroids

- Most commonly used in India Dexamethasone → 24mg

6mg $\xrightarrow{12\ hrs}$ 6mg $\xrightarrow{12\ hrs}$ 6mg $\xrightarrow{12\ hrs}$ 6mg I/M Injection

Best - Betamethasone → 24mg

→ 12mg $\xrightarrow{24\ hrs}$ 12mg I/M Injection

Prevents A. Necrotising enterocolitis

B. Intra cranial haemorrhage

C. Periventricular leucomalacia

.....

- Minimum waiting period → ..... hrs. after ..... dose
- C/I in infection

2. Tocolytic - Drugs to arrest .....

- DOC-..... BCZ it has ..... side effects
- DOC in heart disease – .....
- DOC to prevent pretem labour- .....

Other measures .....

Active space

Mgs04 (mild tocolytic)

Drugs with maximum side effects - beta mimetics

A. Isoxsuprine

B. Ritodrine

C. Salbutamol

D. Terbutaline

MC side effects - tremor

Other side effects - hyperglycemia

Tachycardia, Hypokalemia, Pulmonary edema

Ritodrine to prevent preterm labor cause .....

3. MgSo4 - (Given if gestational age is < 32 weeks)

DOC to prevent neurological complications

4. Fetal monitoring by Serial USG, Nonstress test (NST), Biophysical profile scoring (BPPS)

Antibiotic as prophylaxis against GBS → Not in india

II. - If Gestational age is >=34 weeks (late preterm labor)

no role of tocolytics .....

1. .....

2. .....

Premature Rupture of Membranes (PROM) - old name

Prelabor Rupture of Membranes (PROM) - new name

- Ideally membranes rupture after complete cervical dilatation
- PROM - But If membranes rupture before ..... of labour.

Active space

Preterm Prelabor Rupture of Membranes (PPROM) - But If membranes rupture .....

MC risk factor -

MC Cause → .....

Other causes – Polyhydramnios, Macrosomia, Multifetal Pregnancy

1. Previous H/o PROM, 2. Infections, 3 .Acute trauma, 4. APH

5. Smoking

Features - 1. Tempreture

2. CBC - TLC

3. ..... } Mechanical Stress, Inflammation, Infections ..... Tensile Strength of membranes

- Diagnosis

1. If gross rupture of membranes - complains of .....
..... of cloths

Other presentations :-

1) Uterine Contractions .....

2) Complications

- Cord ..... - Chorioamnionitis - Abruption

2. If Microscopic rupture-

Ask her to .....

Give .....

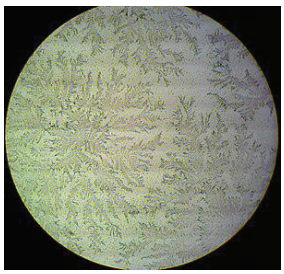
If still .....

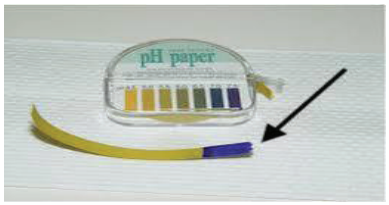
a. .....
.....

B. Cervicovaginal secretion → microscopy →
.....

C. Fetal fibronection in cervico - vaginal secretions

Active space





USG .....

Best tests .....

- Actim PROM : IGFBP (Insulin Growth Factor Binding Protein Assay)
- Amnisure : PAMG (Placenta alpha microglobulin - Protein Assay).

.....

Management

I. If Gestational age is < 34 week

1. Steroids
2. Tocolytics
3. Mgso4 if GA < 32 weeks
4. Antibiotics - DOC -

in india .....

if allergic .....

if resistance .....

5. Fetal monitoring by .....
6. Maternal monitoring for .....

II. - If Gestational age is >=34 weeks

1. .....
2. .....
3. .....

Active space

III. - If Gestational age is >=37 weeks

1. .....

2. .....

Post term pregnancy - Calcifications in placenta - reduced blood supply - ..... - umbilical cord ..... fetal .....

- If Gestational age is >= 42 weeks
- MC cause - mistakes dates → Review menstrual history again

Other causes

- Deficiency of placental sulphatase > aromatase
- Anencephaly
- Sedentary habits

Complication

- Antenatal - .....
- postnatal .....

Mx- .....

Induction of labour - Labor has not begun. we are .....

Methods

- Medical- PGE2 (Dinoprostone) - cervical .....

  GEL Vaginally → May be repeated after .....if required

  Max. Doses ..... Oxytocin is started ..... after last dose

  PGE1 (Misoprostol) - cervical .....

  - uterine .....

  Tablet Vaginally → May be repeated after .....if required

  Max. Doses ..... Oxytocin is started ..... after last dose

Active space

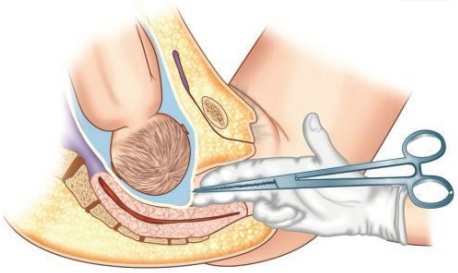
- Surgical

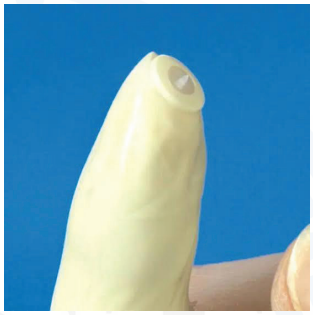
  Stripping (sweeping) of membrane – prostaglandin .....

  digital separation of membrances from .....

  ..... Reflex

Augmentation of labour - Labor has begun. we are .....





- Medical method -DOC .....
- Surgical method - Artificial rupture of membranes

  By Kocher's Forceps

  By amnicot - finger glove with .....

Contraindications OF ARM

- Polyhydraminos → If Mandatory → .....
- HIV
- Herpes - multiple painful vesicles on vulva - Mx - Acyclovir & elective C.S.
- IUFD

Cervical status is Assessed before induction – MC .....

1 Cervical .....- most important- .....

2 Cervical effacement -.....

3 Cervical position -.....

Active space

4 Cervical consistency-

5 Station of head -

If score ..... -induction may be successful

Modified bishop score - ..... Accurate

- Instead of Cervical effacement, check cervical length by .....

Simplified bishop score → Parameters .....

Cervical .....

Cervical .....

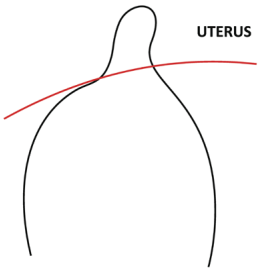
Station .....

Obstructed Labour

- Prolongation of labour despite of good uterine contraction

Features

- Contracted/retracted upper segment
- Bandl's ring/ retraction ring
- Distended lower segment
- Patient – exhausted
- Vagina hot and dry
- Mild bleeding p/v
- Haematuria
- Tachycardia, Tachypnoea, Dehydration
- Fetal distress



DR. AMIT GUPTA
AT 9 PM

P/V - 1. Caput

2. Moulding

Active space

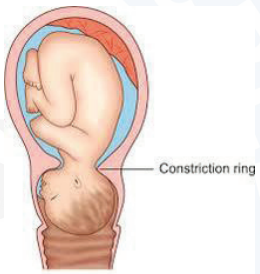
Mx- immediate C-section

Intrauterine death is .....

Risks

- Uterine rupture
- Uterine Prolapse
- Vesicovaginal fistula

Shroeder's Ring/ constriction ring



- Due to injudicious use of oxytocin
- Seen only vaginally
- Physiological

  Mx -May resolve spontaneously→if not → C.S.

Active space

POST PARTUM HEMORRHAGE

Postpartum Haemorrhage

- Excessive bleeding per vaginum after delivery and uptil 12 weeks

MC cause of most .....

Acceptable limit of blood loss after delivery -

- After vaginal delivery - upto .....
- After c- sec delivery - upto .....

PPH-According to ACOG - ACOG →American College of Gynaecology

- Blood loss..... irrespective of mode of delivery

or

- Any blood loss with signs / symptoms of ....

WHO Definition -

PPH - Blood loss .....

Severe PPH - Blood loss .....

1) primary PPH or early PPH

- Within 24 hrs

2) Secondary PPH or late PPH

- After 24 hrs- upto 12 weeks
- Most common- .....PPH
- Most common cause - .....

Active space

Causes of uterine atony :-

- Polyhydramnios
- multifetal pregnancy
- Macrosomia
- Anemia
- Grand multipara
- Prolonged labour
- Precipitate labour - If all stages of labor are completed within ....
- Uterine fibroid
- Injudicious use of oxytocin
- Epidural anaesthesia - For ..... delivery
- H/ O PPH
- After shoulder dystocia drill

Secondary PPH

- MC cause-.....

Other rare causes

- Trauma - MC .....
- Coagulation disturbance

Management protocol :-

- First step .....

  I/v Inj. Tranexemic acid

- Palpate uterus→ If ..... palpable → uterine atony
- DOC to treat PPH- .....

Active space

PPH + Heart disease

DOC = Oxytocin

Which drug is not given - ..... (C/I in heart disease) because it cause Severe transient Hypertension

Uterine massage is given along with .....

- Complication with IV bolus Oxytocin .....

- Misoprostol (PGE_1) .....

- Most potent / Best drug- .....

.....

- Methyl analogue of PGF_2 a
- 0.25 mg I/m x every 15-30 min
- C/I : Asthma
- Tatal : 8 doses in 24 hrs.
- S/E : Significant Diarrhoea

Others:

- Methyl ergometrine- 0.2mg I.M - Kept in brown ampoule because it is .....
- Syntometrine (Oxytocin 5 IU) + Methyl Ergometrine 0.5mg

Carbetocin 100 ug slow I/V in one minute

- Oxytocin+ uterine massage - tried for .....

If no response in 30 mts. .....

↓

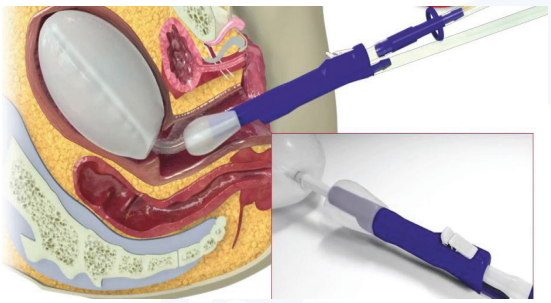
Active space

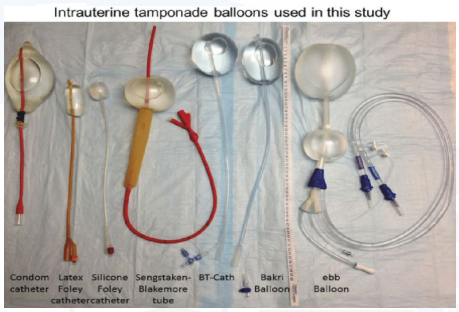
Uterine Tamponade - put a catheter inside uterus & inflate baloon of catheter - compresion of uterine wall

Sengestaken-blackmore Esophageal catheter

MC → Bakri balloon catheter

Condom catheter ..... of saline





If no response

↓

A. If vitals are unstable .....

MC - External aorta compression - shift her to O.T. -

- Military Anti - Shock Garments
- Bimanual compression

Devascularisation of vesseles

↓

1st -..... artery ligation

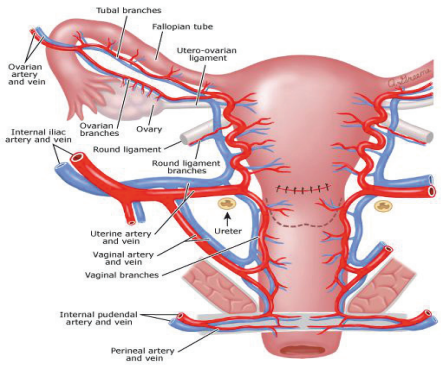
2nd .....

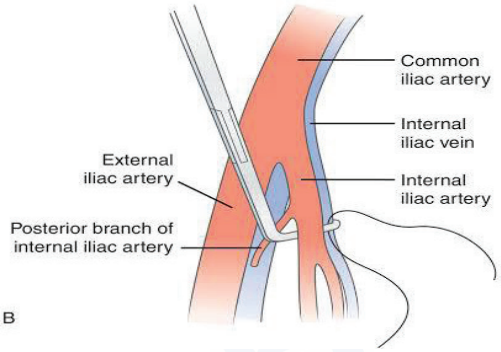
Active space

3rd ..... artery ligation

..... distal to .....

Pulse pressure decreases by approximately ..... percent after ligation of anterior division of internal iliac artery.





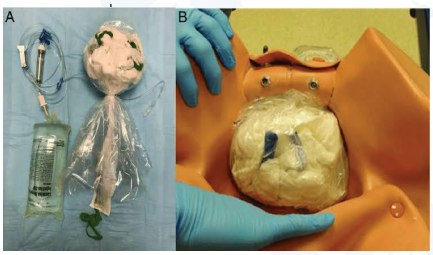
↓

If no response

↓

..... Hysterectomy → Only uterus is removed

↓ Cervix is not removed



If no response

↓

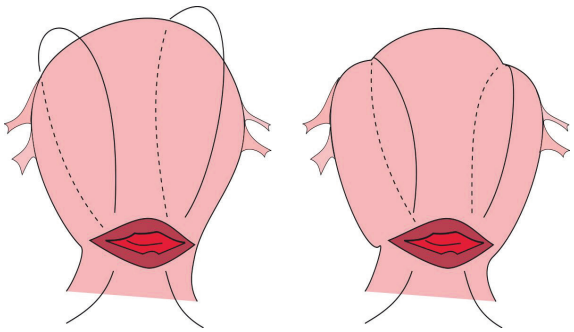
- ..... pack

- ..... pack

B. If vitals are stable after uterine temponade -

1. Uterine artery .....

If not avilable .....



Active space

MC cause of Shock after delivery .....

MC cause of Shock immediately after delivery .....

MC cause of unexplained Shock after delivery .....

AFE is also c/a anaphylactoid syndrome of pregnancy

Sudden onset of breathlessness

Rapidly progresses

Hypotension

Cardiac Arrest

Coma

Amniotic fluid embolism may present as profuse bleeding and features of DIC.

Active space

Puerperium

Puerperium- ..... period after delivery in which all organ / system return back to normal both anatomically & physiologically

- Process of return - involution
- Maximum involution in .....
- Uterine size decrease due to → in ..... of cells

Pulse, BP, S. Fibrogen, hCG, GFR - Normal in approximately .....

Cardiac output.....

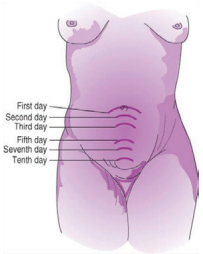
Blood volume .....

Retention of urine and constipation may be seen in puerperium.

Weight of uterus

- Immediately after delivery – .....
- After 1 week- .....
- After 2 week- .....
- After 3 week - .....
- After 6 week- .....

1C M = 1 finger Breadth



Height of uterus

- Immediately after delivery – ⟷
- ..... above pubic symphysis- lower border of umbilicus - ⟷

Active space

weeks pregnancy level

It decreases by .....

- After 24 hrs (on 2nd day post delivery) - ..... below umbilicus ..... Finger breadth
- By ..... pelvic organ

in week

Types of Puerperium

Immediate - upto 24 hours

Early - after 24 hrs - to 1 week

Late - after 1 week to 6 weeks

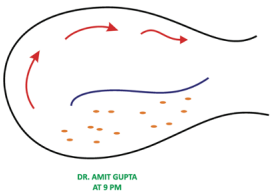
Post partum discharge (Lochia)

- Approx. 250ml
- Fishy odour
- Upto .....
- ..... Red (RBC)- lochia rubra
- ..... Yellow (WBC)- lochia serosa
- ..... White (decidua) – Lochia alba

Correct sequence of lochia

Lochia ..... → Lochia ..... → Lochia .....

Puerperal Sepsis - MC risk factor - .....



- Mc site -.....
- Mc microorganism - .....
- Mc route of spread ..... spread
- Puerperal pyrexia- → ↑in in temperature > 100.4° F on ... occasions .....Apart excluding first .....

MIST

- Management - for mild sepsis .....

  for severe sepsis .....

Post partum blue - in first 10 days due to sudden withdrawal of estrogen/ progesterone

Management - .....

Post Partum Depression –

Occurs between ..... after delivery

Management .....

Pospartum Psychosis

Occurs between ..... after delivery

Management .....

Uterine dehiscence (impending rupture) - Serosa is .....

Most IMP. Risk Factor .....

Symptom - Pain in .....

Sign -

Most consistent sign - fetal .....

Scar .....

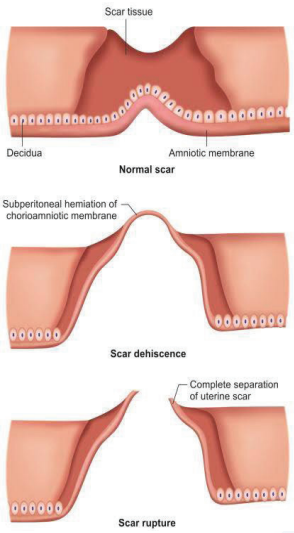
.....

Uterine Contractions .....

Fetal Station .....

Management .....

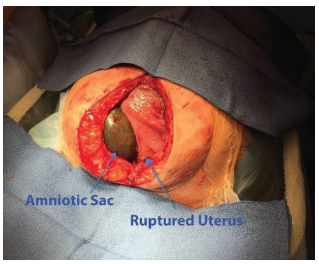
Active space

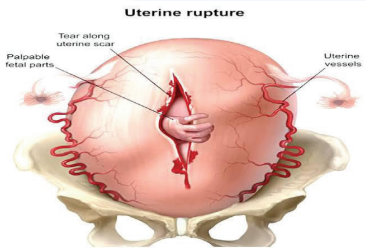


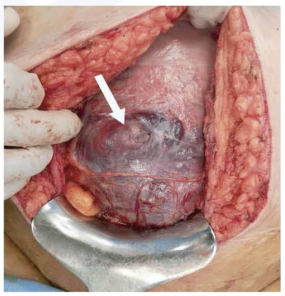
Uterine rupture - Serosa is .....

Signs

- .....
- Fetal parts can be .....
- Most specific sign - loss of .....
- ..... in peritoneal cavity
- Profuse .....
- Profuse .....
- Maternal tachycadia & .....
- Generalised abdominal .....
- Uterine contractions .....







Management Resuscitation & Exploratory .....

Try to .....

If not possible .....

Active space

Postpartum shivering
Cause-
1. ..... - causes fever with chills

Postpartum hot flushes - due to withdrawl of .....

Uterine subinvolution - uterus remains .....
Fever may be associated finding

Peripartum Cardidmyopathy
from ..... antepartum to anytime in .....
postpartum in a patient who has no underlying heart disease.
-Develop heart failure.
Mostly happen in 1st wk after delivery
Treatment- Treat like heart failure
Bromocriptine (Dopamine agonist) may be added

Septic pelvic thrombophlebitis
MC involved = ...
PV exam: cord like structure
Inv- MRI/CT
may show thrombosed vein
Rx - antibiotics + add LMWH.

Active space

HEART DISEASE IN PREGNANCY

Heart Disease in Pregnancy

Most common heart disease in pregnancy : Rheumatic heart disease - .. ..

2nd Most common heart disease in pregnant female : Atrial Septal Defect

Most Common congenital heart disease in pregnancy : .....

Most common congenital valvular heart disease during pregnancy : ....

Most common cyanotic heart disease during pregnancy : .....

Heart disease with the highest risk of maternal mortality or the worst prognosis :

..... syndrome

Most common time of death ..... at the time of delivery or within ..... of delivery.

Most common cause of death : RVF with cardiogenic shock

Which heast disease is MC associated with mortality - .....

Active space

Pathological Symptoms -
Progressive Dyspnoea worsening NYHA Grade
PND
Orthopnea , Hemoptysis , Chest Pain
Cyanosis
Clubbing
Systolic murmur (>Grade 2) Diastolic murmur Physiological only in 20%

Pathological Signs-
Wide split S2
Loud P2
Any evidence of arrhythmia / heart failure Persistantly distended neck veins.

Clarke's classification- on the basis of maternal mortality

I. Low mortality <1%
   (i) ASD (Atrial Septal Defect)
   (ii) VSD (Ventricular Septal Defect)
   (iii) PDA (Patent Ductus Arteriosus)

II. Moderate mortality 15-25%
   (i) AS (Aortic Stenosis)
   (ii) Marfan syndromes Without aortic involvement

III. High mortality 25-50% - These are contra indications for pregnancy
   (i) Pulmonary .....
      A. primary, B. Secondary .....
      .....

Active space

(ii) Marfan syndrome with aortic root involvement

(iii) Coarctation of aorta with aortic valve involvement

- Absolute C/I of pregnancy - .....

Mx

1. Position- Semirecumbent position

   Little lt. lateral

2. Oxygenation
3. Wait for spontaneous labour but if it doesn't start : induction at 37-38 weeks
4. Cut short second stage of labour by .....
   ..... .if it takes>.....
6. ..... uterotonic given. Methyl ergometrine is
   .....
7. ..... immediately after delivery
8. I/V fluid – should not be more than 75 ml/hr due to risk of .....
   or 1ml/kg/HR

Maximum risk of heart failure .....

Anticoagulation protocol

- Upto 12 weeks –

  If dose of warfarin < 5 mg / day - .....
  If dose of warfarin >= 5 mg / day - .....

- After 12 weeks to 36 weeks – .....

- At 36 weeks .....

Active space

- One week before EDD / 3 days before EDD.....
- On ..... all anticoagulants
- If on warfarin on EDD or 2 weeks before it - ..... warfarin and .....
- If on warfarin and in Labour ..... warfarin, do ..... , Give Vitamin K to mother and baby
- .....hrs after vaginal delivering or.....hrs after c-section → May restart anticoagulation .....

Side effect of warfarin (Fetal warfarin syndrome) - Chondrodysplasia Punctata - oromandibular hypogenesis

In Pregnancy :

AHA :- (American Heart Association) - Anticoagulation + Aspirin Also

ESC : (Euroean society of Cardiology) - only anticoagulants

Female with bioprosthetic valves don't need anticoagulation

Give only aspirin during pregnancy (low dose)

Contraception

- Best - .....
- Minilaprotomy – Best at end of .....
- Laparoscopy is C/I in heart disease

Sterility is not achieved immediately after vasectomy.

Use alternative methods for .....

8-12 wk

12-20 wk

-

Active space

III
MISCELLANEOUS

Active space

MIST

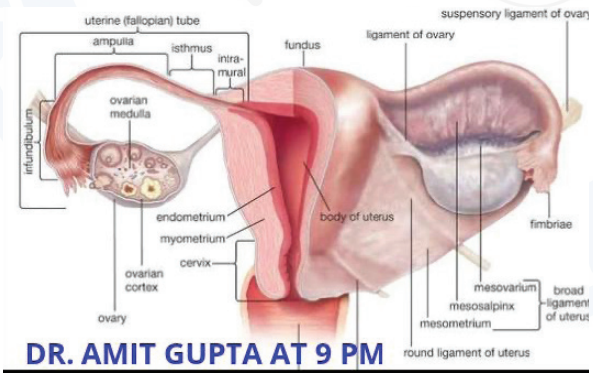
Anatomy

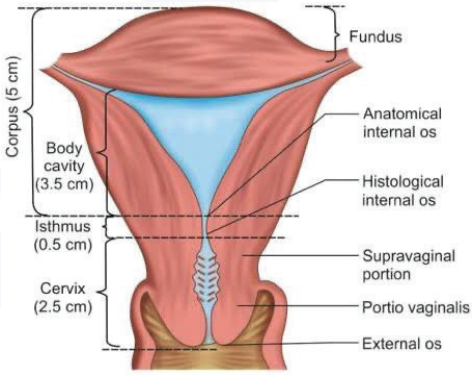
Anatomical internal OS- place where uterine ..... change to cervical .....

Histological Internal OS- place where uterine ..... change to cervical .....

Isthmus- part between ..... and .....

length ..... cm





- Isthmus forms lower uterine segment
- Length of lower uterine segment in pregnancy may be - at term .....

  labor .....
- ..... part of uterus

Active space

- Cervix is between histological internal OS to external OS
- External OS - part of cervix which open in vagina
- Endocervix - Between histological internal OS to upper part of vagina
- Ectocervix - part of cervix which protrudes in vagina or exocervix or portio vaginalis

Angle of anteflexion

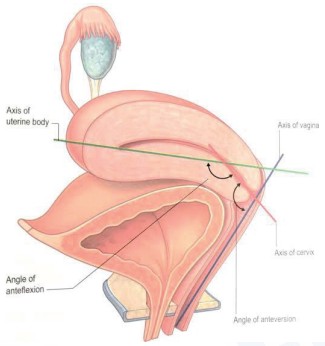
Between uterus and cervix

.....

Angle of anteversion

Between cervix and vagina

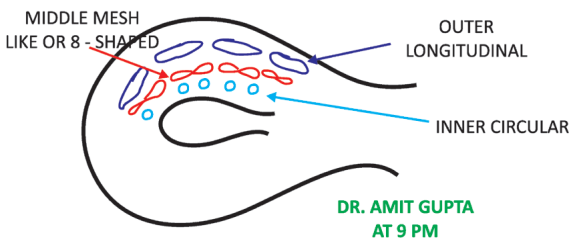
.....

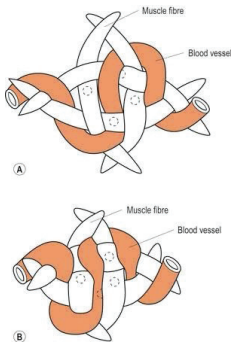


Uterus

1. Outermost layer - serosa/perimetrium
2. Middle part - myometrium
   a) Outer layer - longitudinal
   b) Inner layer - circular
   c) Middle layer .....

Blood vessels run in middle layer and are ligated during uterine contraction→ .....





Active space

Endometrial Thickness

1. Post - Menstrual - ..... mm
2. Around ovulation - ..... mm
3. In secretory phase - ..... mm
4. At implantation - ..... mm

3. Innermost part - endometrium

Shape of uterus - .....

Size .....

Length -

Nulliparae .....

Multiparae .....

Uterus is divided into three parts

Fundus - part above tubal opening

Body - part below tubal opening

Cornuae or angles of uterus - part where fallopian tubes open in uterus

- Weight of nulliparous uterus = ...
- Weight of multiparous uterus = ...
- Weight of pregnant uterus = ...
- Length of nonpregnant uterus = ...
- Length of pregnant uterus = ...
- Capacity of nonpregnant uterus = ...
- Capacity of pregnant uterus = ...

Active space

Blood supply

- By uterine artery (branch of anterior division of internal iliac artery)

↓

Arcuate artery

↓

Radial artery

↓

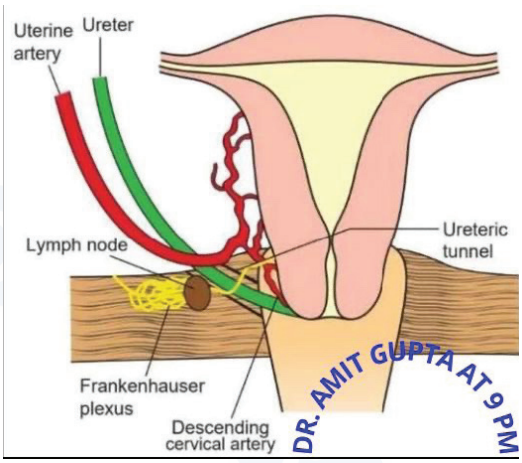
1. Spiral artery

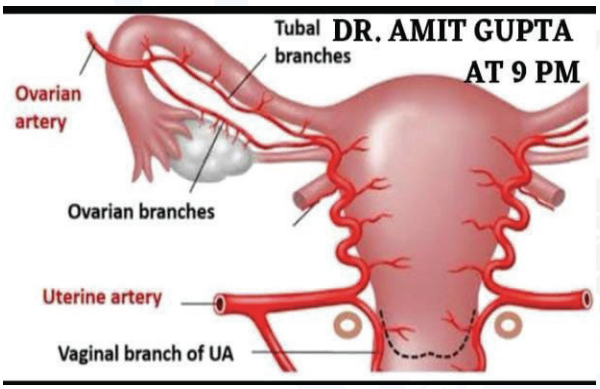
a) Zona compactum

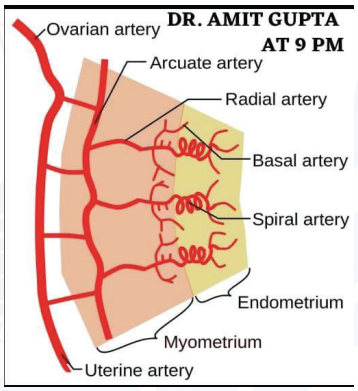
b) Zona spongiosum

2. Basal artery

a) Zona basalis







Lymphatic drainage

- Fundus - paraaortic lymoh node
- Cornuae - superphical inguinal lymph node
- Body - external iliac nodes

Nerve supply T10-L1

Epithelium lining of uterus- columnar

Active space

Cervix

- Endocervix
- Ectocervix
- Length of cervix - in non pregnant ..... - in pregnant.....
- Endocervix length ..... and lined by Tall columnar epithelium (red in colour)
- Ectocervix length ..... and lined by-stratified squamous epithelium (pink in colour)
- Transformation zone or squamo columnar junction (Pale-White)
- External Os - shape circular or conical in nulliparous
  transverse or slit like in multiparous
- Blood supply- by descending cervical Artery
- Nerve supply S2,S3,S4

Lymphatic

Internal iliac lymph nodes

Hypogastric lymph nodes

Obturator lymph nodes

Presacral/paracervical lymph nodes

External iliac lymph nodes

Vagina

- Length of anterior wall .....
- length of posterior wall .....
- Cervix protrude in vagina so fornices are created
- Shallowest fornix - ..... fornix (1)
- Deepest fornix- ..... fornix (1)
- Lateral (2)
- Sample for cytological study is taken from ..... wall

Active space

- sample for hormonal study is taken from ..... wall
- Epithelium - stratified squamous non keratinized

Blood supply

Descending cervicovaginal artery

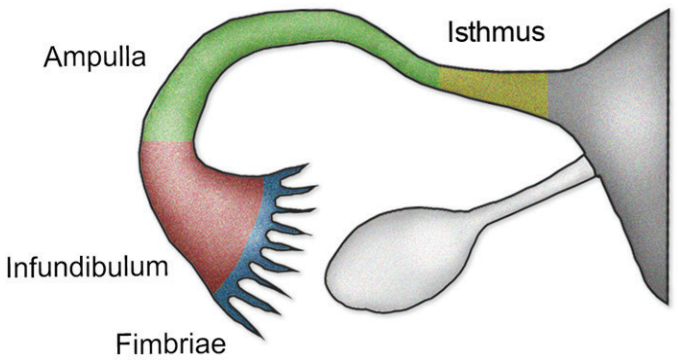
Internal pudendal artery

Middle rectal artery

Lymphatic drainage

1. Upper vagina -
   a) Internal iliac lymph nodes
   b) Hypogastric lymph nodes
   c) Obturator lymph nodes
   d) Presacral/paracervical lymph nodes
   e) External iliac lymph nodes
2. Middle vagina -internal iliac LN
3. Lower vagina- superficial inguinal LN

Fallopian tube



Active space

1. Interstitium or intramural

- Length .....
- Diameter .....
- Narrowest part
- Anatomical sphincter

2. Isthmus

- Length .....
- Diameter .....
- Physiological sphincter

3. Ampulla

- Length .....
- Diameter .....
- Widest and longest part of Fallopian tube
- Most common Site of fertilization
- Most common site of ectopic pregnancy
- Earliest/MC site of genital TB

4. Infundibulum - Fimbriae - Largest Fimbriae - .....

Picks up secondary oocyte after ovulation

- Length .....
- Diameter .....

Active space

3 types of cells in fallopian tube

a. Ciliated columnar

b. Secretory cells - Nutrition to .....

in fallopian tube

c. PEG cells

Blood supply

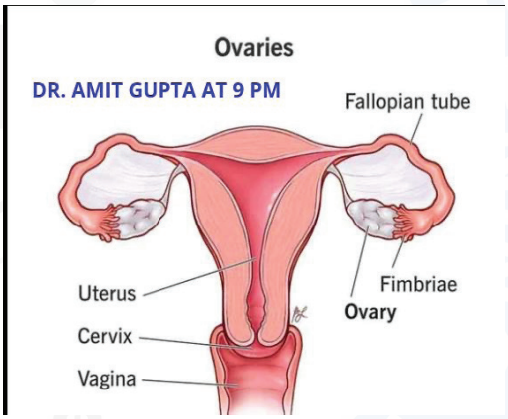
- Medial half - uterine artery
- Lateral half - ovarian artery

Lymphatic drainage

- Medial half- superficial inguinal lymph node
- Lateral half - paraaortic lymph nodes

Ovaries

- 2 in number
- In IUL - at T10
- They descend with gubernaculum and in adult they lie in fossa of waldeyer in lateral pelvic wall
- Size L → ..... cm
  W → ..... cm
  T → ..... cm
  Vol. - Reproductive age .....
  Post menopausal .....
  Wt. - .....



- Blood supply- by ovarian artery
- Lymphatic drainage-paraaortic lymph node
- Nerve supply- ovarian plexus
- Venous drainge

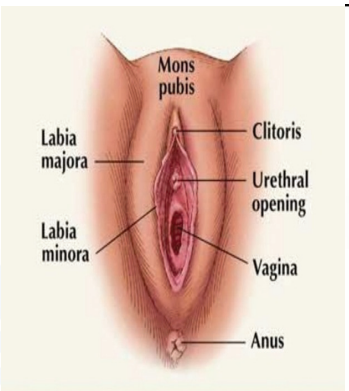
a. Left ovarian vein- into .....

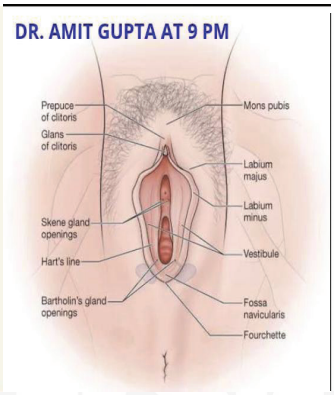
b. Right ovarian vein - into .....

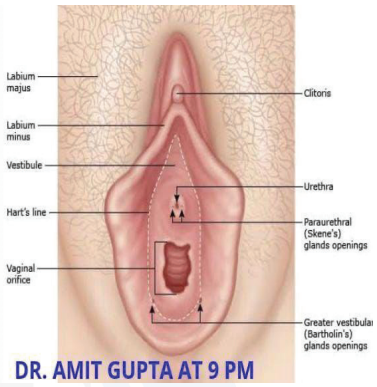
Active space

fallopian tube is - superior to ovarian ligament and round ligament

Female external genetalia - vulva or pudendum







- Labia majora -outer lip of vulva made up of fibrofatty tissue fuse posteriorly to form posterior commissure
- Labia minora - thin folds inside labia majora -inner lips of vulva
- Hart line - between outer keratinsed and inner non keratinsed epithelium of labia minora
- Clitoris -erectile ,muscular body in upper part of vulva
- Most sensitive part of female body because it is richly supplied by nerves
- Labia minora enclose clitoris and fuse above clitoris to form .....and fuse below clitoris to form .....

Active space

Labia minora fuse in lower part to form .....

Vestibule

- Bounded above by clitoris
- Bounded below by fourchette
- Bounded on either side by L. minora

6 openings present in vestibule

1 ..... urethral opening

2 and 3 .....ducts of skene gland (paraurethral glands)

4..... vaginal introitus

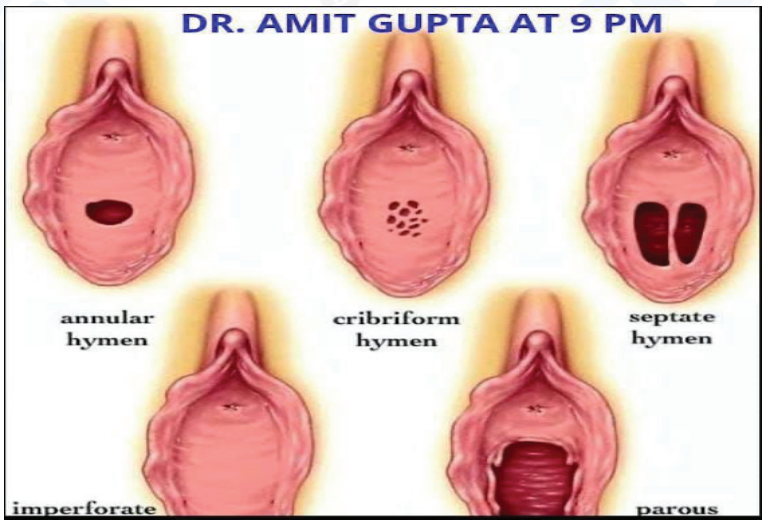
5 and ..... ducts of bartholin glands

..... between vaginal introitus and fourchette

Hymen

- Thin membrane covering vaginal introitus
- A small opening is present in hymen through which mensural blood comes out

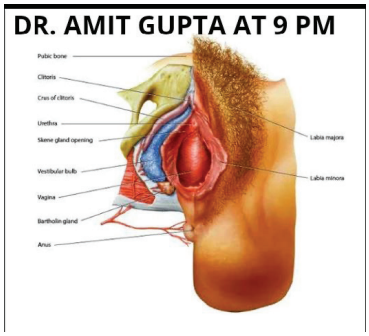
Types of Hymen -
Annular hymen/crescent
Cribriform hymen
Septate hymen
Imperforate hymen
Parous Hymen



Active space

Bartholin Gland or Greater or major Vestibular glands

- Thomas Bartholin - Number .....



Bartholin gland is located in Superphicial Perineal pouch in groove between labia majora & labia minora (on .....)

Bartholin Gland - ..... Secretions

Acidic PH is Spermicidal.

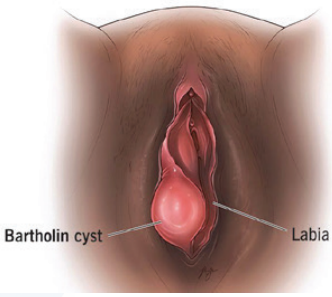
Ducts of Bartholin glands open between ----------and --------

at the junction of ..... and .....at.....

Bartholinitis - MC cause -.....

Bartholin Cyst - On ..... wall of vagina

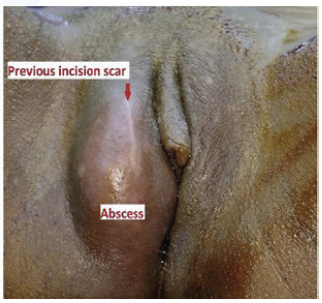
D/D - Gartner duct cyst - on ..... wall of vagina.



MC presentation of bartholin cyst - ..... selling at Labia majora

aggravated during .....

- Intermittent swelling
- Difficulty in .....



Bartholin abscess -

Management of Bartholin cyst

- If Symptomatic .....
- If Asymptomatic - a. if < .....

  b. if >= .....

If I & D is done for 3 times .....

Named criteria for bartholin gland malignancy :- .....

Active space

Blood supply-internal pudendal artery

Nerve supply- pudendal nerve

Lymphatic drainage-superficial inguinal LN → deep inguinal LN

Lymphatic drainage of clitoris -glands of cloquet/Rosenmuller

Mons pubis /veneris -upper part of vulva

Hair pattern -triangular

Cervix : body of uterus ratio

- At Birth .....
- Before Puberty .....
- At Puberty .....
- Reproductive age .....
- Menopause .....

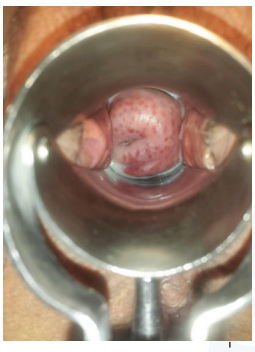
pH of vagina

- At Birth-.....
- Before puberty- .....
- At puberty- changes from alkaline to acidic - due to activation of H. P. O. Axis - estrogen - .....
- Reproductive age - .....
- Menopause - .....
- Menstruation- .....
- Pregnancy- .....

Active space

VAGINITIS

Vaginitis



1. Trichomonal vaginitis
   - By Trichomonas Vaginalis

- Symptoms- Vaginal Discharge - frothy, fowl-smelling, profuse, ...
  Itching .....
  Dysuria .....

  - Sign- numerous tiny ..... on vagina .....
..... Cervix or ..... Vagina or ..... Looking Vagina or .....
  - IOC - Saline Microscopy
- Gold standard- Culture on (Media = Stuart's Media)
- Mx - Tab. metronidazole 2gm Single dose or 500mg BD x 7Days

In pregnancy - in I trimester Not Given
- in II & III trimester - Tab. metronidazole 250mg TDS x 7Days
  - Rx of..... partners.....

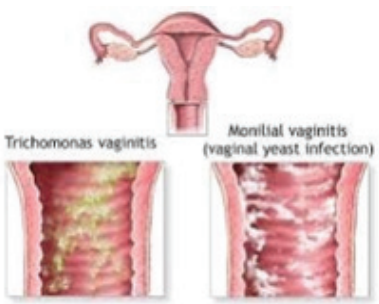
2. Candidal vaginitis - caused by Candida Albicans
   - Symptoms- Vaginal Discharge .....
   .....
   Itching .....
   Dysuria .....

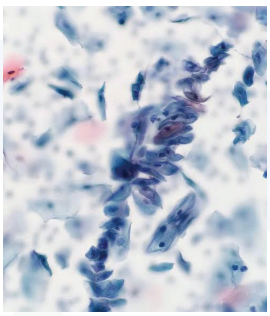
Active space

- IOC - saline microscopy- .....

- Gold standard- ..... (Media = SDA)

- Pap smear .....





- Mx Fluconazole 150mg single dose

  In pregnancy - in I trimester .....

  - in II & III trimester - .....

  - Recurrent candidiasis→

Mx Fluconazole 150mg on day 1, 4,7 x 2 wks

followed by maintenance dose once weekly x 6month

- Predisposing factors

  Immunocompromised

  - HIV +ve

  - H/o prolonged use of steroid & antibiotics

    OCP

    Diabetes

    Pregnancy

  - Rx of male - required only if .....

Active space

3. Chlamydial vaginitis-

- By chlamydia trachomatis

.....

IOC-nucleic acid amplification test (NAAT by PCR)

Gold standard-

Mx – Azithromycin + contact tracing

4. Bacterial vaginosis- ..... vaginosis - Alteration of vaginal flora

- Normal habitant
- Doderlein bacilli Or Lactobacillus acidophilus
- Glycogen→Lactic acid (PH → ACIDIC)
- But if →No. of doderlein bacilli ↓

↑ Gardnerella

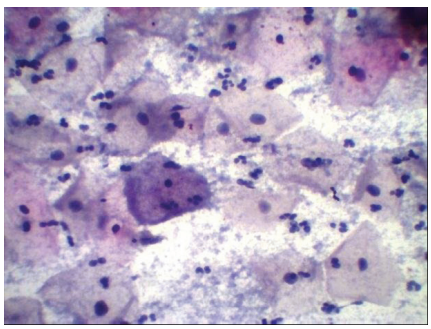
Diagnostic criteria- Amsel criteria 4 parameters -if .....present

1. pH >.....
2. ..... discharge

3. Positive whiff's /amine test→ discharge +.....

4. Clue cells .....

.....



Clue cells (at least 20% of epithelial cells

- Adhere on cell surface

Gold standard - Best test : Gram stain

Management .....

Tab Metronidazole

Active space

(Better) 500mg BD x 7Days

Tab Clindamycin

300mg BD x 7Days

In pregnancy - in I trimester .....

- in II & III trimester Tab, metronidazole

250mg TDS X 7 Days

Treatment of male partner .....

Itching .....

TLC/ESR .....

Risk factor for .....

.....

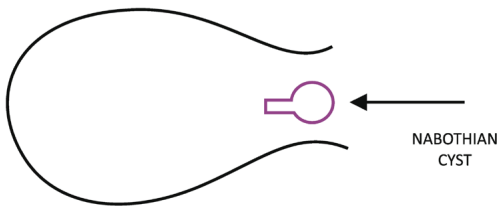
Physiological discharge - Colourless, Odourless, No itching

In Ist half of cycle Thin, Watery, Alkaline, Profuse

In 2nd half of cycle Thick, Scanty, non elastic

Cervicitis

Symptom



DR. AMIT GUPTA
AT 9 PM

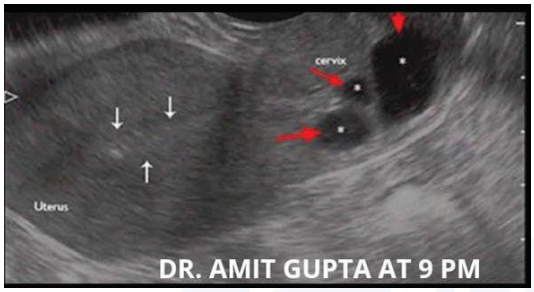
In chronic cervicitis .....

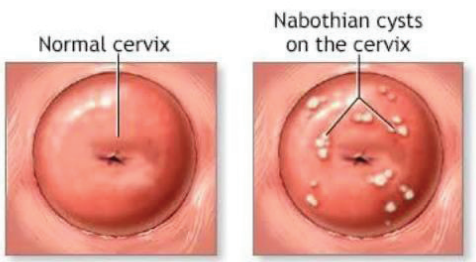
Mx. .....

Cell death occurs at

Mx of cervical discharge .....

Active space





Syndromic management of vaginal discharge .....

Treratment of male partner .....

Active space

TYPES OF HYSTERECTOMY

Types of Hysterectomy

1. Type 1 or Type A or Simple hysterectomy or Total abdominal hysterectomy or Extrafascial hysterectomy

Most commonly performed

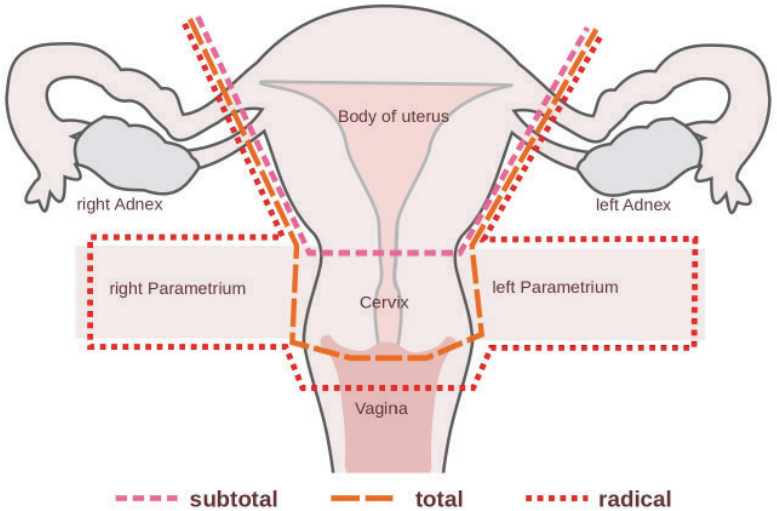
Most common indication - fibroid

MC hysterectomy done for .....

Remove Uterus + Cervix

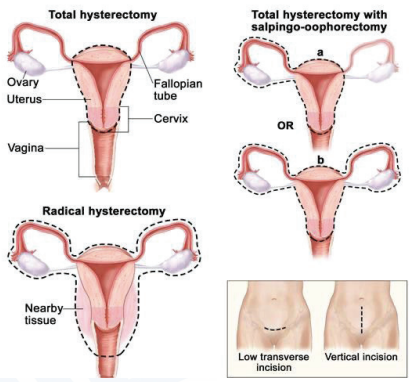
If Uterus+Cervix+Tubes+Ovaries Removed - Panhysterectomy

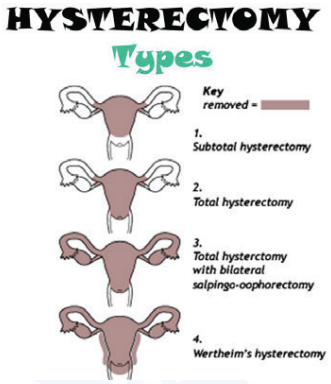
Subtotal Hysterectomy - Only uterus is removed
- Cervix is not removed
- Done in PPH Mx.



Active space

2. Type 2 Hysterectomy or Type B or Wertheims hysterectomy or Modified radical hysterectomy





Structures removed -

Uterus

Cervix

Both ovaries/Tubes

1 cm Vagina

Some Part of uterine artery

medial halves of cardinal and uterosacral ligament

3. Type 3 hysterectomy or Type C or Radical hysterectomy

Structures removed -

Cervix

Both ovaries/tubes

Entire cardinal ligament + entire uterosacral ligament

$\frac{3}{4}$ vagina

Major part of uterine artery

done in endometrial and cervical cancer

Type 4 and 5

Pelvic Exenteration or debulking surgeries

Active space

ENDOMETRIAL CANCER

Endometrial Cancer

- MC Age
- C/O

Causes of Postmenopausal Bleeding

1. MC cause of postmenopausal Bleeding .....

Other causes -

2 Polyp/cancer cervix ?
3. Atrophic endometritis (senile endometritis)
4. Endometrial hyperplasia
5. Atrophic Vaginitis

2. MC cancer responsible for post menopausal bleeding .....

3. & 4. MC Cause/ MC cancer responsible for postmenopausal bleeding in developing countries .....

Risk factors

MC .....

2nd MC .....

- .....menarche/ .....menopause
- ..... socio-economic status
- H/o infertility

Active space

- H/O Rx with .....
- Pregnancy should be avoided for..... after treatment with .....
- Atypical hyperplasia
- Diabetes + Hypertension + Obesity + Ca endometrium →.....
- Genetics

MC-A. HNPCC (Hereditary Non-polyposis colorectal cancer) Syndrome / Lynch Syndrome

due to gene mutation MLH ..... MLH .....MSH.....

MC → Ca .....>..... Ca ..... > Ca .....

B. Mutation in .....

C. .....Syndrome - mutation of .....

- H/O PCOS

Estrogen secreting ovarian tumor Eg. - Granulosa cell tumor

- HRT

Protective factors

Physical .....

Oral .....

.....

Active space

Prevention in HNPCC Syndrome

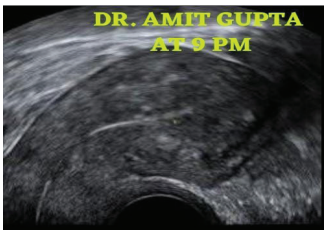
By .....

.....

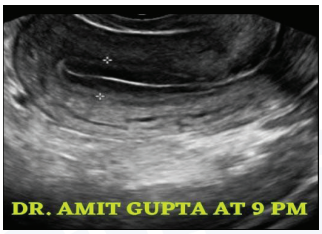
Percentage of females with Postmenopausal bleeding having endometrial cancer .....

If a lady presents with irregular bleeding P/V .....

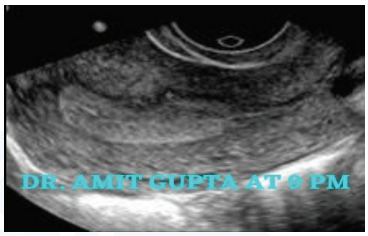
Endometrial .....

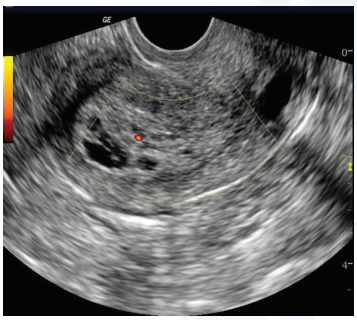


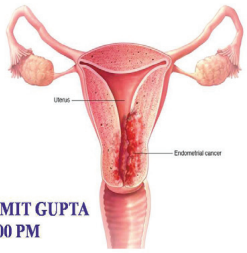
Normal proliferative

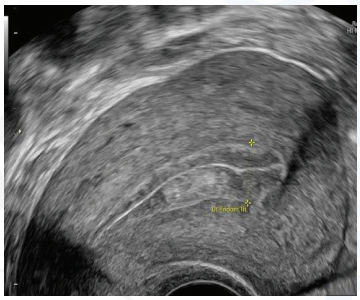


Normal secretory









Endometrial Hyperplasia .....

Endometrial Hyperplasia .....

If USG shows Increased endometrial thickness .....

Screening test ..... No screening for endometrial ca.

A. If diffuse hyperplasia .....
endometreial aspiration .....
by .....
in india by .....

B. If focal Hyperplasia .....

ACOG recommend endometrial assessment in any lady aged >=45years with AUB.

Active space

MIST

If < 45 years-AUB with H/o unopposed estrogen exposure - PCOS, Obesity

AUB - TVS - Endometrial Sampling

Definitive test.....

Old reporting pattern

1. Simple hyperplasia without atypical cells - risk of transformation to frank malignancy later on →

2. Complex hyperplasia without atypical cells - risk of progression into frank malignancy later on→

3. Simple hyperplasia with atypical cells - Risk of progression into frank malignancy later on →

4. Complex hyperplasia with atypical cells - Risk of progression into frank malignancy later on →

New reporting pattern

A. ..... atypical cells (cystic glandular ....)

risk .....

B. ..... atypical cells (endometrial ....)

risk .....

Mx of endometrial hyperplasia

I. If atypical cells are absent- .....

II. If atypical cells are present-

a) If age <45 years→Total abdominal hysterectomy(TAH)

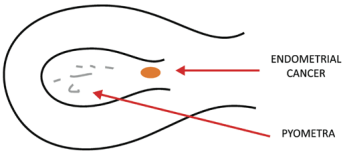
b) If age >45 years→TAH+.....

Active space

- MC symptom of endometrial cancer - .....
- Most specific symptom .....
- Others

  Hydrorrhoea .....

  .....



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AT 9 PM

If report is →Adenocarcinoma

- MC Histopathological finding → .....(adenocarcinoma grade .....) or ..... adenocarcinoma
- Rare but more dangerous → .....(adenocarcinoma grade .....) or ..... adenocarcinoma

MC - Clear cell carcinoma >papillary carcinoma

Type 1	Type 2
- Associated with increased Estrogen - Obesity - White Population - Gene mutation ..... - Age - 50-60 yrs - With endometrial hyperplasia - 80 %	- No relation with Estrogen - No relation with obesity - Black - - ..... - Age - 60-70 yrs - Without hyperplasia - 20 % .....

Gate keeper of endometrial cancer .....

- MC Route of spread →.....

MC site of haematogenous spread .....

Active space

Best investigation to see tissue infiltration .....

Best investigation to see lymph node metastasis .....

- MC Site of recurrence →.....

Grade 1 (.....Differentiated) - Malignant tissue .....

Grade 2 (.....Differentiated) - Malignant tissue .....

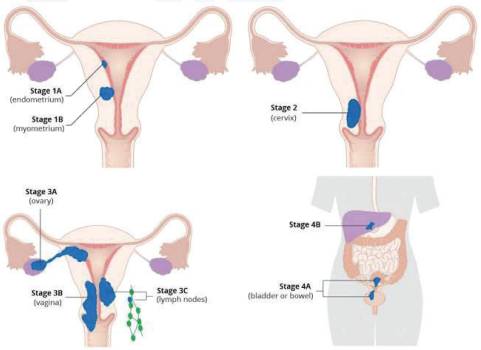
Grade 3 (.....Differentiated) - Malignant tissue .....

Staging

Stage I -

- IA → Endometrium or < .....
- IB → .....myometrium

Stage II - -------



Stage III

- IIIA → Serosa, adnexae, peritoneum .....
- IIIB → .....
- IIIC1→ ..... lymph node

C2 → ..... lymph node

Stage IV - Metastasis

A → Regional metastasis (Urinary Bladder/ Rectum)

B → Distant metastasis .....

Active space

Mx for type 1 endometrial cancer (Adenocarcinoma grade 1 and 2)

If IA grade 1,2 → TAH + BSO

- If tumor size <2cm → no LN dissected
- If tumor size >= 2cm → pelvic lymph node dissection

Post-op treatment .....

IB → TAH+BSO + Pelvic & Paraaortic Lymph node dissection f/b.....

II → ..... hysterectomy + pelvic and paraaortic lymph node dissection F/B .....

III→ ..... hysterectomy + pelvic and paraaortic lymph node dissection F/B .....

IV→ Debulking surgery + pelvic and paraaortic lymph node dissection F/B -.....

Mx for type 2 endometrial cancer (Adenocarcinoma grade 3)

1. Surgery according to .....
2. .....is done in .....

Chemotherapeutic drugs.....

Active space

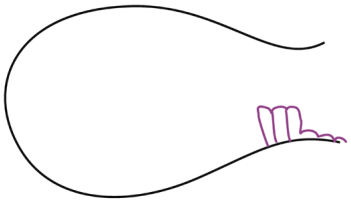
Adnexal mass with endometrial hyperplasia, Atypical cells are present in endometrium .....

Chemotherapeutic drug for distant metastasis or resistant (recurrent) cases .....

Active space

CERVICAL CANCER

Cervical Cancer



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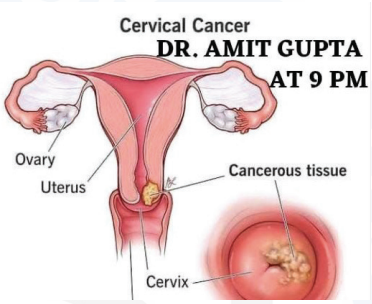


- Normally Transformation of endocervical Columnar to ectocervical Squamous epithelium occurs at squamo - columnar junction .... Metaplasia
- If abnormal cells are formed - .....

Causes

- MC- .....
- MC site .....
- MC and most sensitive-

- Most specific for cervical Ca -

CO- factor- .....

Other risk factors

- Low socio- economic class
- Poor hygiene

Active space

MIST

- Multiparity
- Pregnancy
- Anemia
- Smoking
- Multiple sexual partners
- OCP if taken for .....
- Early age of intercourse .....
- Early age of pregnancy .....

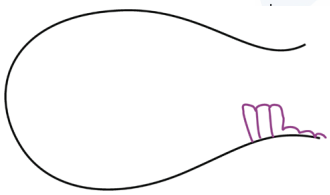
HPV 6, 11 - Genitalwarts (condyloma acuminata)

In neonates - Laryngeal papilloma

treatment podophylline, imiquimod

in pregnancy .....

CIN (Cervical intra epithelial neoplasia)



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- CIN I →If < ..... of epithelium involved
- CIN II → .....but ..... of epithelium
- CIN III → .....but < whole thickness
- Ca in situ – whole thickness
- Invasive cancer – If .....broken and dysplasia invade out of epithelium

BETHSEDA – new classification

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AT 9 PM

- (LSIL) low grade squamous intraepithelial lesion → .....
- (HSIL) high grade squamous intraepithelial lesion → .....
- Invasive cancer

MIST

Age

- CIN I, II& III .....
- Ca situ .....
- Invasive cervix cancer .....
- Median Age. .....

- MC symptom-
- Most specific symptom –

Screening test

Most cost effective VIA (visual inspection with acetic acid 3-5%)

Abnormal area → White (Aceto-white areas)

d/t coagulatⁿ of malignant proteins

VILI -(Obsolete)

According to WHO -

Screen - triage - treat is better than screen - treat

.....

Started at .....

Stop screening at .....

If HIV poitive, start screening at .....

HPV → +ve - another screening test → +ve → Treat

screening test → -ve → Repeat HPV at 24 months

VIA + HPV DNA Every 5 yrs. >

HPV DNA if - ve → repeat after 5 yrs. >

VIA if - ve, repeat after 3 yrs. >

Active space

If HPV DNA + ve & VIA - ve → Repeat HPV After 3yrs

- According to ACOG - Pap smear
- by George papanicolaou
- Old Fixative - 95%ethyl alcohol (old)

New →

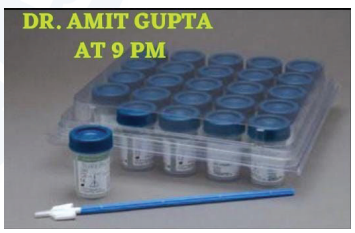
Expensive →liquid based cytology

- Cytobrush (Cytobroom)
- Fixative →liquid methanol

Sample taken in liquid.
advantage: 1. it can be taken when patient is bleeding
2. Same sample May be used for HPV testing

ACOG - Pap smear started at 21 year. from 21-29yrs. Once in 3 year
29 to 65yrs→Co test (PAP+HPV DNA- every 5year)> PAP (every 3 yrs)

- If last 10 years report normal →can stop at 65yrs
- But if abnormal continue till 75 yrs
- In HIV positive- annual screening



Sequential order for collection of pap smear
A. Use posterior vaginal wall retractor
B. Take the sample
C. Make smear on slide
D. Fix the smear

Active space

Instruments used in pap smear are
A. Cusco speculum
B. Cytobroom or ayre spatula etc.
C. Fixative

- Pap's smear - report

1. If normal - Reassurance & Follow Screening protocol
2. Unsatisfactory (If transformation zone is not visible) →repeat after 2-4 months (3 months)
3. Infection- Treat & repeat after 2-4 months (3months)

Diagnosis

- HPV
- Herpes simplex
- Gardnerella
- Trichomonas
- Candida

Can't be diagnosed – Chlamydia, N. Gonorrhoea

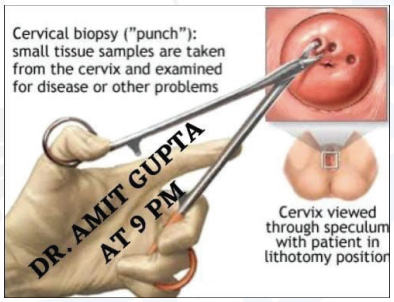
4. Atypical cells of unknown significance (ACUS)

   < 25yrs. → Repeat PAP after 1 year

   ≥ 25yrs. → HPV DNA

5. Atypical glandular cells

   - Endometrial biopsy
   - Endocervical curettage
   - Colposcopy



6. If LSIL

   If <25yrs. → Repeat PAP after 6 months - 1 year

   ≥ 25yrs. → Colposcopy

   A. If lesion is visible – Punch Biopsy

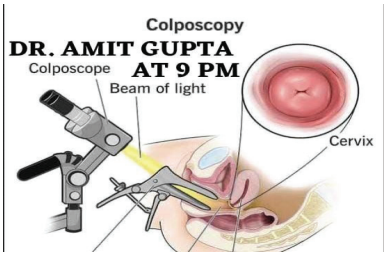
   B. If nothing is visible – coloscopic Biopsy from acetowhite areas→ OPD procedure

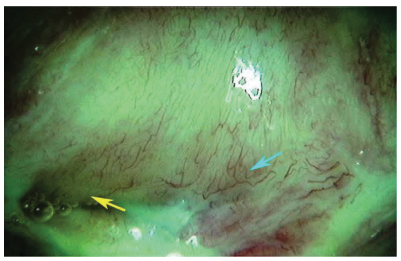
   Local anesthesia/ no anaesthesia

   Has inbuilt ..... filter - Helps to see Vascular pattern.

Microsope Focal Length - .....
Max. magnification - .....

Active space

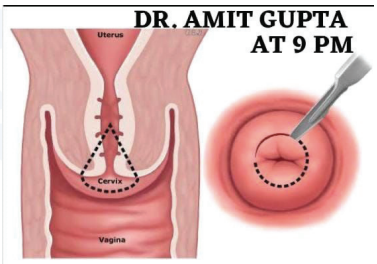




C. If nothing is visible by coloposcopy – Conization → cone Shaped piece Is cut including → Endocervix, squamous columnar junction, ectocervix

IPD Procedure

G.A



7. HSIL →Careful P/S exam
   a. If lesion visible - punch biopsy+ .....
   b. If nothing is visible → colposcopic biopsy+ .....
   c. If nothing visible even by colposcopy →Conization + ...

- MC route of spread-
- Sentinel lymph node – .....
- MC L.N. – .....
- Viral protein required for attachment to epithelium- .....
- Viral protein for replication- .....
- Viral protein for malignant transformation- .....
- E6 Suppress .....
- E7 suppress .....
- Vaccines are prepared from .....

Most important prognostic markers :

stage > Lymph node Status

- M/C route of spread : Direct > lymphatic

Active space

-M/C Site of distant Spread : lungs

Staging----- clinical

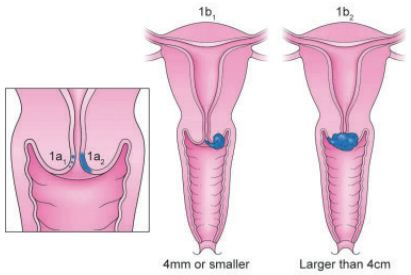
Stage 0 :

Stage I:

I) A

a. A1 → mm

b. A2 → mm

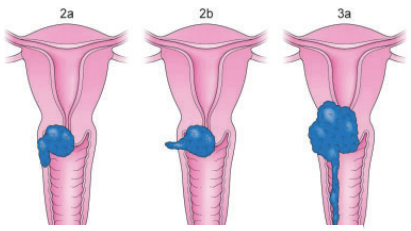


I) B

a. B1 → mm cm

b. B2 → cm cm

c. B3 → cm

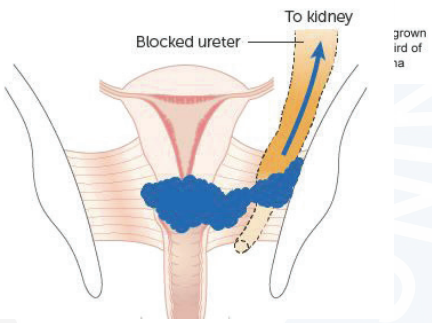


Stage II:

A →

a. A1

b. A2



B → Upper 2/3 Vagina +

Parametrium - all ligaments which surround uterus are together called parametrium

Stage III:

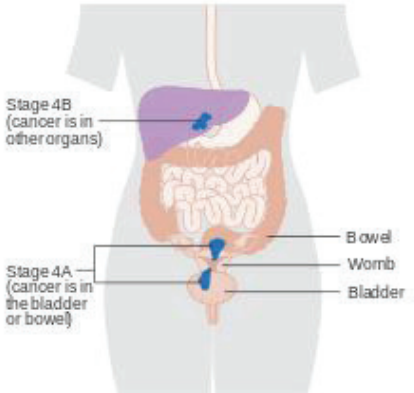
1) A =

2) B = " " +

3) C

a. C1 -

b. C2-



Active space

Stage IV:

A = Metastasis

B = Metastasis

Complications- MC cause of death

- MC- ----------
- 2nd MCC ----------

Others - Pyometra

MC cause of pyometra .....

Mc cancer causing pyometra .....

Other causes

Prevention by vaccine

- Age 9-26 yrs
- Ideal 11-12yrs
- High risk - 27-45 yrs
- WHO age recommendation

9-20 yrs - 1 or 2 doses

>= 21 yrs - 2 doses

HIV positive - 3 doses

- Most effective - CERVIRAX → bivalent-HPV16, 18

0, 1, 6

- India's first cancer cervix vaccine - cervavac

Active space

Quadrivalent -6, 11, 16, 18

Prepared in serum institute of india, Pune

Other

- GARDASIL- Quadrivalent
- HPV 6, 11, 16, 18
- 0, 2, 6

MC complication – syncope

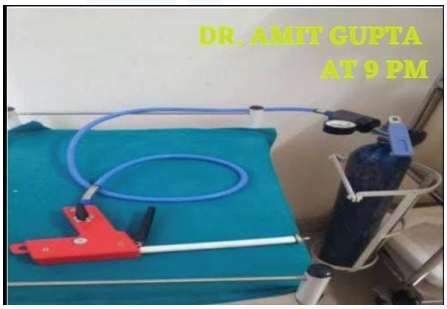
C/I – Preganacy, hypersensitivity

Mx

1. CIN I - risk of malignancy - 1%

   →Observation .....

   If dosen't regress and progress .....



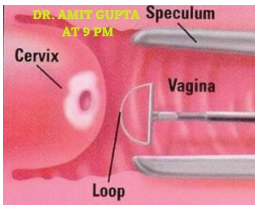
2. CIN II AND III

   Risk of malignancy - in CIN II - 5%

   Risk of malignancy - in CIN III - 20%

   - LEEP- (loop electro surgical excision procedure) or (Large loop excision of tranformation zone)
   - LEEP Electrodes
   - OPD procedure
   - Cut and coagulate at same time

- Upto 10mm depth can be destroyed



3. CIN Extending to lateral fornix- .....

4. Management of Recurrent CIN .....

Principle of Surgery in Cervical Cancer

- If < 2 cm → Vaginal trachelectomy
- If < 4 cm → Surgery including abdominal trachelectomy

5. Ca in situ/IA1 without lymphovascular invasion
   - Young/ nulliparous→ .....
   - OLD/Multiparous→ ..... + ...
6. 1A1 with lympho-vascular space invasion / IA2
   - Young/ nulliparous→ ..... trachelectomy (removal of cervix & parametrium)
   - OLD/Multiparous→ .....
7. IB1
   - Young/ nulliparous→ .....
   - OLD/Multiparous→ .....

8. IB2 → .....

9. IB3 → .....
10. IIA1 - Radical Hysterectomy

Active space

11. IIA2 onwards → chemoradition – cisplatin – Radiosensitiser

Pelvic & Paraaortic Lymph node dissection is done with all sugeries except in .....

Post of management depends on .....

i. No risk factors .....

ii Intermediate risk factors .....

a. Size .....

b. ..... involvement

c. ..... invasion

iii High risk factors .....

a positive .....

b positive .....

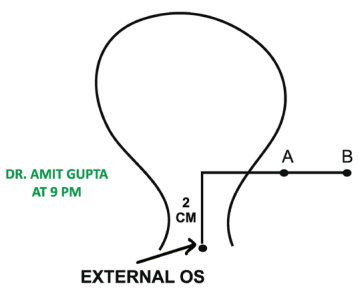
c ..... involvement

Start with external beam radiotherapy with .....

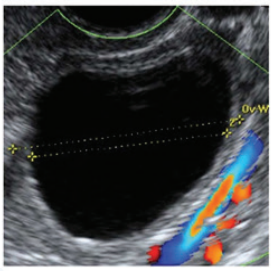
In brachytherapy with .....

I. Point A. ..... cm above & ..... cm lateral to ..... os

II. Point B. ..... cm above & ..... cm lateral to ..... os



Active space



Ovarian Tumor

Ovarian Functional mass

MC.....

Others - corpus luteal cyst -resolves spontaneously

Theca leutin cyst in - Multifetal pregnancy and Molar pregnancy

Ovarian tumor

Risk factors :-

MC- .....

Others-

- Nulliparous
- Early menarche
- Late menopause
- Obesity
- PCOS,
- Workers in asbestos factory
- Dysgenetic gonads

Protective factors -

- OCP
- Multiparity

Active space

- Pregnancy
- Lactation
- Hysterectomy
- Tubal ligation
- Physical Exercise
- Smoking

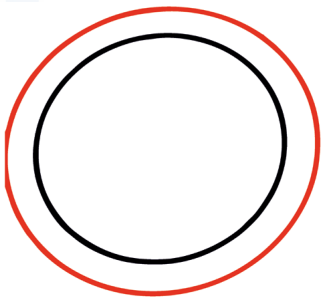
- Hereditary ovarian tumor:
- Mutation- BRCA-1 (chr 17)
  BRCA-2(chr13)
- HNPCC syndrome or Lynch II syndrome
  Hereditary Non- polyposis colorectal cancer
  (colon , breast, endometrium , ovary)

- Best prevention: Prophylactic bilateral Salphingo-oophorectomy by 35 yrs > 40 yrs

- No universal screening.
- Best screening protocol → USG →TVS>TAS. + CA 125

Active space

BENIGN	MALIGANAT
1.-Unilateral 2.-Unilocular 3.<..... 4.Cystic 5.Pain 6.NO septae 7.NO papillary projection 8. More in reproductive age 9. NO ascites 10. NO lymphadenopathy	1. -Bilateral 2. -multilocular 3. -> ..... 4. -solid areas 5. -no pain 6. -mutiple septae 7. -Most important..... 8. -More in extremes of age 9. -Ascites 10. Lymphadenopathy
11. Peripheral vascularity	11. Central vascularity
CT	
12. no omental caking)	12. Omental caking



Classification of ovarian tumor:- approx..

- Most common =epithelial 80 - 90%

  Germ cell 10%

  Sex cord 3-5%

  Metastasic (secondary) 1-5%

Active space

Types

Epithelial MC- 1. ..... Cystadenoma 2. Mucinous Cystadenoma 3. Brenner 4. Endometroid 5. Clear Cell Ca MC in Perimenopausal Mostly bilateral Worst Prognosis	Germ Cell 1. MC - 2. Benign - ..... Malignant - ..... 3. 2nd Mc Malignant - ..... 4. Choriocarcinoma 5. Embronal Cell Ca 6. Mixed 7. Endodermal Sinus or Yolk Sac tumor 8. Polyembryoma Mc Age - 10-30yrs. Mostly unilatereal	- Sex-cord tumor - Reproductive - ..... MC- Granulosa Cell tumor - Estrogen 2. Thecoma 3. Fibroma 4 Sertoli/Leyding 5 Arrhenoblastoma or androblastoma 6 Hilus Cell Best Prognosis

Management of ovarian cyst

I. In non pregnant female

A. In Reproductive age

a. 3-5 cm .....

b. 5-7 cm - follow up with .....

c. > 7 cm ..... due to risk of .....

B. In Pre pubertal age - check .....

..... after further investigations if required

C. in post menopausal age - check .....

..... after further investigations if required

II. In pregnancy

A. If symptomatic - ..... Irrespective of gestational age

B. If asymptomatic -

a. in 1st trimester .....

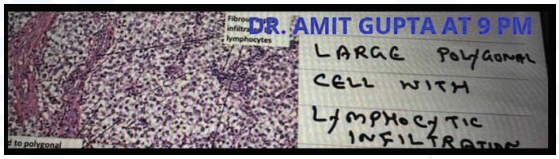
b. in 2nd trimester - If size .....or features of .....

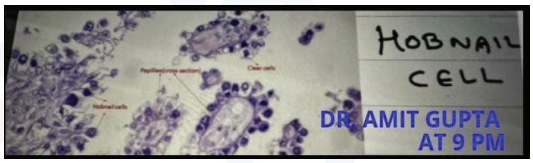
.....

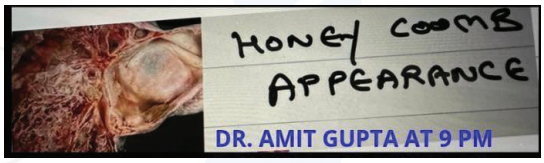
Risk of malignancy index (RMI)

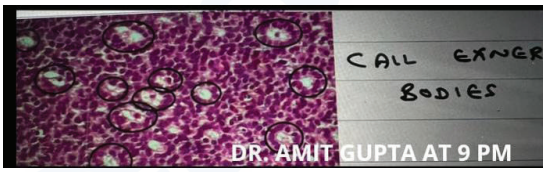
- ultrasound score x Menopausal statusus x CA-125
- if RMI > 200 -high risk of ovarian malignancy

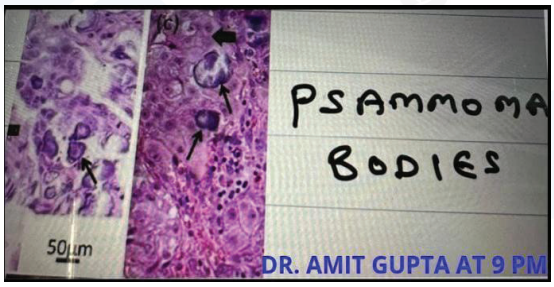
Active space

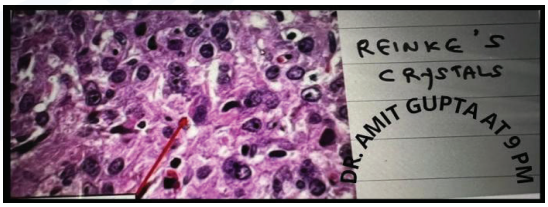


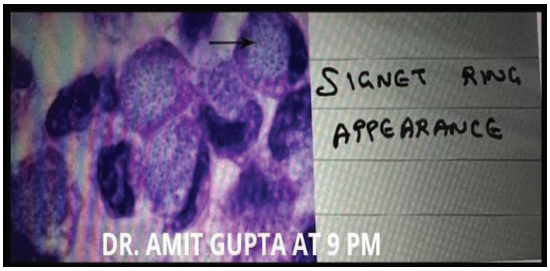






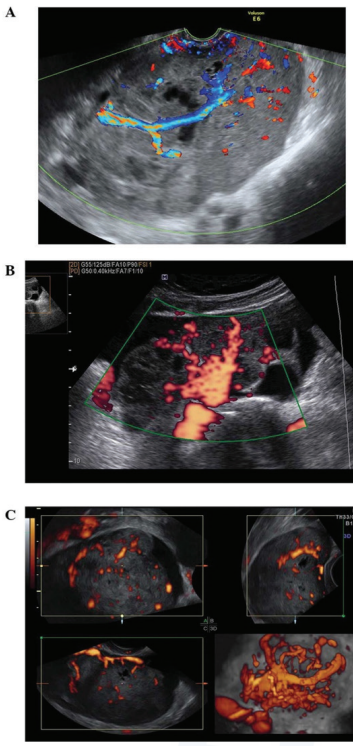


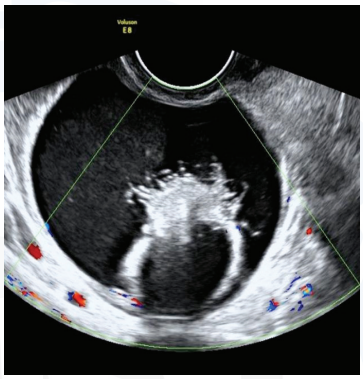




Active space

MIST





Which radiological investigation for early diagnosis of ovarion cancer- .....

Which radiological investigation is done in known case of ovarian ca - ... .

on post op survillence

.....

Investigation - .....

IOC for Adnexal mass :- .....

Tumor Markers :-

1. Serous epithelial- ca125
2. Mucinous epithelial- ca 19-9, cea
3. Dysgerminoma- LDH, placental alkaline phosphatase
4. Endodermal sinus tumor- afp, antitrypsin
5. Choriocarcinoma- hcg

Active space

6. Embryonal cell ca - hcg and afp
7. Granulosa cell tumor :- inhibin

Histopathological Findings-

1. Serous epithelial tumor – psammomma bodies
2. Mucinous epithelial- honey comb appearance
3. Bernner tumor- walthard cell nest and coffee bean nuclei
4. Clear cell ca – hobnail cells
5. Dermoid – hair, teeth, nail, cartilage, calcification
6. Dysgerminoma – large polygonal cell with lymphocytic infiltration
7. Endodermal sinus tumor- shiller – duval bodies
8. Granulosa cell tumor :- call exner bodies
9. HILUS cell tumor :- reinke's crystals
10. Krukenberg tumor :- signet ring apperance

Most common functional ovarian cyst - follicular cyst important pathognomonic features of ovarian malignancy-

Papillary Projection (Most Pathognomonic)

solid areas

Thick Multiple Septae

Most common cause of ovarian tumour - increased number of ovulation

Most common benign tumour - dermoid

Most common benign epithelium tumor - serous cystadenoma

Most common malignancy -serous cystadenocarcinoma

Largest ovarian malignancy- mucinous cystadenocarcinoma

Most common benign tumor in pregnant female - dermoid

Most common malignant tumour in pregnant female - dysgerminoma

Most common solid benign tumour - fibroma

Most deadly, 100% unilateral or most rapidly growing - endodermal

Active space

sinus tumor or yolk sac tumor

MC tumor associated with endometriosis - clear cell T > endometroid

MC tumor associated with endometrial cancer - endometroid, granulosa cell

Germ cell T with best prognosis - Dysgerminoma

Germ cell T with worst prognosis - Yolksact tumor

Only radiosensitive ovarian T - Dysgerminoma

MC Site for Malignancy in dermoid - Rokitansky Proturbance

Pseudomyxoma peritonei – mucinous material in peritoneal cavity

Most common cause-carcinoma of appendix other causes-

Mucinous cystadenocarcinoma

Mucinous cystadenoma

Carcinoma large intestine

Meig syndrome - any 1 - Benign Fibroma, Thecoma, Brenner, Granulosa cell T with

ascites & right sided pleural effusion

Most common complication of ovarian tumours - torsion

Chemotherapeutic drugs used in management of ovarian tumours

carboplatin

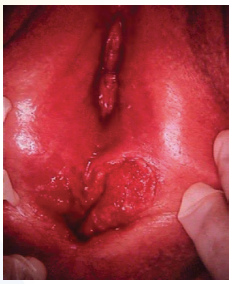
paclitaxel

cisplatin

Pseuodmeig syndrome - Ascites & right sided hydrothorax with any other condition Eg. Fibroid, Mucinous T of ovary

Active space

Ca Vulva



Most common risk factor- ----------

Other risk factors .....

VIN, Smoking, HIV, Vulval warts, Vulval Dystrophy, Paget's ds, p53 mutation

- Most common age group = ----------
- Most common symptoms = ----------
- Most common Histopathological variety - ----------
- Most common site - ----------
- Most important prognostic criteria .....

Sentinel Lymph Node

.....

Mx Stage I, II - ..... + Inguinofemoral Lymph node dissection

Stage III, IV - .....

AUB-O :

Premenopausal AUB-O or metropathia haemorrhogica or ..... Disease

Amenorrhoea ..... followed by .....

Histopathological finding- ..... endometrium or Cystic glandular hyperplasia (cystic dilatation of .....)

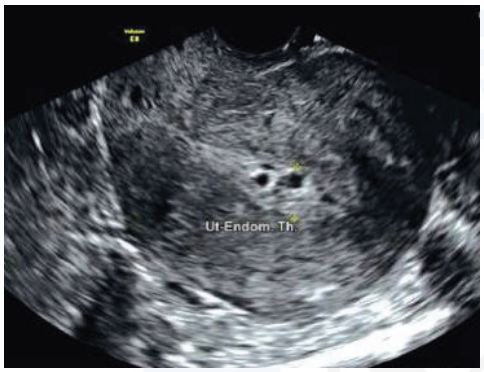
Simple Endometrial Hyperplasia without Atypia

MC age - 40-45yr

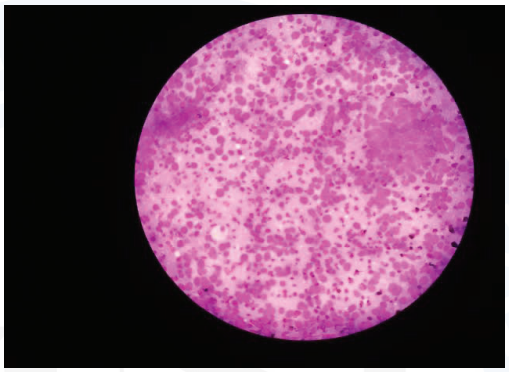
Anovulatory Cycles - Ovaries show cyst

Mangment: - If H/O irregular bleeding = medroxy progesterone acetate

If acute bleeding = norethisterone acetate



USG



Histopathology

Ca Fallopian Tube

MC age -

Malignancy - MC - ..... Adenocarcinoma in .....

..... thirds of fallopian tube

Earlisest / MC symptom → Watery Discharge P/V

> Irregular Bleeding P/V

P/V finding - Adnexal Mass

MC done investigation → USG - Adnexal Mass

Management

Surgery according to stage f/b radiotaerapy or chemotherapy

Active space

RH INCOMPATIBILITY

Rh Incompatibility (Immune Hydrops Fetalis)

- Landsteinor and Wenor-discovered Rh antigens on surface of RBC of Rhesus monkey.

(Rh) - Rhesus antigen - CcDEe →most immunogenic →.....

- If D is present- RhD positive

If fetus is Rh D positive

Fetal hemolysis

↓

Hypoxia

↓

Anoxia

If mother is Rh D negative

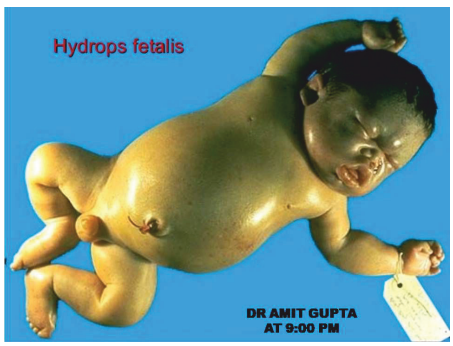
IgM is large so unable to cross placenta

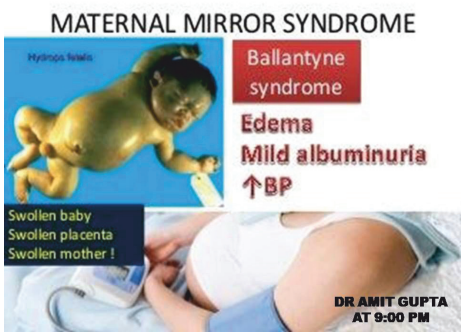
First Pregnancy remain .....

In next pregnancy

↓

IgG smaller



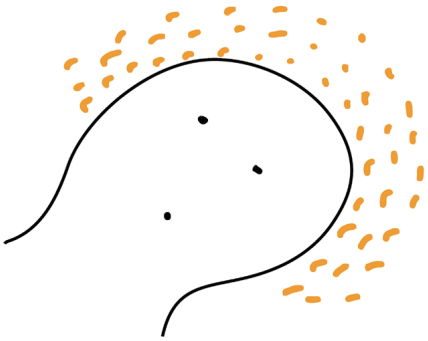


Active space

Effects of Anoxia

- size of placenta increase
- Damage to fetal liver
- Size of fetal liver & spleen increase - for extramedullary erythropoiesis - .....
- Polyhydramnios - Golden yellow
- Hydrops fetalis- If >= 2 of following are present

1. Skin edema
2. Ascites
3. Hydrothorax
4. Pericardial effusion



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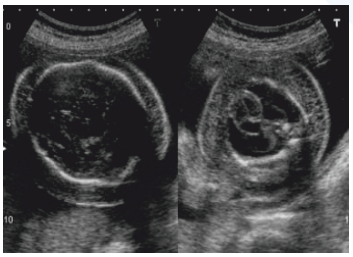
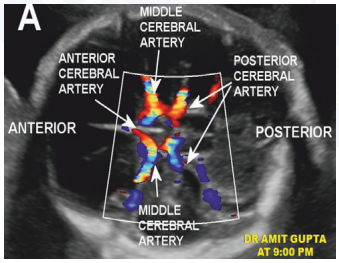
Amniotic fluid -

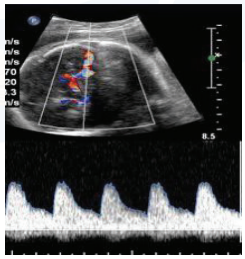
Baby -

Earliest evidence of Hydrops ..... → .....

Risk of PIH →If maternal edema develop → ..... Syndrome or ..... Syndrome







Diagnosis:-

1) Indirect coombs test (ICT)

- ICT is done at 12, 20, 28 weeks → Mandatory at ..... because maximum risk of feto maternal hemorrhage is after..... weeks

Active space

MIST

- ..... fetal blood is sufficient to trigger immune response in mother

- Antibody titre = upto ..... - safe titre

- If ..... iso immunization- (..... titre)

1. If ICT – ve (Rh unsensitized Pregnancy) → give anti-D → ..... (..... IU)

   First dose of Anti - D (Antepartum Prophylaxis) is given at .....

   300 microgram Anti - D can neutralise ....

2. If ICT is +ve (Rh sensitized Pregnancy)

   - If gestational age >= ..... weeks→ .....

   - If gestational age is ..... weeks→ Monitor antibody titre every ..... week→ If ≥ 10IU/L → Old (Amniocentesis → Liley's graph)
   - Latest→ Fetal middle cerebral artery color doppler
   - Increase in peak systolic velocity
   - If increase is .....→ Repeat every 2 week till 37 week

   - If >= 1.5 Multiples of Median (MOM) → if gestational age is >= ..... weeks .....

   - If gestational age is < .....

Active space

If fetal Hb % is >= ..... or hematocrit >= .....→

If fetal Hb % is < ..... or hematocrit < .....→ Intrauterine blood transfusion

After delivery → rosette test - Detects whether fetomaternal haemorrhage is less than ..... or ..... .

- If fetal blood < ..... ml→Anti- D 300 μg (1500IU)

  1 μg = 5 IU

- If > ..... ml→ Kleihauer betke test or acid elution test

  Reagent - citric acid phosphatase→ HbF is resistant to citric acid

For additional 1 ml .....

→ ..... Anti - D is added

30 ml is fetal blood is neutralised by 300 microgram Anti - D

1 ml is fetal blood is neutralised by .....

- Direct coombs test → (DCT) In newborn
  - Hb %
  - bilirubin
  - Blood group

- If DCT is -ve & fetus is Rh +ve →2nd dose of anti-D is given within ..... by

  I/M injection in deltoid (can be given upto .....)

  (Post-partum prophylaxis)

- If abortion occur in Rh -ve lady before 12 week- ..... Anti-D is given

Active space

Non immune hydrops fetalis -(MC hydrops fetalis)

MC - Cause .....

2nd MC .....

Other causes

.....

Cystic hygroma

Twin twin transfusion syndrome

Umbilical vein thrombosis

G.I. Disorder

Renal disorder

Rubella

HIV

NIHF may be caused by chromosome defect, rubella, HIV and not by ..... and HSV

Active space

MATERNAL PELVIS

Maternal Pelvis

- Divided into False pelvis and True pelvis by Imaginary line at pelvic brim – linea terminalis Brim (Inlet)

False Pelvis - ..... linea terminalis

Supports .....

True Pelvis - ..... linea terminalis

True Pelvis - 3 Parts

I. - Inlet

II. - Mid pelvis (Obstetric outlet)

III. - Anatomical Outlet

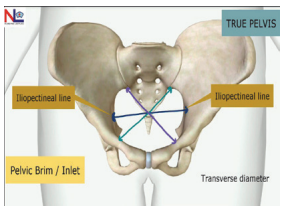
Pelvic Cavity - Between .....

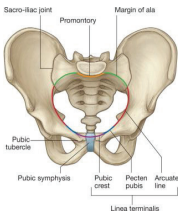
Situated at .....

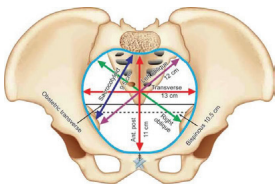
..... Part of pelvis - plane of .....

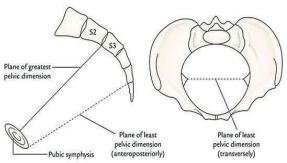
.....

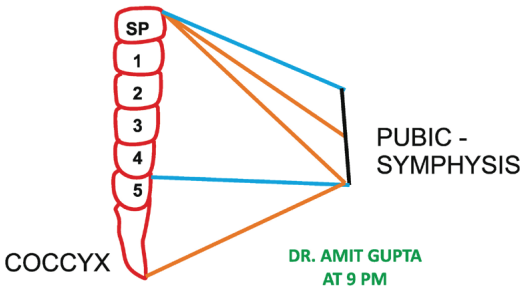
Active space

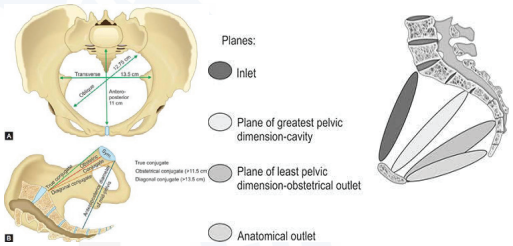


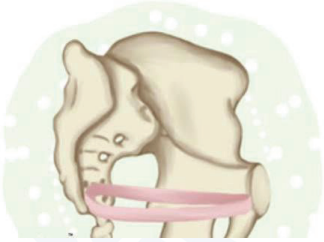


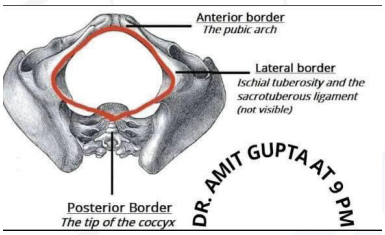


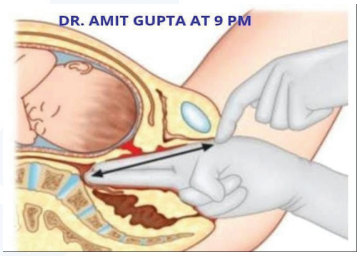












I. Inlet - Anteroposterior diameters .....

1. True conjugate -.....
   - Between midpoint of upper margin of sacral promontory to ..... border of pubic symphysis

2. Obstetric conjugate – ..... cm
   - Between midpoint of upper margin of sacral promontory to ..... pubic symphysis

Active space

3. Diagonal conjugate .....
   - Between midpoint of upper margin of sacral promontory to ..... border of pubic symphysis
   - Can be measured .....

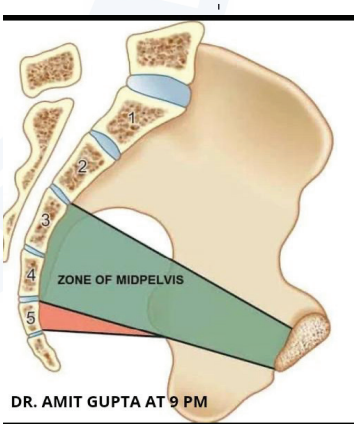
4. Transverse diameter of inlet .....
   - Between two farthest point on ileopectinal line

5. Oblique conjugate of inlet .....
   - From sacroilial joint on one side to opposite ileopubic eminence
     Rt. Oblique is from Rt. S.I. Joint

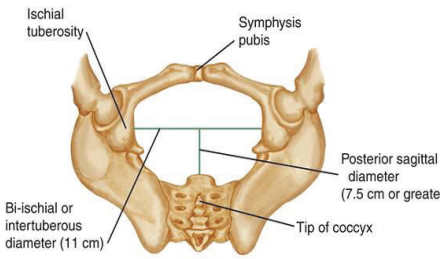
II. Mid pelvis (Obstetric outlet)

1. AP diameter (.....)-between lower border of pubic symphysis to junctions of S4-5 vertebrae
2. Transverse diameter (.....) -between two ischial spine (interspinous or bispinous)

Plane of .....



III. Anatomical Outlet-AP diameter- between lower border of pubic symphysis & tip of cocyyx



View from below (woman is in lithotomy position)

Active space

MIST

1. Anteroposterial diameter -.....cm
2. Transverse diameter- .....cm
   - between ischial .....
   - bituberous or intertuberous

Obstetric A.P. of outlet (at delivery) may reach upto 13 cm.

Posterior Sagittal diameter of Anatomical outlet - between tip of coccyx to intersection

point between AP & intertuberous diameter of outlet- 7 cm

Sacrocotyloid diameter - from centre of .....

to Ipsilateral Ileopectineal .....

- Smallest diameter(most important)of pelvis- .....
- Largest of pelvis -.....
- Can be measured clinically -.....

Angle of inclination - betbeen pelic ..... and .....

Subpubic angle - made by descending rami of .....

In Males .....

In Females .....

Clinical assessment of pelvic diameters is called .....

Done in primi - .....

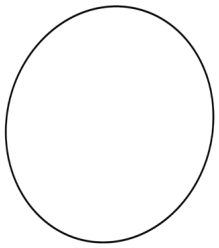
Done in multi - .....

Active space

Types of pelvis→Caldwell moloy - on the basis of shape of .....

1. Gynaecoid -50%

   MC - Inlet- .....



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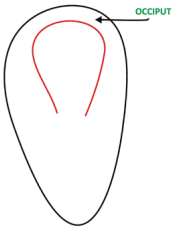
2. Anthropoid- 25%
   - if ..... is much larger
   - ..... like pelvis



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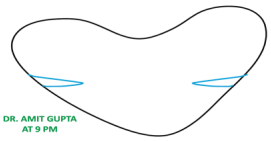
3. Android -20%
   - Fore-pelvis is much smaller

   - Prominent .....
   - ..... shaped
   - .....



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4. Platepelloid pelvis-5%
   - if ..... daimeter is much larger
   - flat pelvis /plate shaped/ bowl shaped

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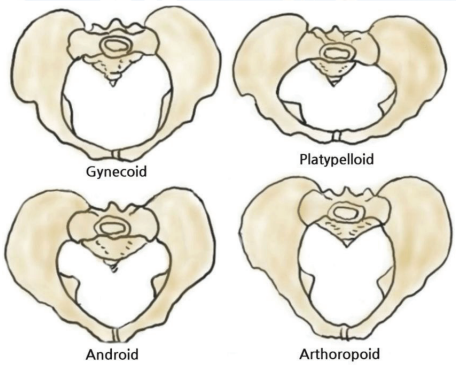


Active space

Contracted pelvis - .....

For diagnosis of

A. Inlet contraction - If obstertric conjugate is .....

B. Mid Pelvis contraction - If Interspinous diameter is .....

C. Anatomical outlet contraction - If intertuberous diameter is .....

.....

Clinical assessment of inlet contraction.....

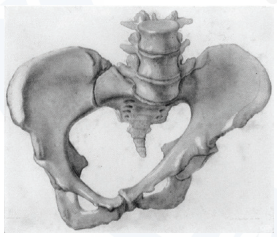
Clinical asessment of mid pelvis contraction - If two fingers of one hand touch .....

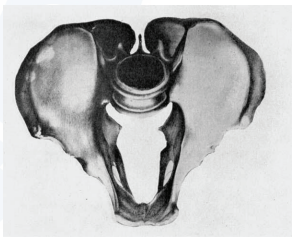
Clinical asessment of anatomical outlet contraction - If not able to introduse ..... of hand betbeen .....

1. Congenitally contracted pelvis

a) Naegle's pelvis - if .....

b) Robert's pelvis -if .....

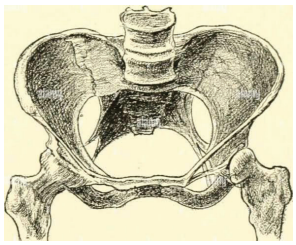


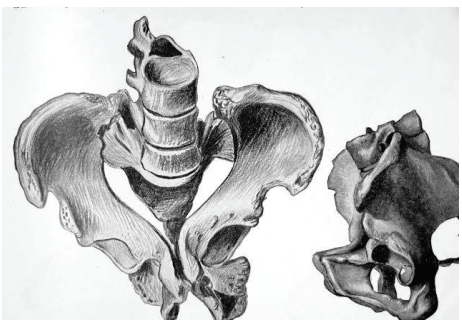


2. Acquired contracted pelvis

a) In children in severe rickets – .....

b) If Adult in osteomalacia→ .....



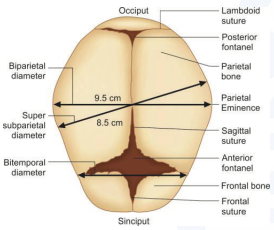


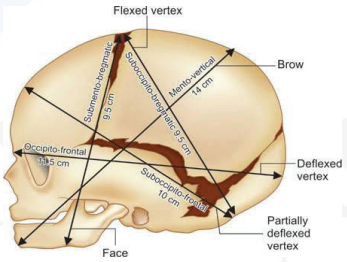
Active space

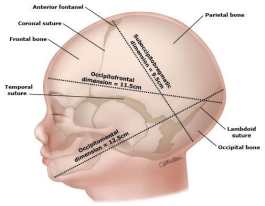
- Best investigation-MRI
- Management - Caesarean section

Cephalopelvic disproportion

- Size of fetal head & Pelvis are out of proportion
- Best dx- .....
- Clinically dx- .....→may be be successful .....

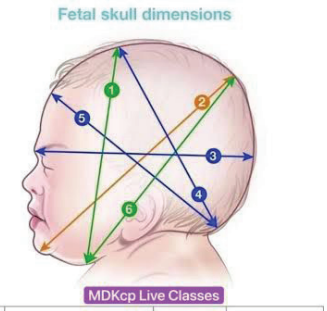






Transverse diameters of fetal skull

- Bimastoid .....
- Bitemporal .....
- Super subparietal .....
- Biparietal ..... cm (MC transverse engaging diameter)



	Engaging diameter	Length	Presentation
1	Submentobregmatic	9.5cm	Face
2	Mentovertical	14cm	Brow
3	Occipitofrontal	11.5cm	Vertex
4	Suboccipitobregmatic	9.5cm	Vertex
5	Suboccipitofrontal	10cm	Vertex
6	Submentovertical	11.5cm	Face

Longitudinal engaging diameter

MC Longitudinal engaging diameter

- Head is .....
- Presenting part - .....

MIST

Largest Longitudinal engaging diameter – .....

Head is in .....

Presenting Part .....

Management .....

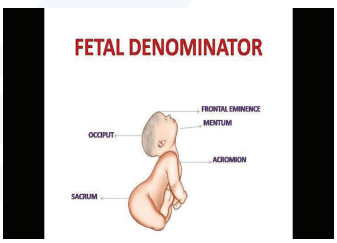
Presentation - Part of fetus which occupies lower pole of uterus after .....weeks

- MC -..... 96.5%
- Podalic or ..... - 3%
- Transverse 0.5%

Presenting part

- Part of ..... which directly ..... internal OS
- MC-Vertex, brow, face

Denominator -Arbitrary bony prominent point ..... part



In Vertex→.....

In Brow →.....

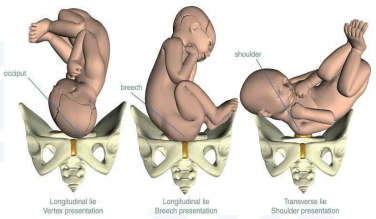
In Face

- .....

Active space

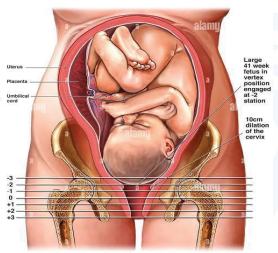
Lie

- Relationship between maternal and fetal spine (Long Axis)
- Mc longitudinal .....
- Transverse .....
- Oblique .....



Station –

- Relationship between presenting part and ischial spine
- At Ischial spine – ..... station
- ..... ischial spine – minus station (-)
- ..... the ischial spine – plus (+) station



Attitude - Relationship between various .....

MC .....

Well flexed head :- on P/A exam : Sinciput

on P/V exam : Occiput

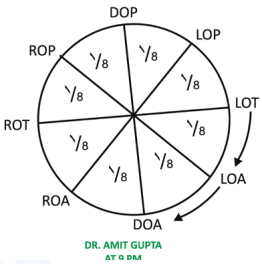
Extended head :- on P/A exam : Occiput

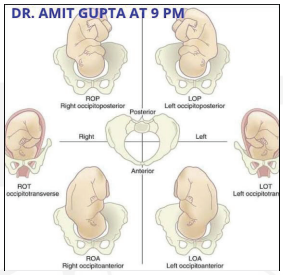
Active space

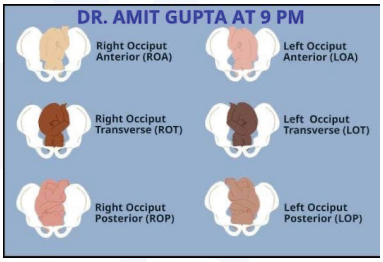
on P/V exam : Face

Position-relationship between denominator & pelvis

- MC→ .....
- MC Position in late labor .....







Unstable lie - If LIE is not fixed even by .....

MC couse -

Other causes -

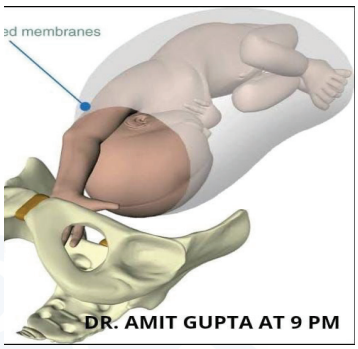
Management -

Active space

MIST

Oxytocin is .....

Compound presentation



Cord Prolapse - If umbilical cord comes below

It is an obstetric emergency

Cord can get compressed by the presenting part (labour)

Temp. Change - intense vasospasm occurs.

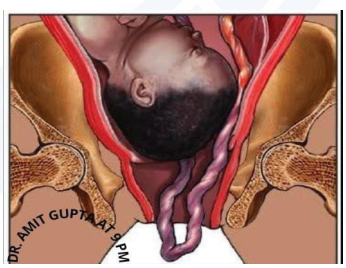
Clinical presentation

severe and prolonged fetal Bradycardia (M/C)

Variable repeated decelerations.

.....

Causes



P/V Finding .....

Active space

Management

-Mx of cord prolapse : Emergency C-Section

-Vaginal Delivery indications :

1) Absent FHR

2) Fully dilated cervix - then do instrumental delivery

Management Algorithm

Call for help / left lateral positioning of female / O2 by mask / if cord lying outside vagina - Gently hold cord with moist gauze & put into vagina - Cord Compression .....

P/V Examination - look for dilation of cervix (labor) - if fully dilated - do instrumental delivery

Manually lift up the presenting part while shifting to OT to improve fetal outcome

To prevent compression of cord.

1. Retrograde filling of the bladder

2. Trendelenburg position (head end is low, butlock are lifted up)

3. Knee chest position - Then do c-section.

Active space

Labour

Regular rhythmic contraction of uterus resultig in proressive .....
.....

Most accepted hypothesis - Due to Activation of fetal Hypothalamo - pituitary - adrenal axis (H.P.A. Axis)

- Fetal hypothalamus secretes ..... → Fetal pituitary secretes ..... → fetal adrenal secretes various mediators including DHEA-S

Effects of mediators -

a) Alteration of Estrogen- progesterone receptor sensitivity

b) Increase in sensitivity of oxytocin receptor in myometrium

c) Cervical sol → Gel component

- Hyaluronic acid .....
- Collagen .....
- Dermatan sulphate .....

Phases of Labor

Phase 0 → Ante partum {Phase of .....}

Phase 1 → Myometrial .....

(a) ↑ Estrogen, ↑ Oxytocin

(b) Progesterone

Phase 2 → ..... Stages of labor

Phase 3 → .....

Active space

Prelabor - events which begin few days before labor

A. Cervical ripening & .....

B. Uterine .....

C. ..... pains

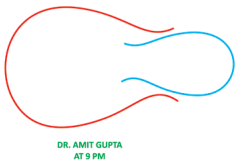
Full uterine shelving

After ..... weeks fundal height is reduced by .....
flanks are ..... (.....)

4 Stages

1. Stage I-From true uterine contraction to complete cervical Dilatation
2. Stage II-From complete cervical dilation to delivery of baby
3. Stage III-placental expulsion
4. Stage IV -..... hr observation after placental expulsion

False pain	True pain
X	- Regular and rhythmic
X	- with cervical dilation
X	- with show thick cervical mucous plug tinged with blood (expelled out)



X	- Bag of ..... dilation

- Radiating to back and thigh
- On sedative-.....
- Analgesics-

d/t pace maker - situated in tubal ostia (Rt> Lt) .....- whole uterus is depolarized in .....

Stage 1

Latent Phase	Active Phase
- Upto ..... cervical dilation	- ..... According to ACOG..... According to WHO labor care guide - from ..... According to WHO modified partogram .....
- Predominant-cervical effacement	- Cervical dilation
- Duration	In Primi -.....
- In Primi→upto.....	In Multi - .....
- In Multi →upto.....	- WHO at least .....
	- Descent of Head In primi ..... In multi .....

- Delayed

  In Primi .....

  In Multi .....

  Causes →

  - Occipito - Posterior position
  - Poor cervix
  - Excessive sedation
  - False pain

- Dealyed or slow or protracted active phase →If cervical dilatⁿ is .....

Active space

MC cause :- CPD (Cephalopelvic Disproportion)

- Occipitoposterior → wait and watch

- Management-Therapeutic

- If not- look for .....

- .....

Do .....

give .....after 30 mts. if required

check after .....

Labour :- slow progress - arrest

MC Cause of Arrest - CPD (Cephalopelvic Disproportion) - No role of instrumental Delivery

Arrest of active phase - If no cervical dilation for ..... despite of good uterine contractions

If no cervical dilation for ..... with inadequate uterine contractions & oxytocin given

Mx - .....

Retraction - ..... property of uterine muscle

After contraction muscle fibres become

....., so fetus is ..... polarity of .....

Lower Uterine segment - Formed by .....

Length - A. In non pregnant .....

B. At term .....

C. During Labour .....

Freidmann -partogram -Best method to assess ..... of labour

Most accepted partogram .....

Active space

In top part of partogram ..... condition

A.

B.

C.

In middle part of partogram .....

In lower part of partogram ..... condition

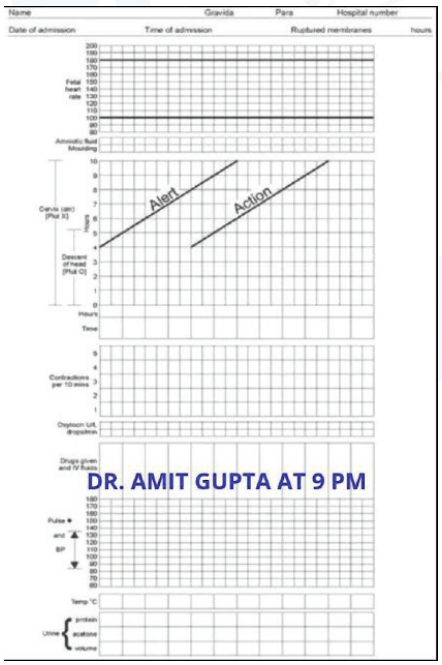
Moulding - Change in shape of ..... in birth canal

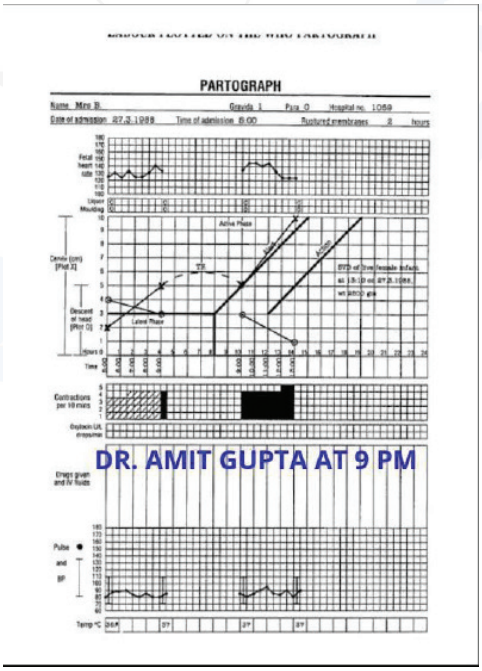
.....

..... Scalp bones ..... each other

..... Scalp bones ..... each other but can be .....

..... Scalp bones ..... each other but cannot be .....





- Cx Dilatation

- Descent of Head (In cm.)

YAxis

X → Time in Hours

Active space

- In WHO modified partogram- monitoring starts at ..... dilation.
- If alert line is crossed-..... labour
- If action line is crossed –Immediate .....
- Time interval between Alert & action line .....

Uterine contractions are monitored every 30 mins
I Square = 1 Contraction

FHR measured every 30 minutes.
I small square in FHR column = 30 min

I big square in FHR column = 1 hour

After compete dilatation of cervix.....

Stage 2- Sequence of Cardinal movments - 7

- Engagement→ Descent →Flexion →Internal Rotation→ Crowning →Extension → Restitution → External Rotation

Engagement – when max transverse engaging diameter enters pelvic brim (Inlet)

Primigravida :- Engagement usually happens at 37 wk
Multigravida :- engagement at onset of labour.

For Diagnosis -

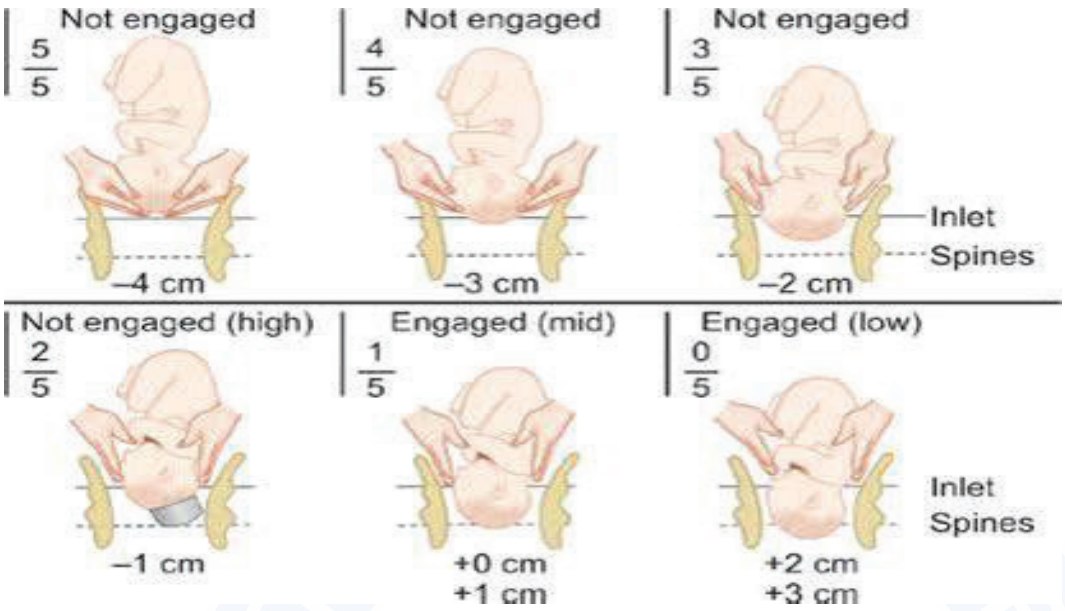
(Abdominal method ..... method .....

If ..... Head is palpable adominally .....

Vaginally - Station is .....

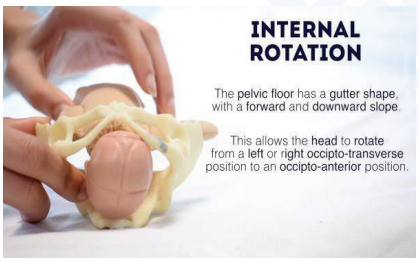
Rule of fifth 2/5 = Engaged Head

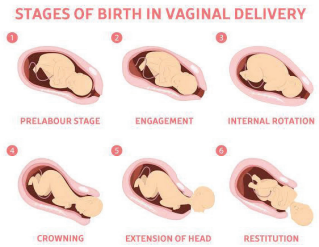
Active space

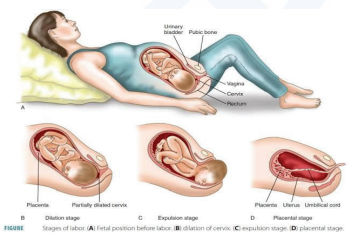


Internal rotation

- Occurs at ischial .....
- Rotation by 2/8 (90°) if LOT
- Downward, forward medially
- Occiput rotate in direction of .....





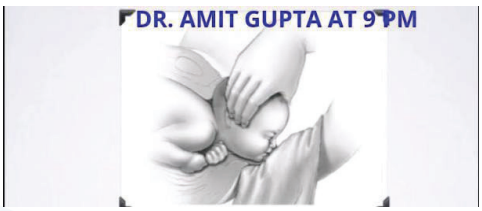


Crowning-permanent visualisation of scalp hair even when uterine contraction is over

Head should be first flexed & then extended with one hand & other hand support perineum

Active space

- WHO→ ..... maneuver
- ..... on perineum



Not mandatory -

- If episiotomy required – may be given
- Fundal pressure - may be given

Extension

Restitution -..... of neck

Initial part of .....

External rotation-..... movement of head outside due to internal rotation of ..... inside.....

Trunk is delivered by - .....

If position is LOT - after delivery neonatal face is towards maternal .. ..

Stage 3 AMTSL ..... Minutes →

Advantage - It prevents .....

- Keep baby tray below maternal level
- Most important aspect - Injection uterotonic→DOC→.....

If oxytocin is not availaible -

i. Methyl ergometring .....

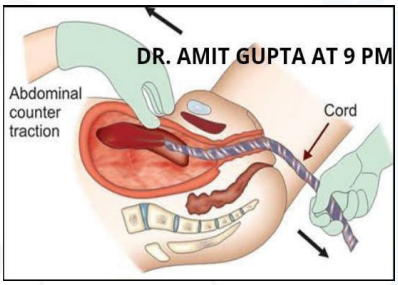
ii. Syntometrine

Active space

iii. Misoprostol 600ug

- ..... cord clamping .....
- Look for the signs of placental separation
- Abdominal- Firm, Globular, Cricket Ball Feel
- Vaginal- most important - (i).....
  (ii) Gush of blood out of vagina
  (iii) Permanent lengthening of umbilical cord

- Controlled cord ..... technique .....



Intermittent assessment of uterine tone

(Old Name - Uterine massage)

Intermittent assessment of uterine tone is done every .....

For .....

Stage 4

- ..... hr observation after placental expulsion
- Physiological chills seen

Most commonly fractured bone during birth is .....

MC nerve injury during labour is .....

Active space

Duration

1. 2nd stage -upto
   - Primi -.....
   - Multi -.....

Arrest of second stage : arrest of descent/rotation

- Primigravida - No descent for 3 hrs (When FHR is normal)
- Multigravida - No descent for 2 hrs (When FHR is normal)

If patient has taken labour analgesia : - (+1hr)

- Primigravida - No descent for 4 hrs (When FHR is normal)
- Multigravida - No descent for 3 hrs (When FHR is normal)

Duration of 3rd stage (Active management of third stage of labour - AMTSL)

- Active mx 5 to 10 min (advantage prevents post partum haemorrhage)

Most important factor in placental separation – .....

Delayed 3rd stage or Retained placenta

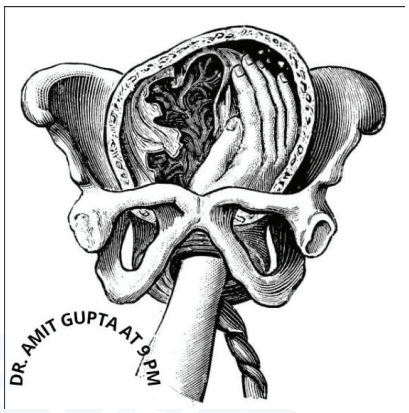
- If >.....
- Mx - first step - .....

If no responce .....in

O.T. in .....

Complication .....

Active space



Indications of early cord clamping

- Asphyxia
- Rh incompatibility (If ICT IS -Ve)
- Heart disease
- HIV +

C/I of uterotonic

- Rh incompatibility
- Maternal Heart disease
- Maternal Hypertension
- Twin-never before delivery of 2nd twin

Pressure

- Stage 1 - .....
- Stage 2/3 -.....
- Montevideo units – no. of uterine contraction in 10min x uterine contraction intensity

- Early labour T10-L1
- Late labour S2,S3,S4

Normally – 3 contractions in 10 mins- each lasting upto 45 sec

Maximum upto .....

Active space

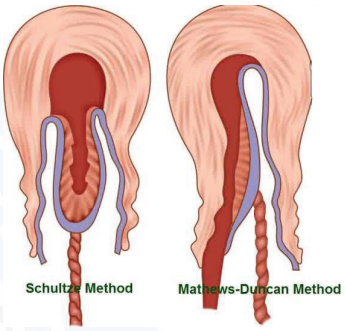
Tachysystole

If tachysystole leads to fetal distress .....

MC method of placental separation→ Schlutz Method > duncan method

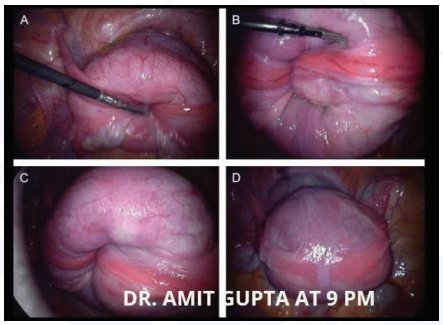
Schlutz - Placenta start separating from .....

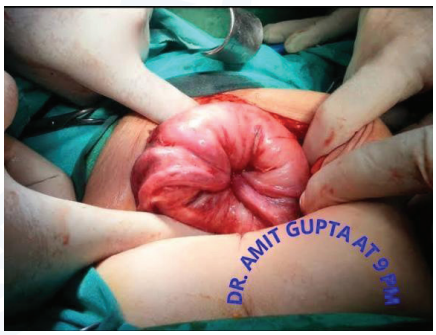
- Retroplacental haematoma forms



Uterine inversion –

- Due to mismanaged .....



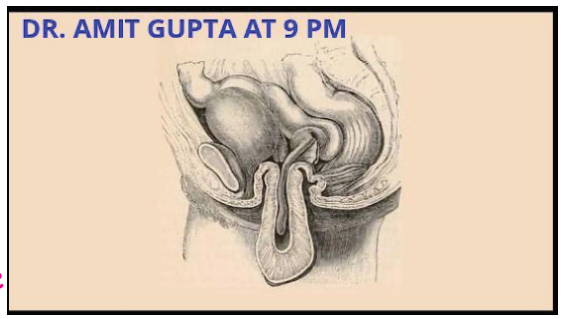


- Diagnosis : Uterus is lying outside vagina.

  P/A Examination - ..... of uterus will not be felt.

Active space

Stage



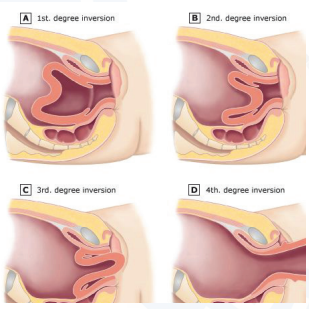
- 1°→ Dimpling of uterus
- 2°→ Upto internal os
- 3°→ Upto external os
- 4°→ Whole uterus turned inside
- Pain d/t Neurogenic shock
- Cause of death → Hypovolemic shock
  d/t excessive hemorrhage

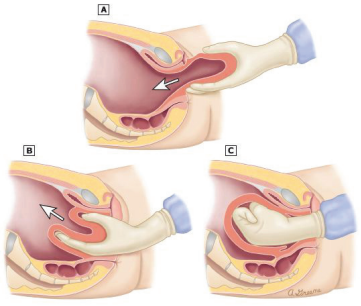
Mx - Immediate resuscitation

Manual repositioning of uterus (Jhonson's method)

→ Oxytocin → then remove placenta.

- O"Sullivan" Hydrostatic method→if no response → surgery → best→ Huntington method





Mx of uterine inversion wih attached placenta - reposition of uterus without .....

Breastfeeding should start

after Vaginal delivery .....

after CS .....

cramping pain during breastfeeding is due to .....

Active space

absolute contraindication for breastfeeding is .....

.....is not contraindication for normal vaginal delivery

MC indirect cause of maternal mortality is .....

Vagitus uterinus .....

Caul - baby is born with .....

History of Vaginal delivery 2 days ago with anterolateral thigh sensory loss. Nerve involved is .....

Active space

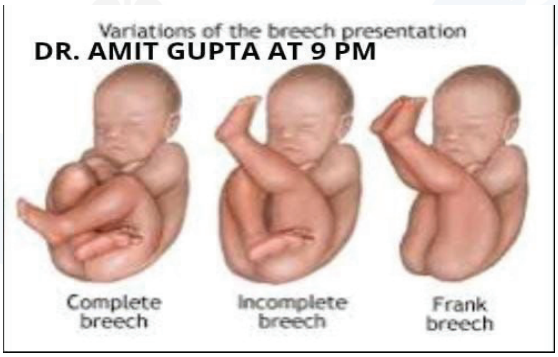
Breech

- MC Cause - .....

  Other causes-

  2. Multiparity
  3. Abnormality of liquor (Oligo/Poly) 4. Placenta Previa
  5. Gross congenital anomalies
     Usvally Hydrocephalus
     Anencephaly
  6. Uterine Anomalies

- Recureent or habitual – if H/o Breech > .....
- MC cause of recurrent breech -.....



Type of breech

1. Complete breech or flexed breech - More common in .....
   - ..... flexed and ..... flexed
   - P/v -.....

Active space

2. Incomplete breech

A. Frank or extended breech

- ..... flexed and ..... extended
- P/v .....
- More common in.....

B. Knee breech

- ..... extended
- ..... flexed
- P/v .....

C. Footling breech

- ..... and ..... are extended
- P/v .....

MC breech in Primigravida = ...
MC breech in multigravida = ...

Most dangerous variety of breech - ..... > ..... > .....

- Danger d/t risk of cord prolapse
- Safest – ..... breech
- Mx- Assisted breech delivery tried in frank > flexed breech
- Most common position- Left Sacroanterior
- Denominator -.....
- Engaging diameter - Bitroachanteric .....
- Engaging diameter of shoulder- Bis-Acromial .....
- Engaging diameter of head - Suboccipitofrontal .....

Active space

Assisted breech delivery

Mx

1 For extended leg - A. In frank breech .....

B. In flexed breech .....

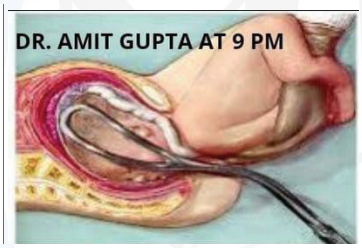
2 For extend arm -.....

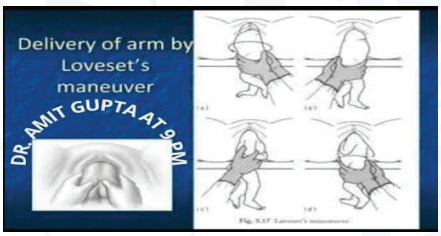
3 To deliver after coming head -

- In developed countries - MC .....
- In developing countries - MC .....

Other methods – Maurice - Smellie - Viet Method

- Bracht Method





..... technique - Wrap umbilical cord & fetus in towel once fetus is delivered till .....

If after coming head is entrapped due to incomplete cervical dilatation

A. If premature→ ..... incisions → on cervix at ..... and ..... o'clock position

B. If term - Zavenelli maneuver→Push whole body inside and do C-section

Active space

MIST

- V. Rare - if stargazer Breech -Hyperextended head
- Mx- C.S.
- V. Rare - if dorsoposterior- Modified Prague Maneuvre

Breech identified for the first time during delivery is a Contra indication for vaginal delivery. Continuous FHR monitoring is required for breech delivery during labor and it should be done by a skilled obstetrician

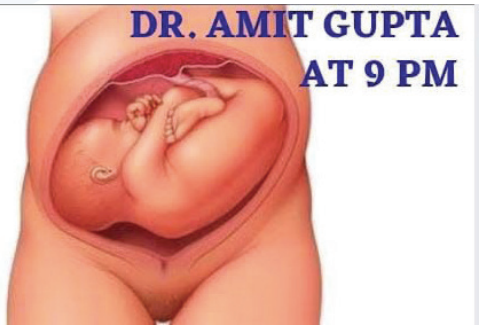
- MC cause of death.....

Zatuchni - Andros score (Breech score) - prognostic index for vaginal delivery in .....

Good score.....

Poor score.....

Transverse



- MC Cause - .....

  MC Cause at term .....

  Other causes
- Multi-parity
- Polyhydraminos
- Platepelloid pelvis
- Contracted pelvis

Active space

- CPD
- Large cervical fibroid

- Most common position- dorso anterior
- ..... shoulder

Denominator .....

Management:

A. if she is in .....

B. if she is not in .....

..... at .....

weeks under USG guidance

If not successful, may retry at .....

Max. risk of cord prolapse is in →Transverse > footling breech

Face /brow

Causes are almost same

If Cause is less severe .....

If Cause is more severe .....

MC cause of face .....

Other Causes

- Platepelloid pelvis
- Prematurity
- Multi-parity

Active space

- Tumor of fetal neck
- Numerous loop of cord around neck

Engaging Diameter :- Submentobregmatic (9.5cm) - Denominator = mentum

Management for Brow - ..... because longitudinal engaging diameter is .....

Management for Brow in early labor .....

Face

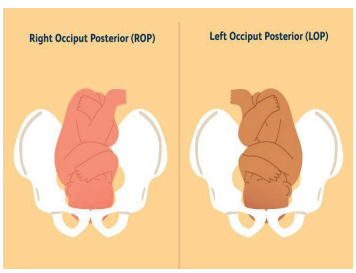
- MC position – left mentoanterior
- If mentoanterior → vaginal delivery
- If mentoposterior position – mx - C-section

Management for mentoposterior in early labor .....

Head in breech and face is delivered by .....

Malposition

- MC Malposition-.....
- MC .....
- More in primigravida

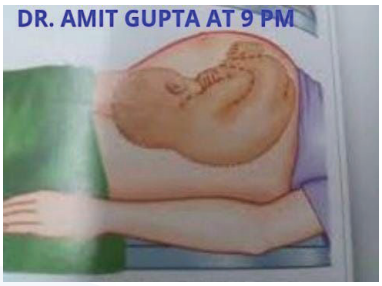


MC Cause .....

Active space

Other Causes

- 2nd MC
- Anthropoid pelvis
- Weak uterine contraction
- Placenta on anterior wall



Infraunbilical .....

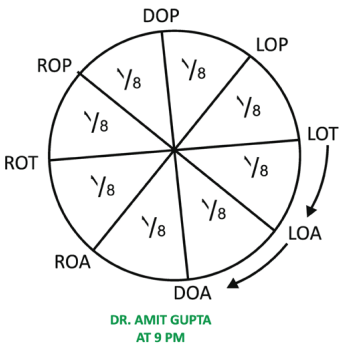
FHS heard on .....

Fetal limbs felt near midline
Fetal back towards the flanks
Anterior shoulder away from midline

Q.1 If a female presents with occipito posterior position → Mx →.....

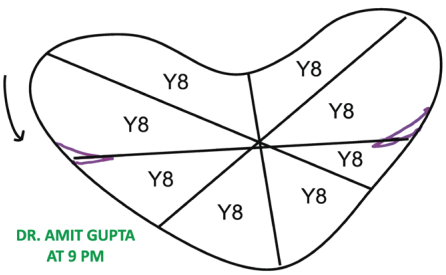
In 90%-Occiput

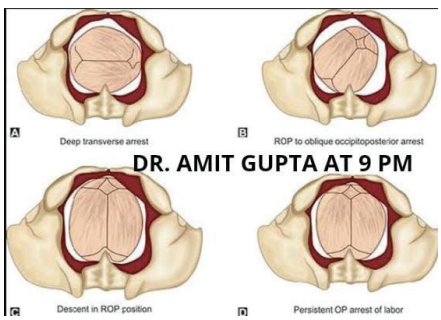
Rotate by 3/8



Q.2. Occipitoposterior position with Android pelvis

Occiput rotate 1/8 & stucked here at ischial spine for > = ...
despite of complete cervical dilation and good uterine contraction →.... .....





Mx

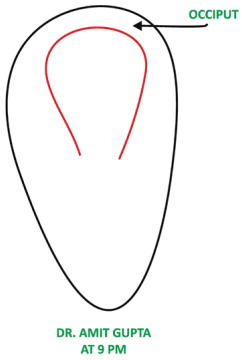
Active space

MIST

In android pelvis- .....

other pelvis- .....

.....



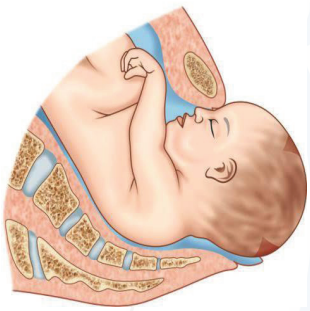
Q3. Occipitoposterior position with Anthropoid pelvis

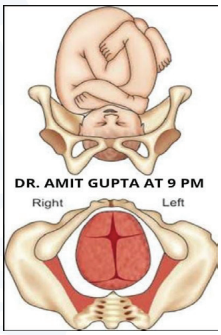
.....

If occiput is stucked at scrum for more than .....

.....

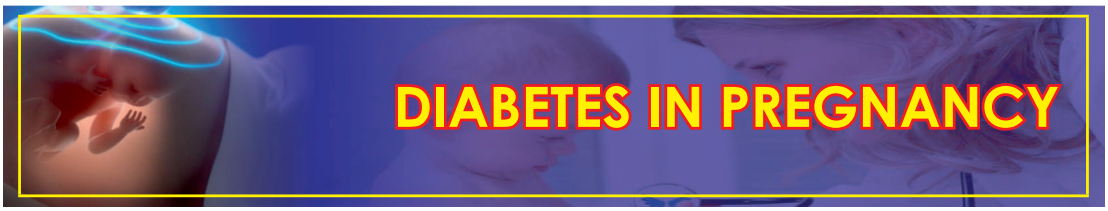
Rotation of occiput is almost impossible in .....





Management

Active space



DIABETES IN PREGNANCY

Diabetes in Pregnancy

Insulin resistance starts at- .....wk

Insuliun resistance max- ..... week

Insulin resistance is due to .....

If a pregnant lady comes for assessment - plasma glucose is checked in .....

ACOG -2 step test - Mandatory between .....

A. First step screening test - Glucose challenge test → with 50 gm glucose→check plasma glucose after 1 hr→

<.....mg%- normal

.....mg%- suspicious → Do GTT

>..... mg%- gestational diabetes

B. 2nd step Glucose tolerance test (Definitive test)

If any 2/4 abnormal .....

With 100 gm glucose

- Fasting .....mg%

Active space

- At end of 1 hr ..... mg%
- At end of 2hr – ..... mg%
- At end of 3hr – ..... mg%

-Single test (Screening & diagnostic) : 75gm OGTT (DIPSI Criteria followed)

Diabetes in pregnancy study group of india

-When & whom ? 1st ANC (All pregnant women must undergo unless she is k/c/o DM) Repeat at 24-28wk.

India- DIPSI criteria (Diabetes in Pregnancy study group of India) - 75 gm glucose irrespective of fasting in 300 ml water in 10 mts.

Check Plasma Glucose after .....

- > ..... mg%- gestational diabetes
- > ..... mg%- Overt diabetes or pre-gestational diabetes

IADPSG (International association of diabetes and pregnancy study groups)

If 1/3 Abnormal

Fasting .....

At end of 1 hr. .....

At end of 2 hr. .....

Active space

Complications

1. Maternal complications

Polyhydramnios

Preterm labour

PROM

PPH

Puerperal sepsis

Malpresentation

Abruption placentae

Cord Pralapse

Placenta Praevia

PIH

Increased risk of diabetes mellitus in future

2. Fetal complication

- Abortion /IUFD
- Macrosomia - if fetal wt. >= ..... or AC.....
  Pederson hypothesis
- Congenital Malformations.....(only in ..... diabetes)

Maximum Oxygen demand
MC period of risk of sudden still birth is 3rd trimester

Pederson Hypothesis -

Maternal .....

↓

Fetal .....

↓

Fetal .....

↓

.....

Active space

Congenital malformations - only in ..... diabetes

- MC system -.....
- 2nd MC system - .....
- Rare but Most specific cardiac malformation- .....
- Rare but MC cardiac finding .....
- Rare but most specific - ..... Regression syndrome or ..... agenesis
- Best method to Diagnose congenital malformation- TIFFA Ultrasound (targeted imaging for fetal anomalies or .....)
- Best method to predict congenital malformation-.....

Neonatal complication

- Most common - .....
- 2nd most common- .....

Polycythemia

Hyperbilirubinemia

↓S. K+, ↓Ca++, ↓Mg++

GDM

A1 - Gestational Diabetes Mellitus Controlled by Diabetic diet alone

A2 - Gestational Diabetes Mellitus Controlled by Pharmacotherapy

- Mx - Initial advice

>= 140

MIST

Active space

MNT (medical nutrition therapy) + Exercise (30mins) x2 wks

>=200
Start 8 units insulin

Termination of pregnancy
Don't deliver well controlled diabetes at less than 39 wks

A1 GDM :- well controlled on diet
- at 39-40^{+6} wks.
A2 GDM :- well controlled on insulin- at 39-39^{+6} wks.
poorly controlled on insulin - at 37-38^{+6} wks.

Shoulder dystocia – if time interval between delivery of head & shoulder.....

1. Clinical suspicion by .....

Mx - Shoulder Dystocia drill

Call for help, Empty bladder, Episiotomy

1. First do McRobert's Maneuver
2. Then apply Suprapubic Pressure
   Fundal pressure is contraindicated because it can cause further impaction of shoulders.

3. Rotational maneuvers
   A. Woods corkscrew maneuver
   B. Rubin maneuver

4. Removal of posterior arm - Jacquemier's maneuver.

5. Roll over maneuver (from supine position to all four limbs) also known as all four's maneuver. (gaskin's all four's maneuver)

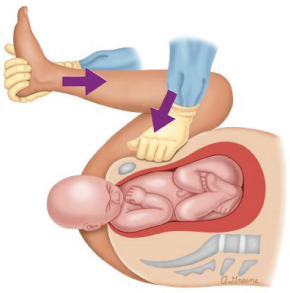
Active space

6. Last resort :- Zavanelli's maneuver - Push back the head into pelvis
Then do c-section.

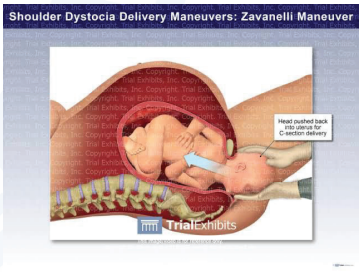
First / best maneuver

- .....→ sudden ..... & ..... of maternal lower limbs
- Complication - injury to .....
(Meralgia Paraesthetica)

- MC maternal complication of shoulder dystocia drill – .....
- MC neonatal complication of shoulder dystocia drill- .....
- Other Complication
  - Clavicle #
  - Humerus #
- Sternocleidomastoid hematoma→torticolis



Gaskin manoeuvre with rotation of posterior shoulder



Active space

INSTRUMENTAL DELIVERY

Instrumental Delivery

Forceps

Prerequisites

- Most important - station >= +2
- Cervix completely dilated
- Membranes ruptured
- No contracted pelvis
- No CPD
- No transverse, brow, mentoposterior
- Lithotomy
- Sometimes -anesthesia required
- Written Consent
- Enema/ catheterization
- Good uterine contractions
- Head should be rotated - if not completely rotated, maximum acceptable deflection from midline is .....

  At ..... - low forceps

  At ..... - outlet forceps - Head on the perineum, scalp visible at introitus, skull on pelvic floor

Old concept (not used now)

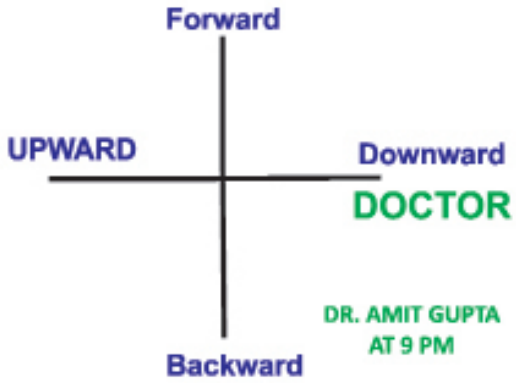
At 0 station ..... forceps

Above ischial spine ..... forceps

Active space

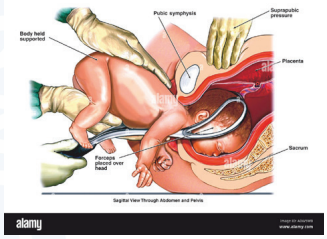
- Traction force - primi ..... kg and multi ..... kg

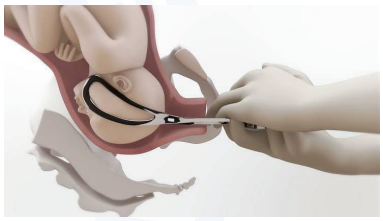
MC used forceps .....



Longest forceps .....

..... blade is introduced first



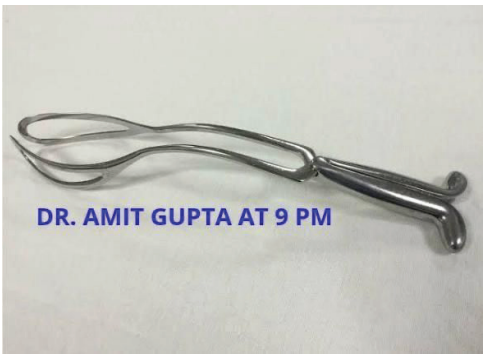


Direction of pull

- In Low forceps-.....

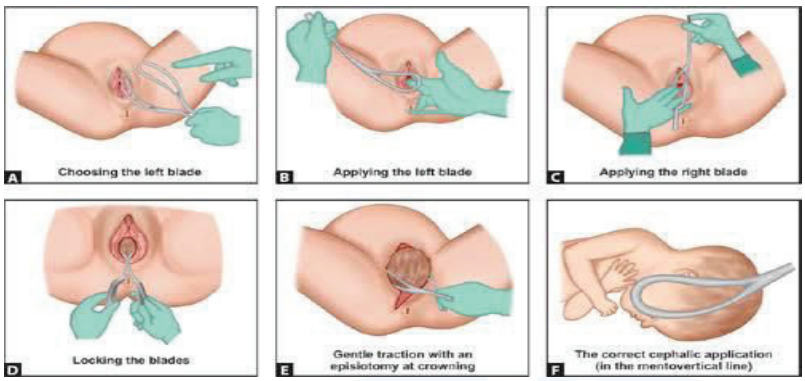
- In outlet forcep-.....

- Maximum pulls- ..... → if fails → failed forceps → C.S



Active space

MIST



Indication

- Maternal distress
- Fetal distress
- Prolonged 2nd Stage of Labour
- Prophylactic use of forceps in heart disease / PIH

Vacuum

- Silastic cup (M/C : Bell shaped Diameter : 5-7 cm)
- Applied on sagittal suture ..... anterior to posterior fontanelle
- Vaccum is attached to suction machine

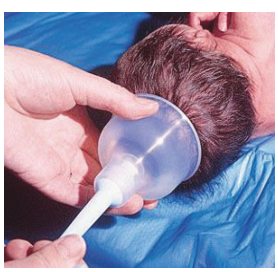
Suction pressure

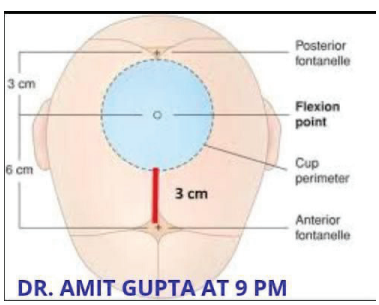
..... →Initial suction pressure

↓ ..... required

..... →Optimum suction pressure

- Max. - ..... fail →failed vacuum - Mx c-section





Active space

advantage of vacuum

- ..... cervical dilation is enough
- Less maternal injury
- No anesthesia required
- No lithotomy position required
- Can be Applied in ..... head, because vacuum promote .....
- Can be used in .....

Disdvantages

- Can be used only in vertex
- Can't be used in prematurity
- More fetal injury
- Cephalhaematoma
- Retinal haemorrhage

Episiotomy

- Deliberate incision on posterior vaginal wall to increase vulval outlet

Indication

- Primi-Rigid perineum (primipara)
- Instrumental delivery
- Assisted breech delivery
- Face to pubis delivery
- Shoulder dystocia drill

Active space

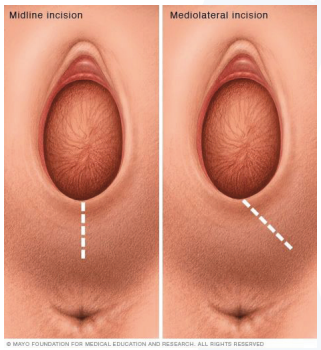
Timings

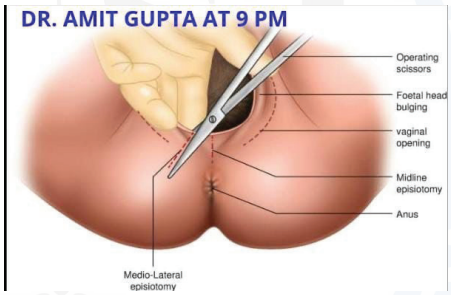
- In cephalic presentation – at .....
- In breech presentation (Mandatory) at .....

Types

1. Median→In midline → risk of anal sphincter
2. Mediolateral → Preferred because inscion can be extended

- Angle 60°>45°





Muscles cut in episiotomy

1. Bulbospongiosus (Bulbocavernosus)
2. Sup. & deep transverse perinei
3. Pubococcygeus

Other Structures whic are cut :-

1. Vaginal mucosa
2. Branch of Pudendal N/A
3. Skin & Subcutaneous tissue.

Active space

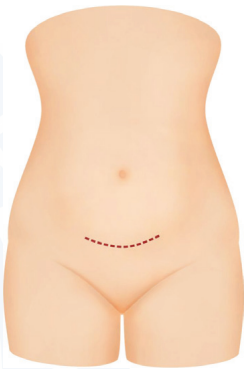
Episiotomy is ..... pereneal tear

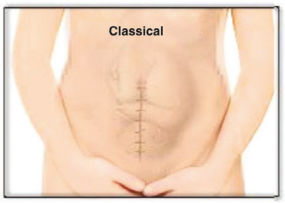
Caesarean Section

- Delivery of baby by abdominal & uterine inscion

Absolute indications

- Contracted pelvis
- CPD
- Transverse, brow, mentoposterior
- Placenta praevia
- Low lying placenta within 1 cm
- Vasa praevia
- 1st twin non - cephalic
- Cord prolapse
- Category 3 FHR
- Active genital herpes / HIV





1. Abdominal incision
   - MC-Low transverse (Pfannenstiel incision)
   - Cosmetically better

Other -paramedian – in fetal distress

2. Uterine incision
   - Lower- segment CS
   - MC-Low transverse (kerr incision)
   - Risk of scar rupture .....

Active space

Modified ..... classifi cation is used for caesarean section. It consists of 10 population categories.

Indication of upper segment C- Section (Classical C. S.)

- Cancer cervix
- Dense adhesions in lower segment
- Repaired Vesicovaginal fistula
- Postmortem
- Morbidly adhered placenta
- .....

Risk of scar rupture – .....

Vaginal birth after caesarean section (VBAC) →Or trial of scar

- Indication- H/O LSCS

C/I for VBAC .....

1. Previous H/O classical cs
2. Previous H/O ruptured uterus
3. Previous 3 LSCS
4. Myomectomy / hysterotomy
5. Cephalopelvic disproportion (CPD)
6. Other indication of CS mentioned above

Best prognostic criteria – H/O Previous VBAC

Contraindication of vaginal delivery - Absolute indications of .....

Obstetric Audit - Dealing with obstetric records of a hospital

Includes no. of vaginal delivery, C. S., indication and complication

Active space

Breastfeeding should start .....

Discharge

After vaginal delivery .....

After caesarean section

WHO define Maternal death as Within 42 Days after delivery from day of fertilization)

Fracturing clavicle to facilitate delivery of baby is .....

Prophylactic Antibiotics before CS - ..... before skin incision

Position of patient :- Supine but place a wedge under Right hip.

M/c Anesthesia :- .....

level of block - .....

Active space

FETAL WELL BEING ASSESSMENT

Fetal Well Being Assessment

Before delivery - Antepartum assessment

During deliver - Intra-partum assessment

Antepartum Assessment:-

First step - 1 . Fetal kick counts :- maximum at .....

Normal = at least ..... hrs of resting condition or at least ..... hrs of working conditions

↓

If they are less than this they are " sluggish fetal movements

↓

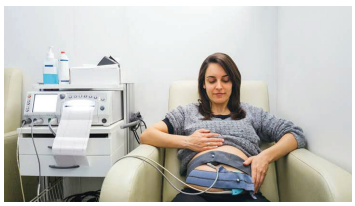
Next step

↓

2. Screening test for fetal well being = Modified biophysical profile score (NST + AFI)

Nonstress test : usvally started at .....

tie a belt around her abdomen and connect it to cardiotocogram



In 20 min - there should be at least 2 episodes of acceleration of FHR by ..... BPM

Active space

Which should sustain for at least ..... sec if not then .....
repeat for another 20 min

↓

in 40 min - if no 2 episodes = nst is non reactive

↓

measure afi by usg (normal= .....)

↓

if modified bpp is abnormal

↓

3. Definitive test for fetal well being--- .....
.....

↓

by using usg for .....

....generalized body movements = at least .....

...NST = at least 3 episodes of increase FHR by ..... bpm sustain for ..... sec

.....atleast one episode of complete extension with return to complete flexion

.....breathing - atleast one episode of 30 sec of movement of fetal diaphragm

.....single deepest pocket - atleast one pocket of 2 cm

every parameter = score is 2

max score = 10

upto 8 is normal

If <= .....

Active space

4. Fetal heart rate
5. Beat to beat variability
6. VAST (Vibroacoustic Stimulation Test)

Artificial larynx - Placed on maternal abdomen.

It delivers high frequency sound waves. If there is no hypoxia ..... HR

(Normal response - immediately Rules out Hypoxia)

Intrapartum Assessment (During Labor) - by continuous electrical ....

M/c used = Intermittent Auscultation of FHR.

Immediately after a contraction x (1 min)

1. Fetal heart rate

normal..... bpm

2. Beat to beat variability ..... bpm

if heart rate is fixed

↓

fixed or sinusoidal heart rate

Causes-

severe fetal anemia

rh incompatibility

twin- twin transfusion syndrome

vasa parevia

Active space

FHR monitoring---

Normally	High Risk
Stage 1 =..... Min Interval	..... Min Interval
Satge 2=..... Min Interval	.....Min Interval

3. Fetal scalp blood ph

Normal .....

If it is between ..... →repeat after 30 min →.....

.....

If ..... – immediate c- section

4. Fetal pulse oxymetry

sensor is put on cheek

Normal .....

If between ..... →repeat after 30 min →if same –

5. Cadiotocography (CTG)..... NST at labor

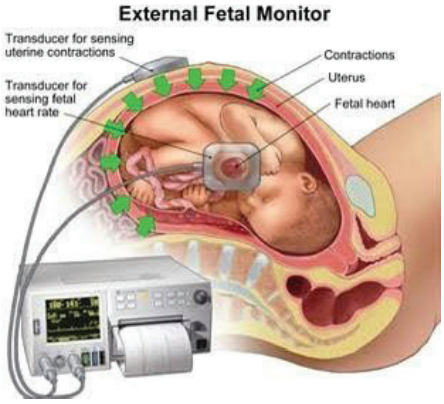
when uterus contracts →blood supply to fetus →

normal fetus overcome reduced blood supply

if hypoxia- ↓ FHR

2 sensor A. One to record FHR

B. One to record uterine contraction



abnormalities

a. Early deceleration → Physiological

in .....

Onset and peak of Dip in FHR Coincides with onset & peak of

.....

Active space

Early decelerations - no hypoxia

b. Late decleration → Pathological

peak of fhr decleration continues even after uterine contraction

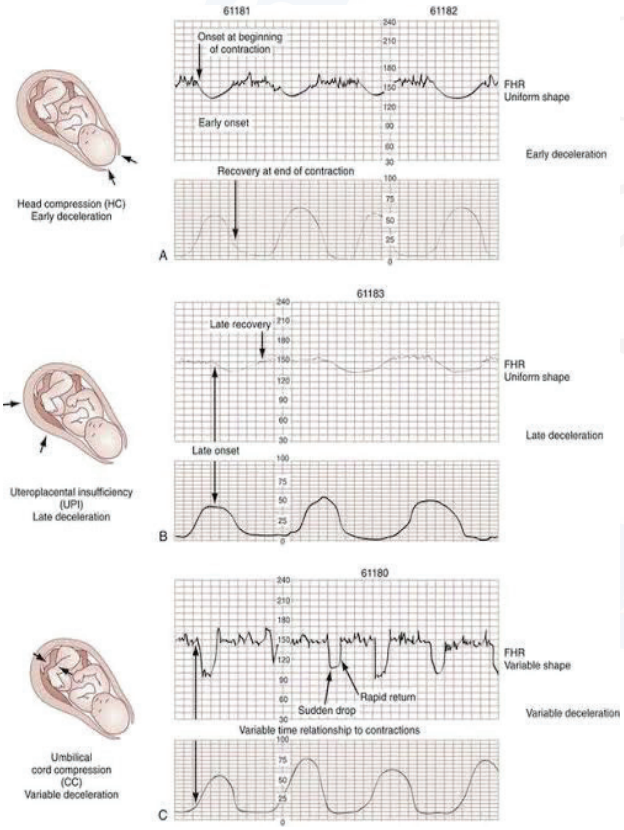
is over seen in ..... pathological

c. Variable decleration

seen in .....

d. Recurrent variable decleration

seen in ..... immediate c- section



Worst Prognosis (Maximum hypoxia)

Sinusoidal > late deceleration > variable deceleration

Category 3 FHR -

1. Sinusdidal HR

Active space

MIST

2. Absent beat to beat variability with any of the following -
   a) Bradycardia
   b) Recurrent late decelerations
   c) Recurrent variable decelerations
      - Worst prognosis
      - Immediate delivery (Emergency CS)

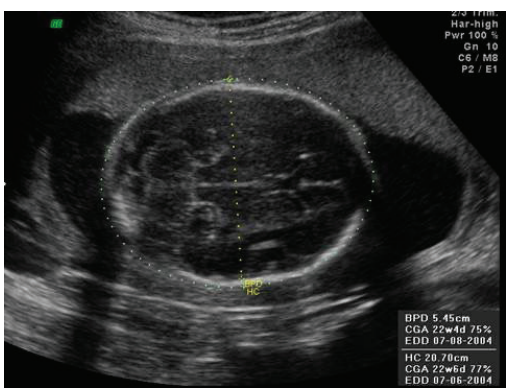
Ultrasound scans at.....Gestational age
Dating scan/ viability scan at 6-8 weeks
NT NB scan at 11-13 weeks
TIFFA scan at 18-20 weeks → Level II or anomaly .....or targeted imaging for fetal anomalies
Growth scan at 30-32 weeks → Fetal weight
Fetal echo at 22-24 weeks
In twins for chorionicity 10-14 weeks
For Twin Twin transfusion syndrome First Trimester , Second trimester
Best time for placental localisation by USG is third trimester- 32-36wks

Best Ultrasound parameter for

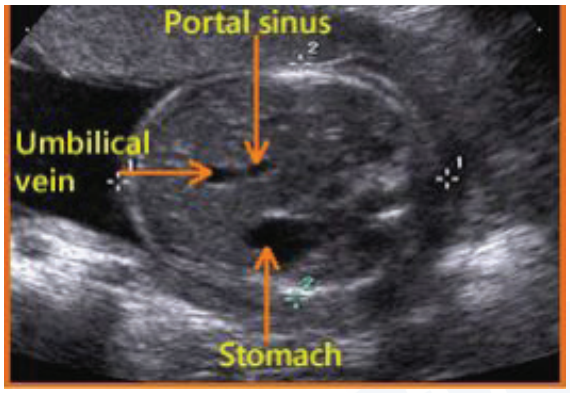
1. Assessment of Gestational age..... (Crown - Rump Length)
2. Assessment of Gestational age in first trimester.....
3. Assessment of Gestational age in second trimester BPD
4. Assessment of Gestational age in third trimester.....
5. Assessment of Fetal weight

BPD (Biparietal Diameter)

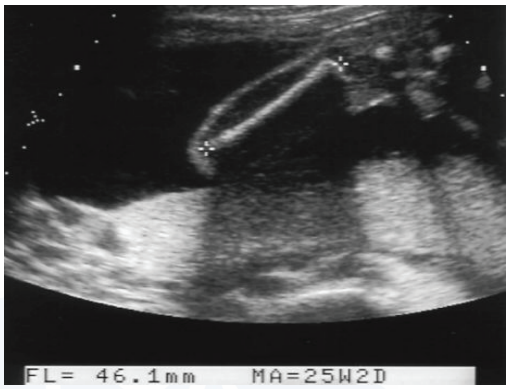
(Crown - Rump Length) CRL



Active space

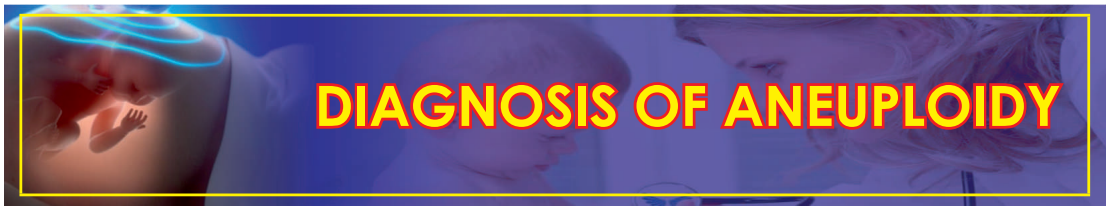


(Abdominal Cercumference) AC



(Femur Length) FL

Active space



DIAGNOSIS OF ANEUPLOIDY

Diagnosis of Aneuploidy

1. In 1st trimester

Screening test..... radiology USG (NT-NB Scan) + Pathology (dual test)

hcg .....

papp-a .....

nt - nuchal translucency

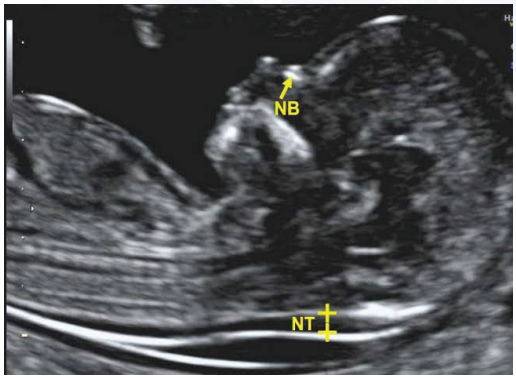
Normal .....

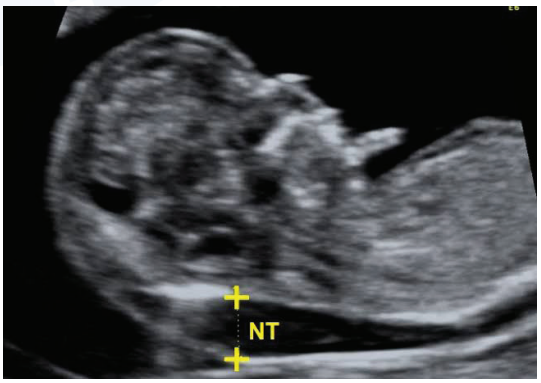
Abnormal .....

nb - nasal bone

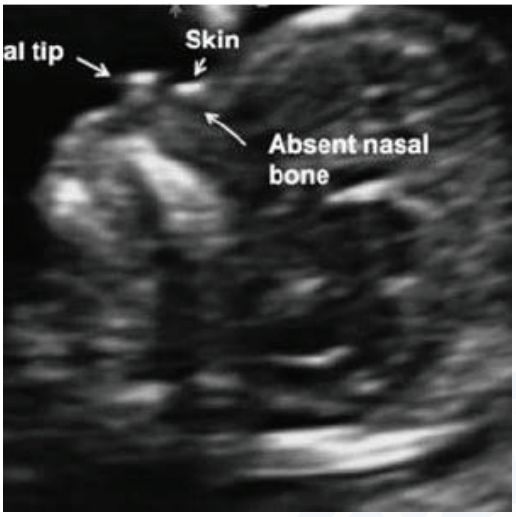
Normal - .....

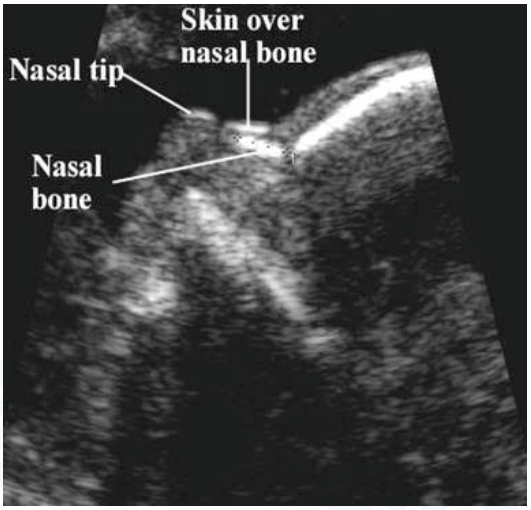
Abnormal - .....





Active space



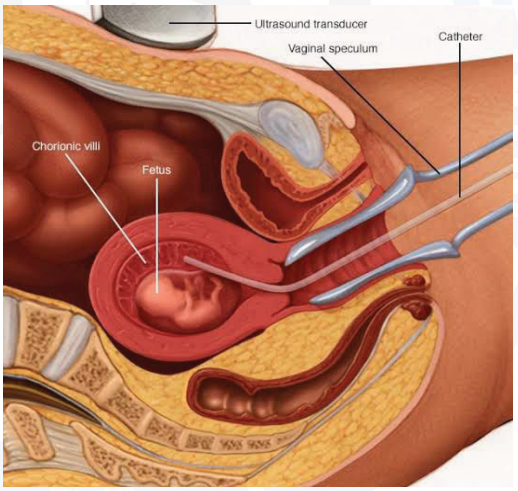


Even if one parameter is abnormal – test is abnormal →definitive test in 1st trimester - .....

Best - 11-13 weeks

Till 10-12 weeks -tvs

After >12 week - tas



Diagnosis

.....dna analysis

.....biochemical analysis

Comlication

mc..... abortion

Complication of cvs if performd before 10 weeks

Limb reduction syndrome

Oromandibular hypogenesis

Active space

2nd trimester 2^0 screening test NIPT (Non invasive prenatal test)
- Cell free fetal DNA Test

Screening test

.....

Triple Test

hcg .....

Alpha feto protein.....

Unconjugated estriol.....

Quadruple test

Triple test + inhibin a .....

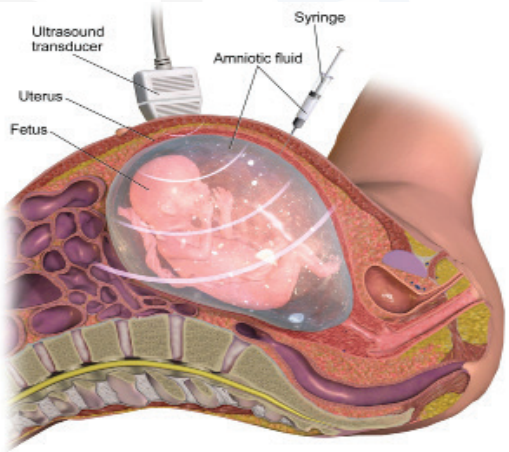
Definitive test - ..... (best done..... week)

can be Done between ..... wks

USG → A. Nuchal fold

Thickness - Normal → .....

B. Short Femur



Integrated Test

→T_1 & T_2 Test

Adv..... neural tube defect can also be diagnosed

diagnosis

.....dna analysis

.....biochemical analysis

.....neural tube defect can also be diagnosed

Best test for NTD -

Diagnostic test - Amniotic fluid ..... > USG.

(Invasive - Diagnostic amniocentesis)

Complication –abortion but risk is less than chorion villous sampling cvs)

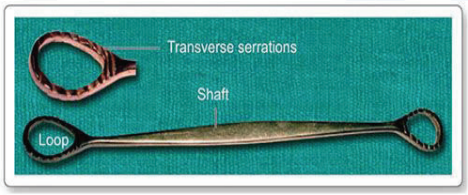
Active space

MIST

Name : ..... speculum
..... double bladed posterior vaginal speculum

Uses : To retract ..... vaginal wall
1. 2 blades
2. ..... difference in size of blades

Name : ..... vaginal wall retractor



Uses : To retract ..... vaginal wal
1. 2 Loop
2. Transverse serrations at loop
3. Loop faces ..... while retracting

Name : Hegar Dilator

Uses : 1. To dilate .....
2. To diagnose incompetent cervix in ..... lady
A. ..... difference on either side
B. .....

Name : Hawkin - .....

Uses : 1. ..... ended
2. To dilate .....
3. Not used now

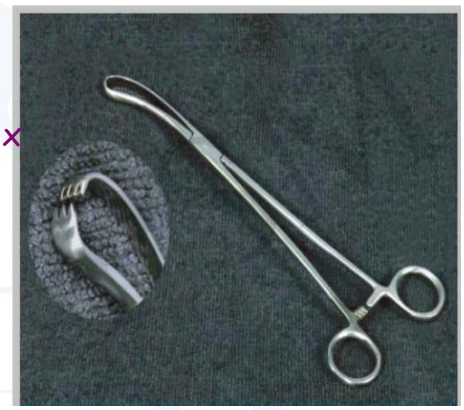
Active space

Name : ..... Self Retaining
..... Bladed Speculum

Uses : To Retract .....
Vaginal Walls Simultaneovsly

Name : .....

Uses : 1. To hold ..... of cervix
2. Concavity faces .....
3. ..... teeth at tip

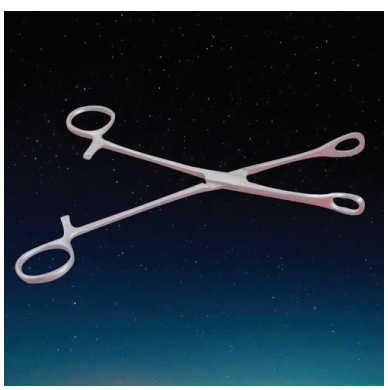


Name : .....

Uses : 1. To hold anterior lip of cervix
in .....
: 2. .....Teeth

Name : .....

Uses : 1. To ..... valva, vaginal, abdominal wall
2. Can provide .....
by compression
- Ring shaped ends c̄ serrations



Active space

Name : .....

Uses : 1. Acts as ..... dilator
2. To measure .....

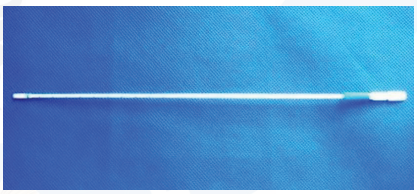
3. To know .....of uterus
4. Can give idea about
A. .....Uterus
B. Misplaced .....

Name : .....

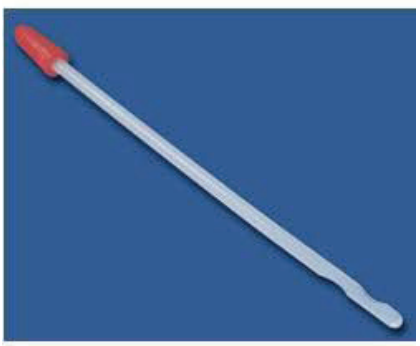
Uses : For .....
In obstetrics - .....
except .....
pregnancy
In gynaecology - ..... end
- Serrations at handle

Name :

Uses : 1. For endometrial .....
- Has round opening
- Has stilette
A. Helps in removing .....
.....
B. Helps in .....



Name : .....



Uses : 1. For .....
2. May be attached to .....
syringe
3. Endometrial
Aspiration Cytology in
..... Plastic
..... Openings

Active space

Name : .....

Uses : 1. To instill radioopaque dye ..... in HSG
2. To instill ..... in laparoscopic chromopertubation
- Funned shaped end c serrations

Name : .....

Uses : 1. To remove products of conception after .....
- Spoon shaped ends
- No lock

Name : .....

Uses : To hold ..... structures
Eg. Fallopian tube round ligament etc.
- Ends are turned inward
- No teeth
- Gap at tip

Name : .....

Uses : For .....
- Blade c shard end is put on ..... vaginal wall
- Length .....
- Has angle

Active space

Name .....

Uses 1. To cut .....
2. Coin-shaped ends c̄ space
3. Length .....

Name .....

Uses Applied approx.
..... from umbilicus

Name .....

Uses 1. For artificial rupture of membranes .....
- Has ..... teeth on one side
- Has ..... teeth on other side

2. Used to hold tough structures

Name .....

Uses To hold edges of .....
- Has flat triangular, broad tip with serrations

Active space

Name .....

Uses To hold .....
- Hanble : Blade ratio
.....
- Has longitudinal
Groove {Eye} c̄ Serrations

Name .....

Uses To retract .....

Name .....

Uses To hold .....

Active space

Name .....

Uses For .....

A. Ayre spatula .....

B. Endocervical brush .....

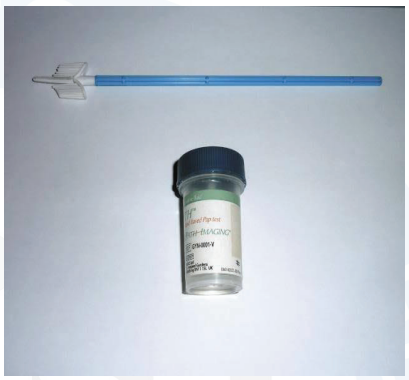
C. .....

Name .....

Uses For .....

cytobroom .....

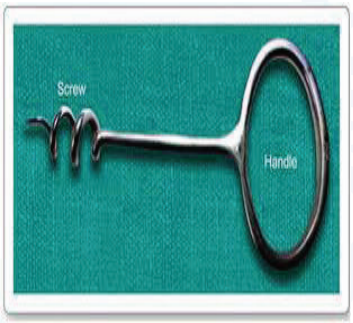
.....



Name .....

Uses To hold .....

during .....



Name .....

Uses 1. To release blood supply to

.....

Active space

ANTENATAL ADVICE (PSM & OBG)

Ideal number of antenatal visits =

Minimum as recommended by WHO = ...

Minimum as recommended by GOI. .....

Investigations in fisrt antenatal visit

1. Blood group -ABO /Rh
2. Hb / CBC
3. VDRL
4. HIV (opt out screening)
5. HBsAg
6. Urine routine microscopy
7. Indian Guidelines - DIPSI criteria - screening at .....
   - If normal - repeat test @ .....

Inj. Dt1 0.5 ml i/m
2nd dose after .....
only booster is given if she received vaccine within last 3 yrs

Active space

Nutritional Advice

1. Add = ...

2. Rich protein / iron diet

3. Take iodised salt

1st Trimester
Folic acid (Prophylactic Dose) - ..... start 1 month before pregnancy till 3 months after pregnancy

Therapeutic Dose - ..... If H/O Neural Tube Defects (NTD) - start 3 month before pregnancy till 3 months after pregnancy

Dose of folic acid in
Diabetes .....
On anti-epileptic drugs - Before pregnancy .....
- After pregnancy .....

For treatment of megaloblastic anemia .....
In sickle cell anemia .....

2nd Trimester
IFA tablets - 60 mg + 0.5 mg folic acid
1 tablet / day x 6 months during preganancy to 6 months after preganancy

Calcium - RDA 1000 mg
- Dose 500 mg x 2 tablets / day

Time gap between IFA & calcium .....

Active space

Active space